

Human review packet: MSVV07AdWords

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: MSVV07AdWords
Paper id: MSVV07AdWords
Source version: Journal of the ACM, 2007
Source record: audit/paper_statement_map.json
Rows: 50
Generated: 2026-09-27
Source URL: [official source](#)

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper MSVV07AdWords --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

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Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.Online.AdWordsInstance

Declaration location: reusable library

Lean declaration body

```
field budget:
  {Advertiser : Type u_1} →
  {Query : Type u_2} → AppliedModelingLib.Online.AdWordsInstance Advertiser Query →  $\mathbb{R}$ 

field bid:
  {Advertiser : Type u_1} →
  {Query : Type u_2} → AppliedModelingLib.Online.AdWordsInstance Advertiser Query → Advertiser → Query →  $\mathbb{R}$ 
```

AppliedModelingLib.Online.AdWordsInstance.Assignment

Declaration location: reusable library

Lean declaration body

```
type:
  Type u_3 → Type u_4 → Type (max u_4 u_3)

value:
  fun (Advertiser : Type u_3) (Query : Type u_4) => Query → Option Advertiser
```

AppliedModelingLib.Online.AdWordsInstance.ChoiceRule

Declaration location: reusable library

Lean declaration body

```
type:
  Type u_3 → Type u_4 → Type (max u_4 u_3)

value:
  fun (Advertiser : Type u_3) (Query : Type u_4) =>
    AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Query → Option Advertiser
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)

AppliedModelingLib.Online.AdWordsInstance.HistoryState

Declaration location: reusable library

Lean declaration body

```
field assignment:
  {Advertiser : Type u_3} →
  {Query : Type u_4} →
    AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query →
    AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query

field seen:
  {Advertiser : Type u_3} →
  {Query : Type u_4} → AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query → Finset Query
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)

AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} → {Query : Type u_2} → AppliedModelingLib.Online.AdWordsInstance Advertiser Query → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query) =>
  ∀ (a : Advertiser) (q : Query), 0 ≤ I.bid a q
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)

AppliedModelingLib.Online.AdWordsInstance.NonnegativeBudgets

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} → {Query : Type u_2} → AppliedModelingLib.Online.AdWordsInstance Advertiser Query → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query) =>
  ∀ (a : Advertiser), 0 ≤ I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)

AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} → {Query : Type u_2} → AppliedModelingLib.Online.AdWordsInstance Advertiser Query → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query) =>
  ∀ (a : Advertiser), 0 < I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)

AppliedModelingLib.Online.AdWordsInstance.assignQuery

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[DecidableEq Query] →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query →
Query → Advertiser → Query → Option Advertiser

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [DecidableEq Query]
  (A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) (q : Query) (a : Advertiser)
  (a_1 : Query) =>
  if a_1 = q then some a else A a_1
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)

AppliedModelingLib.Online.AdWordsInstance.balanceDiscount

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (x :  $\mathbb{R}$ ) => 1 - Real.exp (x - 1)
```

AppliedModelingLib.Online.AdWordsInstance.emptyAssignment

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} → {Query : Type u_2} → Query → Option Advertiser

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (a : Query) => none
```

AppliedModelingLib.Online.AdWordsInstance.historyFinset

Declaration location: reusable library

Lean declaration body

```
type:
{Query : Type u_2} → [DecidableEq Query] → List Query → Finset Query

value:
fun {Query : Type u_2} [DecidableEq Query] (a : List Query) =>
  List.brecOn a AppliedModelingLib.Online.AdWordsInstance.historyFinset._f
```

AppliedModelingLib.Online.AdWordsInstance.initialHistoryState

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} → [DecidableEq Query] → AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [DecidableEq Query] =>
  { assignment := AppliedModelingLib.Online.AdWordsInstance.emptyAssignment, seen :=  $\emptyset$  }
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance.HistoryState](#)
- [AppliedModelingLib.Online.AdWordsInstance.emptyAssignment](#)

AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Advertiser] →
[Nonempty Advertiser] → AppliedModelingLib.Online.AdWordsInstance Advertiser Query → Query →  $\mathbb{R}$ 

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [Nonempty Advertiser]
  (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query) (q : Query) =>
  let bids : Finset  $\mathbb{R}$  := Finset.image (fun (a : Advertiser) => I.bid a q) Finset.univ;
  have hbids : bids.Nonempty := AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery._proof_1 I q;
  max 0 (bids.max' hbids)
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)

AppliedModelingLib.Online.AdWordsInstance.revenue

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query]
  (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) =>
  ∑ q : Query,
  match A q with
  | none => 0
  | some a => I.bid a q
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)

AppliedModelingLib.Online.AdWordsInstance.spend

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) (a : Advertiser) =>
  ∑ q : Query,
  match A q with
  | none => 0
  | some a' => if a' = a then I.bid a q else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)

AppliedModelingLib.Online.AdWordsInstance.CanAssign

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Query → Advertiser → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) (q : Query) (a : Advertiser) =>
  I.spend A a + I.bid a q ≤ I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.spend](#)

AppliedModelingLib.Online.AdWordsInstance.Feasible

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
(A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) =>
∀ (a : Advertiser), I.spend A a ≤ I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.spend](#)

AppliedModelingLib.Online.AdWordsInstance.IsOptimalAssignment

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
(A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) =>
I.Feasible A ∧
∀ (B : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query),
I.Feasible B → I.revenue B ≤ I.revenue A
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.Feasible](#)
- [AppliedModelingLib.Online.AdWordsInstance.revenue](#)

AppliedModelingLib.Online.AdWordsInstance.offlineOptimumAssignment

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Advertiser] →
[Fintype Query] →
[DecidableEq Advertiser] →
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query) →
I.NonnegativeBudgets → Query → Option Advertiser

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query) (hbudget : I.NonnegativeBudgets) (a : Query) =>
Classical.choose (AppliedModelingLib.Online.AdWordsInstance.exists_optimalAssignment I hbudget) a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.IsOptimalAssignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBudgets](#)

AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Advertiser] →
[Fintype Query] →
[DecidableEq Advertiser] →
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query) → I.NonnegativeBudgets → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query) (hbudget : I.NonnegativeBudgets) =>
I.revenue (I.offlineOptimumAssignment hbudget)
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBudgets](#)
- [AppliedModelingLib.Online.AdWordsInstance.offlineOptimumAssignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.revenue](#)

AppliedModelingLib.Online.AdWordsInstance.spentFraction

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
(A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) (a : Advertiser) =>
I.spend A a / I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.spend](#)

AppliedModelingLib.Online.AdWordsInstance.balanceScore

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Advertiser → Query → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) (a : Advertiser) (q : Query) =>
  I.bid a q * AppliedModelingLib.Online.AdWordsInstance.balanceDiscount (I.spentFraction A a)
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.balanceDiscount](#)
- [AppliedModelingLib.Online.AdWordsInstance.spentFraction](#)

AppliedModelingLib.Online.AdWordsInstance.IsBalanceChoice

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Advertiser] →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Query → Advertiser → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (A : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) (q : Query) (a : Advertiser) =>
  I.CanAssign A q a ∧ ∀ (b : Advertiser), I.CanAssign A q b → I.balanceScore A b q ≤ I.balanceScore A a q
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.CanAssign](#)
- [AppliedModelingLib.Online.AdWordsInstance.balanceScore](#)

AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Advertiser] →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query → Query → Option Advertiser

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (a : AppliedModelingLib.Online.AdWordsInstance.Assignment Advertiser Query) (a_1 : Query) =>
  if h : ∃ (a_2 : Advertiser), I.CanAssign a a_1 a_2 then
    some
      (Classical.choose (AppliedModelingLib.Online.AdWordsInstance.exists_balanceChoice_of_exists_canAssign I a a_1 h))
  else none
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.CanAssign](#)
- [AppliedModelingLib.Online.AdWordsInstance.IsBalanceChoice](#)

AppliedModelingLib.Online.AdWordsInstance.stepHistoryState

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
[DecidableEq Query] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.ChoiceRule Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query →
Query → AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser] [DecidableEq Query]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
(rule : AppliedModelingLib.Online.AdWordsInstance.ChoiceRule Advertiser Query)
(S : AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query) (q : Query) =>
if q ∈ S.seen then { assignment := S.assignment, seen := insert q S.seen }
else
match rule S.assignment q with
| none => { assignment := S.assignment, seen := insert q S.seen }
| some a =>
{ assignment := AppliedModelingLib.Online.AdWordsInstance.assignQuery S.assignment q a, seen := insert q S.seen }
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.ChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.HistoryState](#)
- [AppliedModelingLib.Online.AdWordsInstance.assignQuery](#)

AppliedModelingLib.Online.AdWordsInstance.runHistoryStateFrom

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
[DecidableEq Query] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.ChoiceRule Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query →
List Query → AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser] [DecidableEq Query]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
(rule : AppliedModelingLib.Online.AdWordsInstance.ChoiceRule Advertiser Query)
(a : AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query) (a_1 : List Query) =>
List.brecOn (motive := fun (x : List Query) =>
AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query)
a_1 (AppliedModelingLib.Online.AdWordsInstance.runHistoryStateFrom._f I rule) a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.ChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.HistoryState](#)
- [AppliedModelingLib.Online.AdWordsInstance.stepHistoryState](#)

AppliedModelingLib.Online.AdWordsInstance.runHistoryState

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[FIntype Query] →
[DecidableEq Advertiser] →
[DecidableEq Query] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.ChoiceRule Advertiser Query →
List Query → AppliedModelingLib.Online.AdWordsInstance.HistoryState Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [FIntype Query] [DecidableEq Advertiser] [DecidableEq Query]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
(rule : AppliedModelingLib.Online.AdWordsInstance.ChoiceRule Advertiser Query) (history : List Query) =>
I.runHistoryStateFrom rule AppliedModelingLib.Online.AdWordsInstance.initialHistoryState history
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.ChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.HistoryState](#)
- [AppliedModelingLib.Online.AdWordsInstance.initialHistoryState](#)
- [AppliedModelingLib.Online.AdWordsInstance.runHistoryStateFrom](#)

AppliedModelingLib.Online.AdWordsInstance.runAssignment

Declaration location: reusable library

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[FIntype Query] →
[DecidableEq Advertiser] →
[DecidableEq Query] →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
AppliedModelingLib.Online.AdWordsInstance.ChoiceRule Advertiser Query →
List Query → Query → Option Advertiser

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [FIntype Query] [DecidableEq Advertiser] [DecidableEq Query]
(I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
(rule : AppliedModelingLib.Online.AdWordsInstance.ChoiceRule Advertiser Query) (history : List Query) (a : Query) =>
(I.runHistoryState rule history).assignment a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.ChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.HistoryState](#)
- [AppliedModelingLib.Online.AdWordsInstance.runHistoryState](#)

AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
(Advertiser → Query → ℝ) → AppliedModelingLib.Online.AdWordsInstance Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
(effectiveBid : Advertiser → Query → ℝ) =>
{ budget := I.budget, bid := effectiveBid }
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)

AppliedModelingLib.Online.AdWordsInstance.withAdvertiserWeights

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
  (Advertiser → ℝ) → AppliedModelingLib.Online.AdWordsInstance Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (weight : Advertiser → ℝ) =>
  I.withEffectiveBids fun (a : Advertiser) (q : Query) => weight a * I.bid a q
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids](#)

AppliedModelingLib.Online.AdWordsInstance.withAvailability

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
  (active : Advertiser → Query → Prop) →
  [(a : Advertiser) → (q : Query) → Decidable (active a q)] →
  AppliedModelingLib.Online.AdWordsInstance Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (active : Advertiser → Query → Prop) [(a : Advertiser) → (q : Query) → Decidable (active a q)] =>
  I.withEffectiveBids fun (a : Advertiser) (q : Query) => if active a q then I.bid a q else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids](#)

AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  AppliedModelingLib.Online.AdWordsInstance Advertiser Query →
  (Advertiser → Query → ℝ) → AppliedModelingLib.Online.AdWordsInstance Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (I : AppliedModelingLib.Online.AdWordsInstance Advertiser Query)
  (ctr : Advertiser → Query → ℝ) =>
  I.withEffectiveBids fun (a : Advertiser) (q : Query) => ctr a q * I.bid a q
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids](#)

AppliedModelingLib.Online.BMatchingIntegralPrefixChoice

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \text{Type}$ 

value:
fun (N :  $\mathbb{N}$ ) => (Fin N  $\rightarrow$  Finset (Fin N))  $\rightarrow$  Fin N  $\rightarrow$  Option (Fin N)
```

AppliedModelingLib.Online.BMatchingIntegralPrefixChoice.Feasible

Declaration location: paper-local

Lean declaration body

```
type:
{N :  $\mathbb{N}$ }  $\rightarrow$  AppliedModelingLib.Online.BMatchingIntegralPrefixChoice N  $\rightarrow$  Prop

value:
fun {N :  $\mathbb{N}$ } (choice : AppliedModelingLib.Online.BMatchingIntegralPrefixChoice N) =>
 $\forall$  (obs : Fin N  $\rightarrow$  Finset (Fin N)) (round bidder : Fin N), choice obs round = some bidder  $\rightarrow$  bidder  $\in$  obs round
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingIntegralPrefixChoice](#)

AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \text{Type}$ 

value:
fun (N :  $\mathbb{N}$ ) => { choice : AppliedModelingLib.Online.BMatchingIntegralPrefixChoice N // choice.Feasible }
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingIntegralPrefixChoice](#)
- [AppliedModelingLib.Online.BMatchingIntegralPrefixChoice.Feasible](#)

AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation

Declaration location: paper-local

Lean declaration body

```
type:
{N :  $\mathbb{N}$ }  $\rightarrow$  AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N  $\rightarrow$  (Fin N  $\rightarrow$  Finset (Fin N))  $\rightarrow$  Fin N  $\rightarrow$  Fin N  $\rightarrow \mathbb{R}$ 

value:
fun {N :  $\mathbb{N}$ } (algorithm : AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N) (obs : Fin N  $\rightarrow$  Finset (Fin N))
(round bidder : Fin N) =>
if  $\uparrow$  algorithm obs round = some bidder then 1 else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm](#)
- [AppliedModelingLib.Online.BMatchingIntegralPrefixChoice](#)
- [AppliedModelingLib.Online.BMatchingIntegralPrefixChoice.Feasible](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaU

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun ( $\kappa$  :  $\mathbb{R}$ ) =>  $\kappa^{-1}$ 
```

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaY

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ : ℝ) => AppliedModelingLib.Online.MSVV07Appendix.derivedKappaU κ ^ 2
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaU](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaZ

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ : ℝ) => Real.exp (1 - AppliedModelingLib.Online.MSVV07Appendix.derivedKappaU κ)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaU](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ : ℝ) =>
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaZ κ + AppliedModelingLib.Online.MSVV07Appendix.derivedKappaY κ
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaY](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaZ](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaB

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ : ℝ) =>
  (AppliedModelingLib.Online.MSVV07Appendix.derivedKappaZ κ - 1) /
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator κ
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaZ](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ :  $\mathbb{R}$ ) =>
  (AppliedModelingLib.Online.MSVV07Appendix.derivedKappaZ κ - 1 +
   AppliedModelingLib.Online.MSVV07Appendix.derivedKappaU κ) /
   AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator κ
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaU](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaZ](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaT

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ :  $\mathbb{R}$ ) => (AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator κ)-1
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaX

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ :  $\mathbb{R}$ ) =>
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaY κ /
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator κ
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaDenominator](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaY](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ :  $\mathbb{R}$ ) =>
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaB κ + AppliedModelingLib.Online.MSVV07Appendix.derivedKappaX κ +
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaT κ
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaB](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaT](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaX](#)

AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (κ :  $\mathbb{R}$ ) =>
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaB κ +
  κ * AppliedModelingLib.Online.MSVV07Appendix.derivedKappaX κ
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaB](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaX](#)

AppliedModelingLib.Online.MSVV07Appendix.instDecidable_mSVV07AdWords

Declaration location: paper-local

Lean declaration body

```
type:
(p : Prop) → Decidable p

value:
fun (p : Prop) => Classical.propDecidable p
```

AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R}$ 

value:
f (_bidder :  $\mathbb{R}$ ) in 0..1, 1
```

AppliedModelingLib.Online.MSVV07Appendix.printedIndividualBudget

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (_bidder :  $\mathbb{R}$ ) => 1
```

AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{R} \rightarrow \mathbb{R}$ ) → Prop

value:
fun (spend :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
  IntervalIntegrable spend MeasureTheory.volume 0 1 ∧
  ∀ bidder ∈ Set.Icc 0 1,
    0 ≤ spend bidder ∧ spend bidder ≤ AppliedModelingLib.Online.MSVV07Appendix.printedIndividualBudget bidder
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.printedIndividualBudget](#)

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneAllocationRate

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (r :  $\mathbb{R}$ ) => (9 / 10 - r)-1
```

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneEligible

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \text{Prop}$ 

value:
fun (r bidder :  $\mathbb{R}$ ) => 1 / 10 + r ≤ bidder ∧ bidder ≤ 1
```

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneRawBid

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (r bidder :  $\mathbb{R}$ ) => if AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneEligible r bidder then 1 else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.instDecidable_mSVV07AdWords](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneEligible](#)

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneSpentBefore

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (r bidder :  $\mathbb{R}$ ) =>
  ∫ (s :  $\mathbb{R}$ ) in 0..r,
    if AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneEligible s bidder then
      AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneAllocationRate s
    else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.instDecidable_mSVV07AdWords](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneAllocationRate](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneEligible](#)

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeRawBid

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (owner bidder :  $\mathbb{R}$ ) => if bidder = owner then 1 else 0
```

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R}$ 

value:
3 / 5 - Real.log (9 / 5)
```

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeAllocationRate

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (progress owner bidder :  $\mathbb{R}$ ) =>
  if
    0 ≤ progress ∧
    progress ≤ AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail ∧ bidder = owner then
    1
  else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail](#)

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeServedMass

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R}$ 

value:
1 / 2 * AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail](#)

AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R}$ 

value:
2 / 5 + 2 * (1 / 10) + AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeServedMass
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeServedMass](#)

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoAllocationDensity

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (bidder :  $\mathbb{R}$ ) => if 1 / 2 ≤ bidder ∧ bidder ≤ 1 then 2 else 0
```

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoRawBid

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (offset bidder :  $\mathbb{R}$ ) => if  $1 / 2 \leq \text{bidder} \wedge \text{bidder} \leq 1$  then 2 else if bidder = offset then 1 else 0
```

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoSpentBefore

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (offset bidder :  $\mathbb{R}$ ) =>
  AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneSpentBefore (2 / 5) bidder +
   $\int (s : \mathbb{R}) \text{ in } 0..offset,$ 
    AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoRawBid s bidder *
    AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoAllocationDensity bidder
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneSpentBefore](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoAllocationDensity](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoRawBid](#)

AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeSpentBefore

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (progress owner :  $\mathbb{R}$ ) =>
  AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoSpentBefore (1 / 10) owner +
   $\int (s : \mathbb{R}) \text{ in } 0..progress,$ 
    AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeAllocationRate s owner owner *
    AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeRawBid owner owner
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeAllocationRate](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeRawBid](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoSpentBefore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.instDecidable_mSVV07AdWords

Declaration location: paper-local

Lean declaration body

```
type:
(p : Prop) → Decidable p

value:
fun (p : Prop) => Classical.propDecidable p
```

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3GreedyTwoWordRatio

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (c epsilon :  $\mathbb{R}$ ) => (c + epsilon) / (2 * c + epsilon)
```

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5ExactTypeSpend

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} → (k :  $\mathbb{N}$ ) → (Advertiser →  $\mathbb{R}$ ) → (Advertiser → Fin k) → Prop

value:
fun {Advertiser : Type u_1} (k :  $\mathbb{N}$ ) (spentFraction : Advertiser →  $\mathbb{R}$ ) (finalType : Advertiser → Fin k) =>
   $\forall (a : \text{Advertiser}), \text{spentFraction } a = \uparrow(\uparrow(\text{finalType } a) + 1) / \uparrow k$ 
```

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6BetaLowerBounds

Declaration location: paper-local

Lean declaration body

```
type:
{k :  $\mathbb{N}$ } →  $\mathbb{R} \rightarrow (\text{Fin } k \rightarrow \mathbb{R}) \rightarrow (\text{Fin } k \rightarrow \mathbb{R}) \rightarrow \text{Prop}$ 

value:
fun {k :  $\mathbb{N}$ } (alphaTotal :  $\mathbb{R}$ ) (alpha beta : Fin k →  $\mathbb{R}$ ) =>
   $\forall (i : \text{Fin } k), (\text{alphaTotal} - \sum j \in \text{Finset.lic } i, \text{alpha } j) / \uparrow k \leq \text{beta } i$ 
```

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8WeightedAlphaBeta

Declaration location: paper-local

Lean declaration body

```
type:
{k :  $\mathbb{N}$ } → (Fin k →  $\mathbb{R}$ ) → (Fin k →  $\mathbb{R}$ ) → (Fin k →  $\mathbb{R}$ ) → Prop

value:
fun {k :  $\mathbb{N}$ } (psi alpha beta : Fin k →  $\mathbb{R}$ ) =>  $\sum i : \text{Fin } k, \text{psi } i * \text{alpha } i \leq \sum i : \text{Fin } k, \text{psi } i * \text{beta } i$ 
```

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment

Declaration location: paper-local

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_2 u_1)

value:
fun (Advertiser : Type u_1) (Query : Type u_2) => Query → Option Advertiser
```

AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment

Declaration location: paper-local

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_2 u_1)

value:
fun (Advertiser : Type u_1) (Query : Type u_2) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance

Declaration location: paper-local

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_1 u_2)

value:
fun (Advertiser : Type u_1) (Query : Type u_2) => AppliedModelingLib.Online.AdWordsInstance Advertiser Query
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance

Declaration location: paper-local

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_1 u_2)

value:
fun (Advertiser : Type u_1) (Query : Type u_2) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperMsvvRatio

Declaration location: paper-local

Lean declaration body

```
type:
ℝ

value:
1 - 1 / Real.exp 1
```

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperRevenue

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[FIntype Query] →
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [FIntype Query]
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query)
  (A : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query) =>
  ∑ q : Query,
  match A q with
  | none => 0
  | some a => I.bid a q
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperSmallBids

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} → AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query → ℝ → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) (epsilon : ℝ) =>
  ∀ (a : Advertiser) (q : Query), l.bid a q ≤ epsilon * l.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperSpend

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query] [DecidableEq Advertiser]
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query)
  (A : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query) (a : Advertiser) =>
  ∑ q : Query,
  match A q with
  | none => 0
  | some a' => if a' = a then l.bid a q else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperAssignment](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query →
(alive : Advertiser → Query → Prop) →
[(a : Advertiser) → (q : Query) → Decidable (alive a q)] → Advertiser → Query → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [DecidableEq Advertiser]
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query)
  (alive : Advertiser → Query → Prop) [(a : Advertiser) → (q : Query) → Decidable (alive a q)] (a : Advertiser)
  (q : Query) =>
  have others : Finset Advertiser := {b ∈ Finset.univ.erase a | alive b q};
  if h : others.Nonempty then max 0 (others.sup' h fun (b : Advertiser) => l.bid b q) else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all

Declaration location: paper-local

Lean declaration body

```
type:
  {Advertiser : Type u_1} →
  {Query : Type u_2} →
  [Fintype Advertiser] →
  [DecidableEq Advertiser] →
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query → Advertiser → Query → ℝ

value:
  fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [DecidableEq Advertiser]
    (I : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) (a : Advertiser)
    (q : Query) =>
    have others : Finset Advertiser := Finset.univ.erase a;
    if h : others.Nonempty then max 0 (others.sup' h fun (b : Advertiser) => I.bid b q) else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets

Declaration location: paper-local

Lean declaration body

```
type:
  {Advertiser : Type u_3} →
  {Query : Type u_4} → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query → Prop

value:
  fun {Advertiser : Type u_3} {Query : Type u_4}
    (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) =>
    ∀ (a : Advertiser), I.budget a = 1
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult

Declaration location: paper-local

Lean declaration body

```
field winner:
  {Advertiser : Type u_3} →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser → Option Advertiser

field candidateTests:
  {Advertiser : Type u_3} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser → ℕ

field scoreComparisons:
  {Advertiser : Type u_3} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser → ℕ
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser

Declaration location: paper-local

Lean declaration body

```
type:
  ℕ → Type

value:
  fun (m : ℕ) => Fin ((m + 1) ^ m)
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision

Declaration location: paper-local

Lean declaration body

```
field query:
{Advertiser : Type u_3} →
{Query : Type u_4} →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query → Query

field winner:
{Advertiser : Type u_3} →
{Query : Type u_4} →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query → Option Advertiser

field spentBefore:
{Advertiser : Type u_3} →
{Query : Type u_4} →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query → Advertiser → ℝ
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState

Declaration location: paper-local

Lean declaration body

```
field spent:
{Advertiser : Type u_3} →
{Query : Type u_4} →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query → Advertiser → ℝ

field revenue:
{Advertiser : Type u_3} →
{Query : Type u_4} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query → ℝ

field decisionsRev:
{Advertiser : Type u_3} →
{Query : Type u_4} →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
  List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query)

field candidateTests:
{Advertiser : Type u_3} →
{Query : Type u_4} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query → ℕ

field scoreComparisons:
{Advertiser : Type u_3} →
{Query : Type u_4} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query → ℕ
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative

Declaration location: paper-local

Lean declaration body

```
type:
Type u_3 → ℕ → Type (max u_3 0)

value:
fun (Advertiser : Type u_3) (m : ℕ) => Advertiser × Fin m
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
  [DecidableEq Advertiser] →
    AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
    Query →
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser →
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
  (scan : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser) =>
  have charge : Advertiser → ℝ := fun (a : Advertiser) => I.bid a q;
  {
```

```

spent := fun (a : Advertiser) =>
  match scan.winner with
  | none => S.spent a
  | some winner => if a = winner then S.spent a + charge winner else S.spent a,
revenue :=
  match scan.winner with
  | none => S.revenue
  | some winner => S.revenue + charge winner,
decisionsRev := { query := q, winner := scan.winner, spentBefore := S.spent } :: S.decisionsRev,
candidateTests := S.candidateTests + scan.candidateTests,
scoreComparisons := S.scoreComparisons + scan.scoreComparisons }

```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.availableFromEntry

Declaration location: paper-local

Lean declaration body

```

type:
{Advertiser : Type u_1} → {n : ℕ} → (Advertiser → Fin (n + 1)) → Advertiser → Fin n → Prop

value:
fun {Advertiser : Type u_1} {n : ℕ} (entryTime : Advertiser → Fin (n + 1)) (a : Advertiser) (a_1 : Fin n) =>
  entryTime a ≤ a_1.castSucc

```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff

Declaration location: paper-local

Lean declaration body

```

type:
(k : ℕ) → Fin k → ℝ

value:
fun (k : ℕ) (i : Fin k) => 1 - Real.exp (-(1 - ↑(↑i + 1) / ↑k))

```

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff

Declaration location: paper-local

Lean declaration body

```

type:
(k : ℕ) → Fin k → ℝ

value:
fun (k : ℕ) (i : Fin k) => AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff k i

```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.endpointRoundingErrorAt

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
(k : ℕ) →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) →
(Fin history.length → Option Advertiser) →
(Advertiser → Fin k) → (Fin history.length → Advertiser → Fin k) → Fin history.length → ℝ

value:
fun {Advertiser : Type u_3} {Query : Type u_4} (k : ℕ)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
  (opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k)
  (currentSlab : Fin history.length → Advertiser → Fin k) (t : Fin history.length) =>
match opt t with
| none => 0
| some a => if ↑(finalType a) < ↑(currentSlab t a) then l.bid a (history.get t) / ↑k else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℕ → ℝ

value:
fun (m : ℕ) (N : ℝ) (stage : ℕ) => N * (1 - 1 / ↑(m + 1)) ^ stage
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFinalSpendFraction

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) → Fin m → ℝ

value:
fun (m : ℕ) (i : Fin m) => ↑(↑i + 1) / ↑(m + 1)
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFullMass

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ

value:
fun (m : ℕ) (N : ℝ) => AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass m N m
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → ℕ

value:
fun (m stage : ℕ) => (m + 1) ^ (m - stage) * m ^ stage
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery

Declaration location: paper-local

Lean declaration body

```
type:
  ℕ → Type

constructor AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.stage:
  {m : ℕ} →
    (stage : Fin m) →
      Fin (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m ↑ stage) →
        AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m

constructor AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.final:
  {m : ℕ} →
    Fin (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m) →
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m

constructor AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.reserve:
  {m : ℕ} →
    Fin (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m) →
      Fin m → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEligible

Declaration location: paper-local

Lean declaration body

```
type:
  (m : ℕ) →
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m →
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m → Prop

value:
fun (m : ℕ) (a : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
  (q : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m) =>
  match q with
  | AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.stage stage target =>
    ↑ a < AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m ↑ stage
  | AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.final bidder =>
    ↑ a < AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m
  | AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.reserve bidder copy =>
    ↑ a < AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEmbedActive

Declaration location: paper-local

Lean declaration body

```
type:
  (m stage : ℕ) →
    stage ≤ m →
      Fin (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m stage) → Fin ((m + 1) ^ m)

value:
fun (m stage : ℕ) (hstage : stage ≤ m)
  (a : Fin (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m stage)) =>
  (↑ a, AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEmbedActive._proof_1 m stage hstage a)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightFinalHistory

Declaration location: paper-local

Lean declaration body

```
type:
(m r : ℕ) →
r ≤ AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m →
List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)

value:
fun (m r : ℕ) (a : r ≤ AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m) =>
Nat.brecOn (motive := fun (x : ℕ) =>
x ≤ AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m →
List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m))
r (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightFinalHistory._f m) a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightQueryTarget

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m

value:
fun (m : ℕ) (a : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m) =>
match a with
| AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.stage stage a =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEmbedActive m (↑ stage)
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightQueryTarget._proof_1 m stage) a
| AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.final a =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEmbedActive m m le_rfl a
| AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.reserve a copy =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEmbedActive m m le_rfl a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEmbedActive](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightReserveHistory

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) → List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)

value:
fun (m : ℕ) =>
List.map
(fun (p : Fin (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m)) × Fin m =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery.reserve p.1 p.2)
(List.ofFn ↑ finProdFinEquiv.symm)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightStageHistory

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) →
(stage : Fin m) →
(r : ℕ) →
r ≤ AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m ↑ stage →
List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)

value:
fun (m : ℕ) (stage : Fin m) (r : ℕ)
(a : r ≤ AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m ↑ stage) =>
Nat.brecOn (motive := fun (x : ℕ) =>
x ≤ AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m ↑ stage →
List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m))
r (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightStageHistory._f m stage) a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightPreterminalHistory

Declaration location: paper-local

Lean declaration body

```
type:
(m s : ℕ) → s ≤ m → List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)

value:
fun (m s : ℕ) (a : s ≤ m) =>
Nat.brecOn (motive := fun (x : ℕ) =>
x ≤ m → List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)) s
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightPreterminalHistory._f m) a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightStageHistory](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineCoreHistory

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) → List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)

value:
fun (m : ℕ) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightPreterminalHistory m m le_rfl ++
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightFinalHistory m
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount m m)
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineCoreHistory._proof_1 m)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightActiveCount](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightFinalHistory](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightPreterminalHistory](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineHistory

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) → List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)

value:
fun (m : ℕ) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineCoreHistory m ++
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightReserveHistory m
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineCoreHistory](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightReserveHistory](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidHarmonicBefore

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → ℝ

value:
fun (N r : ℕ) => ∑ round ∈ Finset.range r, 1 / ↑(N - round)
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → ℝ

value:
fun (N r : ℕ) => min 1 (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidHarmonicBefore N r)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidHarmonicBefore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceFinalSpend

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) → Fin N → ℝ

value:
fun (N : ℕ) (bidder : Fin N) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N (↑bidder + 1)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceRoundIncrement

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → ℝ

value:
fun (N r : ℕ) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N (r + 1) -
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N r
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) → Fin N → Fin N → ℝ

value:
fun (N : ℕ) (round bidder : Fin N) =>
  if ↑round ≤ ↑bidder then AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceRoundIncrement N ↑round
  else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceRoundIncrement](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) → Fin N → Fin N → ℝ

value:
fun (N : ℕ) (round bidder : Fin N) =>
  if ↑round ≤ ↑bidder then AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N ↑round
  else AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N (↑bidder + 1)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} =>
{ spent := fun (x : Advertiser) => 0, revenue := 0, decisionsRev := [], candidateTests := 0, scoreComparisons := 0 }
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentIsTotal

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} → {Query : Type u_2} → (history : List Query) → (Fin history.length → Option Advertiser) → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (history : List Query) (A : Fin history.length → Option Advertiser) =>
  ∀ (t : Fin history.length), ∃ (a : Advertiser), A t = some a
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceBalanceScore

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query → Query → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
  (a : Advertiser) =>
  I.bid a q * (1 - Real.exp (S.spent a / I.budget a - 1))
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
  Query → Advertiser → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
  (a : Advertiser) =>
  S.spent a + I.bid a q ≤ I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  (k : ℕ) →
  0 < k →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
  Query → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ) (hk : 0 < k)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
  (a : Advertiser) =>
  I.bid a q *
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff k
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent a) (I.budget a))
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (history : List Query) → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser (Fin history.length)

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query) =>
  { budget := I.budget, bid := fun (a : Advertiser) (t : Fin history.length) => I.bid a (history.get t) }
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (history : List Query) → (Fin history.length → Option Advertiser) → Fin history.length → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
  (A : Fin history.length → Option Advertiser) (t : Fin history.length) =>
  match A t with
  | none => 0
  | some a => I.bid a (history.get t)
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenue

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (history : List Query) → (Fin history.length → Option Advertiser) → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
  (A : Fin history.length → Option Advertiser) =>
  ∑ t : Fin history.length, AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt I history A t
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) → (Fin history.length → Option Advertiser) → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [DecidableEq Advertiser]
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(A : Fin history.length → Option Advertiser) (a : Advertiser) =>
Σ t : Fin history.length,
match A t with
| none => 0
| some owner => if owner = a then l.bid a (history.get t) else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceTradeoffScore

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(ℝ → ℝ) →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (psi : ℝ → ℝ)
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
(a : Advertiser) =>
l.bid a q * psi (S.spent a / l.budget a)
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScore

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
{m : ℕ} →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ) →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → Query → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} {m : ℕ}
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(rankWeight : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ)
(r : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) (q : Query) =>
l.bid r.1 q * rankWeight r
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
List Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
(step :
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
(a : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
(a_1 : List Query) =>
List.brecOn (motive := fun (x : List Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
a_1 (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom._f step) a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend

Declaration location: paper-local

Lean declaration body

```
type:
{k : ℕ} → ℝ → (Fin k → ℝ) → Fin k → ℝ

value:
fun {k : ℕ} (N : ℝ) (x : Fin k → ℝ) (i : Fin k) => N / ↑k - (∑ j ∈ Finset.lto i, x j) / ↑k
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5OptTypeFiberRevenue

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
{k : ℕ} →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) → (Fin history.length → Option Advertiser) → (Advertiser → Fin k) → Fin k → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} {k : ℕ}
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k) (i : Fin k) =>
∑ t : Fin history.length,
match opt t with
| none => 0
| some a => if finalType a = i then l.bid a (history.get t) else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgSlabFiberRevenue

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) →
Fin k → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ) (hk : 0 < k)
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(decisions : List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query))
(i : Fin k) =>
(List.map
(fun (d : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) =>
if AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab k hk l d = some i then
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue l d
else 0)
decisions).sum
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType

Declaration location: paper-local

Lean declaration body

```
type:
{k : ℕ} →
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype (Fin k)] →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) → (Fin history.length → Option Advertiser) → (Advertiser → Fin k) → ℝ

value:
fun {k : ℕ} {Advertiser : Type u_3} {Query : Type u_4} [Fintype (Fin k)]
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k) =>
Σ i : Fin k,
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueByFinalType l history opt finalType i
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueByFinalType](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabLower

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query → Fin k → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ)
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (i : Fin k) (a : Advertiser) =>
l.budget a * ↑ i / ↑ k
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabUpper

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query → Fin k → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (i : Fin k) (a : Advertiser) =>
  l.budget a * ↑(↑i + 1) / ↑k
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteTradeoff

Declaration location: paper-local

Lean declaration body

```
type:
(k : ℕ) → Fin k → ℝ

value:
fun (k : ℕ) (i : Fin k) => 1 - (1 - 1 / ↑k) ^ (k - (↑i + 1))
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceTheorem8DiscreteScore

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ) (hk : 0 < k)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
  (a : Advertiser) =>
  l.bid a q *
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteTradeoff k
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent a) (l.budget a))
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteTradeoff](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8OptWeightedRevenueAt

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
(k : ℕ) →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (history : List Query) →
  (Fin history.length → Option Advertiser) → (Advertiser → Fin k) → Fin history.length → ℝ

value:
fun {Advertiser : Type u_3} {Query : Type u_4} (k : ℕ)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
  (opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k) (t : Fin history.length) =>
  match opt t with
  | none => 0
  | some a =>
    l.bid a (history.get t) *
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteTradeoff k (finalType a)
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteTradeoff](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8PreterminalAlphaFromFinalTypes

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} → [Fintype Advertiser] → (m : ℕ) → (Advertiser → Fin (m + 1)) → Fin m → ℝ

value:
fun {Advertiser : Type u_3} [Fintype Advertiser] (m : ℕ) (finalType : Advertiser → Fin (m + 1)) (a : Fin m) =>
  ∑ a_1 : Advertiser, if finalType a_1 = a.castSucc then 1 else 0
```

AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio

Declaration location: paper-local

Lean declaration body

```
type:
ℝ

value:
AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperMsvvRatio
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperMsvvRatio](#)

AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Query]
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (A : AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperRevenue l A
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperRevenue](#)

AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query → ℝ → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (epsilon : ℝ) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperSmallBids I epsilon
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperSmallBids](#)

AppliedModelingLib.Online.MSVV07PaperFacing.paperSpend

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[FIntype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [FIntype Query] [DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (A : AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) (a : Advertiser) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperSpend I A a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.paperSpend](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.ExhaustsEveryBudget

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[FIntype Query] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query → Prop

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [FIntype Query] [DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (A : AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) =>
  ∀ (a : Advertiser), AppliedModelingLib.Online.MSVV07PaperFacing.paperSpend I A a = I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperSpend](#)

AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(alive : Advertiser → Query → Prop) →
[(a : Advertiser) → (q : Query) → Decidable (alive a q)] → Advertiser → Query → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [DecidableEq Advertiser]
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (alive : Advertiser → Query → Prop)
[(a : Advertiser) → (q : Query) → Decidable (alive a q)] (a : Advertiser) (q : Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive l alive a q
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive](#)

AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query → Advertiser → Query → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [DecidableEq Advertiser]
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (a : Advertiser) (q : Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all l a q
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteCoeff

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → Fin m → Fin m → ℝ

value:
fun {m : ℕ} (i j : Fin m) => 1 + (↑ i - ↑ j) / ↑ (m + 1)
```

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → Fin m → ℝ

value:
fun {m : ℕ} (i : Fin m) => 1 / ↑ (m + 1) * (1 - 1 / ↑ (m + 1)) ^ (m - (↑ i + 1))
```

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteIdealBeta

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → ℝ → (Fin m → ℝ) → Fin m → ℝ

value:
fun {m : ℕ} (N : ℝ) (alpha : Fin m → ℝ) (i : Fin m) => N / ↑(m + 1) - (∑ j ∈ Finset.lio i, alpha j) / ↑(m + 1)
```

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → Fin m → Fin m → ℝ

value:
fun {m : ℕ} (i j : Fin m) => if ↑j ≤ ↑i then AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteCoeff i j else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteCoeff](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrefix

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → Fin m → Finset (Fin m)

value:
fun {m : ℕ} (i : Fin m) => {j : Fin m | ↑j ≤ ↑i}
```

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDelta

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (Fin m → ℝ) → (Fin m → ℝ) → Fin m → ℝ

value:
fun {m : ℕ} (alpha beta : Fin m → ℝ) (i : Fin m) =>
  ∑ j ∈ AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrefix i, (alpha j - beta j)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrefix](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (Fin m → ℝ) → Fin m → ℝ

value:
fun {m : ℕ} (x : Fin m → ℝ) (i : Fin m) =>
  ∑ j ∈ AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrefix i,
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteCoeff i j * x j
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteCoeff](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrefix](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → ℝ → Fin m → ℝ

value:
fun {m : ℕ} (N : ℝ) (i : Fin m) => N / ↑(m + 1) * (1 - 1 / ↑(m + 1)) ^ ↑i
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) → ℝ → Fin m → ℝ

value:
fun (m : ℕ) (N : ℝ) (i : Fin m) => AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate N i
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ

value:
fun (m : ℕ) (N : ℝ) =>
  ∑ i : Fin m,
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass m N i *
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFinalSpendFraction m i +
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFullMass m N
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFinalSpendFraction](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFullMass](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → Fin m → ℝ

value:
fun {m : ℕ} (i : Fin m) => ↑(m - ↑i) / ↑(m + 1)
```

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (Fin m → ℝ) → ℝ

value:
fun {m : ℕ} (x : Fin m → ℝ) =>
  ∑ i : Fin m, AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff i * x i
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → ℝ → Fin m → ℝ

value:
fun {m : ℕ} (N : ℝ) (i : Fin m) => ↑(↑i + 1) / ↑(m + 1) * N
```

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualObjective

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → ℝ → (Fin m → ℝ) → ℝ

value:
fun {m : ℕ} (N : ℝ) (y : Fin m → ℝ) => ∑ i : Fin m, AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs N i * y i
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalFeasible

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → ℝ → (Fin m → ℝ) → Prop

value:
fun {m : ℕ} (N : ℝ) (x : Fin m → ℝ) =>
  (∀ (i : Fin m),
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow x i ≤
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs N i) ∧
  ∀ (i : Fin m), 0 ≤ x i
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteSuffix

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → Fin m → Finset (Fin m)

value:
fun {m : ℕ} (i : Fin m) => {j : Fin m | ↑i ≤ ↑j}
```

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualRow

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (Fin m → ℝ) → Fin m → ℝ

value:
fun {m : ℕ} (y : Fin m → ℝ) (i : Fin m) =>
  ∑ j ∈ AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteSuffix i,
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteCoeff j i * y j
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteCoeff](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteSuffix](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualFeasible

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (Fin m → ℝ) → Prop

value:
fun {m : ℕ} (y : Fin m → ℝ) =>
  (∀ (i : Fin m),
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff i ≤
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualRow y i) ∧
  ∀ (i : Fin m), 0 ≤ y i
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualRow](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (Fin m → ℝ) → Fin m → ℝ

value:
fun {m : ℕ} (y : Fin m → ℝ) (i : Fin m) => ∑ j ∈ AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteSuffix i, y j
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteSuffix](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → Fin m → ℝ

value:
fun {m : ℕ} (a : Fin m) =>
  AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual](#)

AppliedModelingLib.Online.theorem9BidderSpendUpperBound

Declaration location: paper-local

Lean declaration body

type:
(N : ℕ) → Fin N → ℝ

value:
fun (N : ℕ) (bidder : Fin N) => min 1 (∑ round : Fin N, if ↑round ≤ ↑bidder then 1 / ↑(N - ↑round) else 0)

AppliedModelingLib.Online.theorem9EligibleBidders

Declaration location: paper-local

Lean declaration body

type:
(N : ℕ) → Fin N → Finset (Fin N)

value:
fun (N : ℕ) (round : Fin N) => {bidder : Fin N | ↑round ≤ ↑bidder}

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution

Declaration location: paper-local

Lean declaration body

field allocation:
∀ {N : ℕ} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N → Fin N → Fin N → ℝ

field spentBefore:
∀ {N : ℕ} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N → Fin N → Fin N → ℝ

field finalSpend:
∀ {N : ℕ} → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N → Fin N → ℝ

field zero_of_ineligible:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N}
(round bidder : Fin N), ¬ ↑round ≤ ↑bidder → self.allocation round bidder = 0

field eligible_equal_least_spend:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N}
(round bidder bidder' : Fin N),
↑round ≤ ↑bidder → ↑round ≤ ↑bidder' → self.spentBefore round bidder = self.spentBefore round bidder'

field allocation_nonneg:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N}
(round bidder : Fin N), 0 ≤ self.allocation round bidder

field budget_feasible:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N}
(round bidder : Fin N), self.spentBefore round bidder + self.allocation round bidder ≤ 1

field spentBefore_eq_prefix:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N}
(round bidder : Fin N),
self.spentBefore round bidder = ∑ previous : Fin N, if ↑previous < ↑round then self.allocation previous bidder else 0

field state_update:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N}
(round next bidder : Fin N),
↑next = ↑round + 1 → self.spentBefore next bidder = self.spentBefore round bidder + self.allocation round bidder

field round_conservation:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N} (round : Fin N),
∑ bidder ∈ AppliedModelingLib.Online.theorem9EligibleBidders N round, self.allocation round bidder ≤ 1

field work_conserving_or_saturated:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N} (round : Fin N),
∑ bidder ∈ AppliedModelingLib.Online.theorem9EligibleBidders N round, self.allocation round bidder = 1 ∨
∀ bidder ∈ AppliedModelingLib.Online.theorem9EligibleBidders N round,
self.spentBefore round bidder + self.allocation round bidder = 1

field finalSpend_eq_sum:
∀ {N : ℕ} {self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N} (bidder : Fin N),
self.finalSpend bidder = ∑ round : Fin N, self.allocation round bidder

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9EligibleBidders](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.nestedSuffixFluidBalanceExecution

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N

value:
fun (N : ℕ) =>
{ allocation := AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N,
  spentBefore := AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N,
  finalSpend := AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceFinalSpend N,
  zero_of_ineligible :=
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation_zero_of_ineligible N,
  eligible_equal_least_spend :=
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_eligible_equal_least_spend N,
  allocation_nonneg := AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation_nonneg N,
  budget_feasible := AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_budget_feasible N,
  spentBefore_eq_prefix :=
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_spentBefore_eq_prior_allocations N,
  state_update := AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_state_update N,
  round_conservation :=
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_round_allocation_conservation N,
  work_conserving_or_saturated :=
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_round_full_or_all_eligible_saturated N,
  finalSpend_eq_sum :=
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_finalSpend_eq_sum_allocation N }
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceFinalSpend](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore](#)
- [AppliedModelingLib.Online.theorem9EligibleBidders](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9BaseEqualSpreadBalanceRevenue

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ

value:
fun (N : ℕ) =>
  ∑ bidder : Fin N,
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.nestedSuffixFluidBalanceExecution N).finalSpend bidder
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.nestedSuffixFluidBalanceExecution](#)

AppliedModelingLib.Online.theorem9ActualEligibleBidders

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) → Equiv.Perm (Fin N) → Fin N → Finset (Fin N)

value:
fun (N : ℕ) (permutation : Equiv.Perm (Fin N)) (round : Fin N) =>
  Finset.image (↑ permutation) (AppliedModelingLib.Online.theorem9EligibleBidders N round)
```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9EligibleBidders](#)

AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (N :  $\mathbb{N}$ ) => ( $\sum$  bidder : Fin N, AppliedModelingLib.Online.theorem9BidderSpendUpperBound N bidder) /  $\uparrow$  N
```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9BidderSpendUpperBound](#)

AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
field normalizedRevenue:
{N :  $\mathbb{N}$ } →
{Algorithm : Type u_1} →
{ratio :  $\mathbb{R}$ } →
AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio →
Algorithm → Equiv.Perm (Fin N) →  $\mathbb{R}$ 

field allocation:
{N :  $\mathbb{N}$ } →
{Algorithm : Type u_1} →
{ratio :  $\mathbb{R}$ } →
AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio →
Algorithm → Equiv.Perm (Fin N) → Fin N → Fin N →  $\mathbb{R}$ 

field normalizedRevenue_le_cappedAllocationSpend:
 $\forall$  {N :  $\mathbb{N}$ } {Algorithm : Type u_1} {ratio :  $\mathbb{R}$ }
(self : AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
(algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
self.normalizedRevenue algorithm permutation  $\leq$ 
( $\sum$  bidder : Fin N, min 1 ( $\sum$  round : Fin N, self.allocation algorithm permutation round bidder)) /  $\uparrow$  N

field allocation_zero_of_ineligible:
 $\forall$  {N :  $\mathbb{N}$ } {Algorithm : Type u_1} {ratio :  $\mathbb{R}$ }
(self : AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
(algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
 $\neg \uparrow$  round  $\leq \uparrow$  bidder  $\rightarrow$  self.allocation algorithm permutation round bidder = 0

field round_allocation_sum_le_one:
 $\forall$  {N :  $\mathbb{N}$ } {Algorithm : Type u_1} {ratio :  $\mathbb{R}$ }
(self : AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
(algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round : Fin N),
 $\sum$  bidder  $\in$  AppliedModelingLib.Online.theorem9EligibleBidders N round,
self.allocation algorithm permutation round bidder  $\leq$ 
1

field relabelInput:
{N :  $\mathbb{N}$ } →
{Algorithm : Type u_1} →
{ratio :  $\mathbb{R}$ } →
AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio →
Fin N → Fin N → Fin N → Equiv.Perm (Fin N)  $\approx$  Equiv.Perm (Fin N)

field allocation_eq_of_relabel_eligible:
 $\forall$  {N :  $\mathbb{N}$ } {Algorithm : Type u_1} {ratio :  $\mathbb{R}$ }
(self : AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
(algorithm : Algorithm) (round bidder bidder' : Fin N),
 $\uparrow$  round  $\leq \uparrow$  bidder  $\rightarrow$ 
 $\uparrow$  round  $\leq \uparrow$  bidder'  $\rightarrow$ 
 $\forall$  (permutation : Equiv.Perm (Fin N)),
self.allocation algorithm ((self.relabelInput round bidder bidder') permutation) round bidder =
self.allocation algorithm permutation round bidder'

field revenueBound_le_ratio:
 $\forall$  {N :  $\mathbb{N}$ } {Algorithm : Type u_1} {ratio :  $\mathbb{R}$ }
(self : AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio),
AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N  $\leq$  ratio
```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9EligibleBidders](#)
- [AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound](#)

AppliedModelingLib.Online.theorem9ObservedPrefix

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) → Equiv.Perm (Fin N) → Fin N → Fin N → Finset (Fin N)

value:
fun (N : ℕ) (permutation : Equiv.Perm (Fin N)) (currentRound query : Fin N) =>
  if ↑query ≤ ↑currentRound then AppliedModelingLib.Online.theorem9ActualEligibleBidders N permutation query else ∅
```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9ActualEligibleBidders](#)

AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
field normalizedRevenue:
{N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → Equiv.Perm (Fin N) → ℝ

field prefixAllocation:
{N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → (Fin N → Finset (Fin N)) → Fin N → Fin N → ℝ

field normalizedRevenue_le_cappedPrefixAllocationSpend:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  self.normalizedRevenue algorithm permutation ≤
  (∑ bidder : Fin N,
    min 1
      (∑ round : Fin N,
        self.prefixAllocation algorithm (AppliedModelingLib.Online.theorem9ObservedPrefix N permutation round) round
        (permutation bidder))) /
  ↑N

field prefixAllocation_zero_of_not_visible:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (obs : Fin N → Finset (Fin N)) (round bidder : Fin N),
  bidder ∉ obs round → self.prefixAllocation algorithm obs round bidder = 0

field prefixAllocation_sum_le_one:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (obs : Fin N → Finset (Fin N)) (round : Fin N),
  ∑ bidder ∈ obs round, self.prefixAllocation algorithm obs round bidder ≤ 1

field revenueBound_le_ratio:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio),
  AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound](#)
- [AppliedModelingLib.Online.theorem9ObservedPrefix](#)

AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
field normalizedRevenue:
{N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → Equiv.Perm (Fin N) → ℝ

field allocation:
{N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → Equiv.Perm (Fin N) → Fin N → Fin N → ℝ
```

```

field prefixAllocation:
  {N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → (Fin N → Finset (Fin N)) → Fin N → Fin N → ℝ

field allocation_eq_prefix:
  ∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
  self.allocation algorithm permutation round bidder =
  self.prefixAllocation algorithm (AppliedModelingLib.Online.theorem9ObservedPrefix N permutation round) round
  (permutation bidder)

field normalizedRevenue_le_cappedAllocationSpend:
  ∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  self.normalizedRevenue algorithm permutation ≤
  (Σ bidder : Fin N, min 1 (Σ round : Fin N, self.allocation algorithm permutation round bidder)) / ↑N

field allocation_zero_of_ineligible:
  ∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
  → ↑round ≤ ↑bidder → self.allocation algorithm permutation round bidder = 0

field round_allocation_sum_le_one:
  ∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round : Fin N),
  Σ bidder ∈ AppliedModelingLib.Online.theorem9EligibleBidders N round,
  self.allocation algorithm permutation round bidder ≤
  1

field revenueBound_le_ratio:
  ∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio),
  AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio

```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9EligibleBidders](#)
- [AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound](#)
- [AppliedModelingLib.Online.theorem9ObservedPrefix](#)

AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.toObservedPrefixAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```

type:
  {N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio →
  AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio

value:
  fun {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (C : AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio) =>
  { normalizedRevenue := C.normalizedRevenue,
    allocation := fun (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N) =>
      C.prefixAllocation algorithm (AppliedModelingLib.Online.theorem9ObservedPrefix N permutation round) round
      (permutation bidder),
    prefixAllocation := C.prefixAllocation,
    allocation_eq_prefix :=
      AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.toObservedPrefixAllocationRevenueCertificate._proof_1
      C,
    normalizedRevenue_le_cappedAllocationSpend := C.normalizedRevenue_le_cappedPrefixAllocationSpend,
    allocation_zero_of_ineligible :=
      AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.toObservedPrefixAllocationRevenueCertificate._proof_2
      C,
    round_allocation_sum_le_one :=
      AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.toObservedPrefixAllocationRevenueCertificate._proof_3
      C,
    revenueBound_le_ratio := C.revenueBound_le_ratio }

```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate](#)

- [AppliedModelingLib.Online.theorem9EligibleBidders](#)
- [AppliedModelingLib.Online.theorem9ObservedPrefix](#)

AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.toRelabelSymmetricPointwiseAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```

type:
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio →
AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio

value:
fun {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
(C : AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio) =>
{ normalizedRevenue := C.normalizedRevenue, allocation := C.allocation,
normalizedRevenue_le_cappedAllocationSpend := C.normalizedRevenue_le_cappedAllocationSpend,
allocation_zero_of_ineligible := C.allocation_zero_of_ineligible,
round_allocation_sum_le_one := C.round_allocation_sum_le_one,
relabelInput := fun (x bidder bidder' : Fin N) => Equiv.mulRight (Equiv.swap bidder bidder'),
allocation_eq_of_relabel_eligible :=
AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.toRelabelSymmetricPointwiseAllocationRevenueCertificate._proof_1
C,
revenueBound_le_ratio := C.revenueBound_le_ratio }

```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate](#)

AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy

Declaration location: paper-local

Lean declaration body

```

field allocationCount:
{N b : ℕ} →
AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b →
(Fin N → Finset (Fin N)) → Fin N → Fin N → Fin (b + 1)

field allocationCount_zero_of_not_visible:
∀ {N b : ℕ} (self : AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b) (obs : Fin N → Finset (Fin N))
(round bidder : Fin N), bidder ∉ obs round → self.allocationCount obs round bidder = 0

field allocationCount_round_le_capacity:
∀ {N b : ℕ} (self : AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b) (obs : Fin N → Finset (Fin N))
(round : Fin N), ∑ bidder ∈ obs round, ↑(self.allocationCount obs round bidder) ≤ b

field allocationCount_bidder_le_capacity:
∀ {N b : ℕ} (self : AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b) (permutation : Equiv.Perm (Fin N))
(bidder : Fin N),
∑ round : Fin N,
↑(self.allocationCount (AppliedModelingLib.Online.theorem9ObservedPrefix N permutation round) round
(permutation bidder)) ≤
b

```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9ObservedPrefix](#)

AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy.allocationFraction

Declaration location: paper-local

Lean declaration body

```

type:
{N b : ℕ} →
AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b → (Fin N → Finset (Fin N)) → Fin N → Fin N → ℝ

value:
fun {N b : ℕ} (policy : AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b) (obs : Fin N → Finset (Fin N))
(round bidder : Fin N) =>
↑ ↑ (policy.allocationCount obs round bidder) / ↑ b

```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy](#)

AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy.normalizedRevenue

Declaration location: paper-local

Lean declaration body

```
type:
{N b : ℕ} → AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b → Equiv.Perm (Fin N) → ℝ

value:
fun {N b : ℕ} (policy : AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b)
  (permutation : Equiv.Perm (Fin N)) =>
  (∑ bidder : Fin N,
    ∑ round : Fin N,
    policy.allocationFraction (AppliedModelingLib.Online.theorem9ObservedPrefix N permutation round) round
    (permutation bidder)) /
  ↑ N
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy](#)
- [AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy.allocationFraction](#)
- [AppliedModelingLib.Online.theorem9ObservedPrefix](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → Type

value:
fun (N b : ℕ) => AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceNormalizedRevenue

Declaration location: paper-local

Lean declaration body

```
type:
{N b : ℕ} → AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy N b → Equiv.Perm (Fin N) → ℝ

value:
fun {N b : ℕ} (policy : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy N b)
  (permutation : Equiv.Perm (Fin N)) =>
  AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy.normalizedRevenue policy permutation
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy.normalizedRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceRandomizedOnlineAlgorithm

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → Type

value:
fun (N b : ℕ) => PMF (AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy N b)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralFiniteCertificate

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) →
  AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N
  (AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N) 1

value:
fun (N : ℕ) =>
{
  normalizedRevenue :=
    fun (algorithm : AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N)
      (permutation : Equiv.Perm (Fin N)) =>
      (∑ bidder : Fin N,
        min 1
          (∑ round : Fin N,
            algorithm.prefixAllocation (AppliedModelingLib.Online.theorem9ObservedPrefix N permutation round) round
            (permutation bidder))) /
        ↑N,
  prefixAllocation :=
    fun (algorithm : AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N) (obs : Fin N → Finset (Fin N))
      (round bidder : Fin N) =>
    algorithm.prefixAllocation obs round bidder,
  normalizedRevenue_le_cappedPrefixAllocationSpend :=
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralFiniteCertificate._proof_1 N,
  prefixAllocation_zero_of_not_visible :=
    fun (algorithm : AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N) (obs : Fin N → Finset (Fin N))
      (round bidder : Fin N) (hnot : bidder ∉ obs round) =>
    AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation_zero_of_not_visible algorithm obs
      round bidder hnot,
  prefixAllocation_sum_le_one :=
    fun (algorithm : AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N) (obs : Fin N → Finset (Fin N))
      (round : Fin N) =>
    AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation_sum_le_one algorithm obs round,
  revenueBound_le_ratio :=
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralFiniteCertificate._proof_2 N }
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm](#)
- [AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation](#)
- [AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound](#)
- [AppliedModelingLib.Online.theorem9ObservedPrefix](#)

AppliedModelingLib.Online.instFintypeBMatchingTheorem9SourceHardPolicy

Declaration location: paper-local

Lean declaration body

```
type:
(N b : ℕ) → Fintype (AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b)

value:
fun (N b : ℕ) =>
  Fintype.ofInjjective
    (fun (policy : AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy N b) => policy.allocationCount)
    (AppliedModelingLib.Online.instFintypeBMatchingTheorem9SourceHardPolicy._proof_1 N b)
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingTheorem9SourceHardPolicy](#)

AppliedModelingLib.Optimization.IsMinimizerOn

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → (α → Prop) → (α → ℝ) → α → Prop

value:
fun {α : Type u_1} (feasible : α → Prop) (objective : α → ℝ) (x : α) =>
feasible x ∧ ∀ (y : α), feasible y → objective x ≤ objective y
```

AppliedModelingLib.Optimization.StandardMaxLP

Declaration location: reusable library

Lean declaration body

```
field A:
{ι : Type u_1} → {κ : Type u_2} → AppliedModelingLib.Optimization.StandardMaxLP ι κ → κ → ι → ℝ

field b:
{ι : Type u_1} → {κ : Type u_2} → AppliedModelingLib.Optimization.StandardMaxLP ι κ → κ → ℝ

field c:
{ι : Type u_1} → {κ : Type u_2} → AppliedModelingLib.Optimization.StandardMaxLP ι κ → ι → ℝ
```

AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) → ℝ → AppliedModelingLib.Optimization.StandardMaxLP (Fin m) (Fin m)

value:
fun (m : ℕ) (N : ℝ) =>
{ A := AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff,
  b := AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs N,
  c := AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff }
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs](#)
- [AppliedModelingLib.Optimization.StandardMaxLP](#)

AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (Fin m → ℝ) → (Fin m → ℝ) → AppliedModelingLib.Optimization.StandardMaxLP (Fin m) (Fin m)

value:
fun {m : ℕ} (c l : Fin m → ℝ) =>
{ A := AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff, b := l, c := c }
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff](#)
- [AppliedModelingLib.Optimization.StandardMaxLP](#)

AppliedModelingLib.Optimization.StandardMaxLP.DualFeasible

Declaration location: reusable library

Lean declaration body

```
type:
{ι : Type u_1} → {κ : Type u_2} → [Fintype κ] → AppliedModelingLib.Optimization.StandardMaxLP ι κ → (κ → ℝ) → Prop

value:
fun {ι : Type u_1} {κ : Type u_2} [Fintype κ] (P : AppliedModelingLib.Optimization.StandardMaxLP ι κ) (y : κ → ℝ) =>
(∀ (k : κ), 0 ≤ y k) ∧ ∀ (i : ι), P.c i ≤ ∑ k : κ, P.A k i * y k
```

Direct retained dependencies

- [AppliedModelingLib.Optimization.StandardMaxLP](#)

AppliedModelingLib.Optimization.StandardMaxLP.PrimalFeasible

Declaration location: reusable library

Lean declaration body

```
type:
  {ι : Type u_1} → {κ : Type u_2} → [Fintype ι] → AppliedModelingLib.Optimization.StandardMaxLP ι κ → (ι → ℝ) → Prop
value:
  fun {ι : Type u_1} {κ : Type u_2} [Fintype ι] (P : AppliedModelingLib.Optimization.StandardMaxLP ι κ) (x : ι → ℝ) =>
    (∀ (i : ι), 0 ≤ x i) ∧ ∀ (k : κ), ∑ i : ι, P.A k i * x i ≤ P.b k
```

Direct retained dependencies

- [AppliedModelingLib.Optimization.StandardMaxLP](#)

AppliedModelingLib.Optimization.StandardMaxLP.dualObjective

Declaration location: reusable library

Lean declaration body

```
type:
  {ι : Type u_1} → {κ : Type u_2} → [Fintype κ] → AppliedModelingLib.Optimization.StandardMaxLP ι κ → (κ → ℝ) → ℝ
value:
  fun {ι : Type u_1} {κ : Type u_2} [Fintype κ] (P : AppliedModelingLib.Optimization.StandardMaxLP ι κ) (y : κ → ℝ) =>
    ∑ k : κ, y k * P.b k
```

Direct retained dependencies

- [AppliedModelingLib.Optimization.StandardMaxLP](#)

AppliedModelingLib.Optimization.StandardMaxLP.primalObjective

Declaration location: reusable library

Lean declaration body

```
type:
  {ι : Type u_1} → {κ : Type u_2} → [Fintype ι] → AppliedModelingLib.Optimization.StandardMaxLP ι κ → (ι → ℝ) → ℝ
value:
  fun {ι : Type u_1} {κ : Type u_2} [Fintype ι] (P : AppliedModelingLib.Optimization.StandardMaxLP ι κ) (x : ι → ℝ) =>
    ∑ i : ι, P.c i * x i
```

Direct retained dependencies

- [AppliedModelingLib.Optimization.StandardMaxLP](#)

AppliedModelingLib.Sequence.SeqTendsTo

Declaration location: reusable library

Lean declaration body

```
type:
  (ℕ → ℝ) → ℝ → Prop
value:
  fun (x : ℕ → ℝ) (L : ℝ) => ∀ (δ : ℝ), 0 < δ → ∃ (N : ℕ), ∀ (k : ℕ), N ≤ k → |x k - L| ≤ δ
```

AppliedModelingLib.pmfExp

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → PMF α → (α → ℝ) → ℝ

value:
fun {α : Type u_1} [Fintype α] (μ : PMF α) (f : α → ℝ) => ∑ a : α, (μ a).toReal * f a
```

AppliedModelingLib.uniformPMF

Declaration location: reusable library

Lean declaration body

```
type:
(α : Type u_1) → [Fintype α] → [Nonempty α] → PMF α

value:
fun (α : Type u_1) [Fintype α] [Nonempty α] =>
  PMF.ofFintype (fun (x : α) => (↑(Fintype.card α))-1) (AppliedModelingLib.uniformPMF_proof_1 α)
```

AppliedModelingLib.Online.uniformPermutationDistribution

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) → PMF (Equiv.Perm (Fin N))

value:
fun (N : ℕ) => AppliedModelingLib.uniformPMF (Equiv.Perm (Fin N))
```

Direct retained dependencies

- [AppliedModelingLib.uniformPMF](#)

AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
field normalizedRevenue:
{N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → Equiv.Perm (Fin N) → ℝ

field allocation:
{N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → Equiv.Perm (Fin N) → Fin N → Fin N → ℝ

field normalizedRevenue_le_cappedAllocationSpend:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  self.normalizedRevenue algorithm permutation ≤
  (∑ bidder : Fin N, min 1 (∑ round : Fin N, self.allocation algorithm permutation round bidder)) / ↑N

field expectedAllocation le:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (round bidder : Fin N),
  (AppliedModelingLib.pmfExp (AppliedModelingLib.Online.uniformPermutationDistribution N)
    fun (permutation : Equiv.Perm (Fin N)) => self.allocation algorithm permutation round bidder) ≤
  if ↑round ≤ ↑bidder then 1 / ↑(N - ↑round) else 0

field revenueBound_le_ratio:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio),
  AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound](#)
- [AppliedModelingLib.Online.uniformPermutationDistribution](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
field normalizedRevenue:
  {N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → Equiv.Perm (Fin N) → ℝ

field expectedRoundBidderAllocation:
  {N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate N Algorithm ratio →
  Algorithm → Fin N → Fin N → ℝ

field deterministicAverage_le_cappedExpectedSpend:
  ∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm),
  (AppliedModelingLib.pmfExp (AppliedModelingLib.Online.uniformPermutationDistribution N)
    fun (permutation : Equiv.Perm (Fin N)) => self.normalizedRevenue algorithm permutation) ≤
  (Σ bidder : Fin N, min 1 (Σ round : Fin N, self.expectedRoundBidderAllocation algorithm round bidder)) / ↑ N

field expectedRoundBidderAllocation_le:
  ∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (round bidder : Fin N),
  self.expectedRoundBidderAllocation algorithm round bidder ≤ if ↑ round ≤ ↑ bidder then 1 / ↑ (N - ↑ round) else 0

field revenueBound_le_ratio:
  ∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate N Algorithm ratio),
  AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound](#)
- [AppliedModelingLib.Online.uniformPermutationDistribution](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate.toRoundAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
type:
  {N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio →
  AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate N Algorithm ratio

value:
  fun {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (C : AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio) =>
  { normalizedRevenue := C.normalizedRevenue,
    expectedRoundBidderAllocation := fun (algorithm : Algorithm) (round bidder : Fin N) =>
      AppliedModelingLib.pmfExp (AppliedModelingLib.Online.uniformPermutationDistribution N)
        fun (permutation : Equiv.Perm (Fin N)) => C.allocation algorithm permutation round bidder,
    deterministicAverage_le_cappedExpectedSpend :=
      AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate.toRoundAllocationRevenueCertificate._proof_1
      C,
    expectedRoundBidderAllocation_le := C.expectedAllocation_le, revenueBound_le_ratio := C.revenueBound_le_ratio }
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.uniformPermutationDistribution](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
field normalizedRevenue:
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio →
Algorithm → Equiv.Perm (Fin N) → ℝ

field allocation:
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio →
Algorithm → Equiv.Perm (Fin N) → Fin N → Fin N → ℝ

field normalizedRevenue_le_cappedAllocationSpend:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
(self : AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
(algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
self.normalizedRevenue algorithm permutation ≤
(∑ bidder : Fin N, min 1 (∑ round : Fin N, self.allocation algorithm permutation round bidder)) / ↑N

field allocation_zero_of_ineligible:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
(self : AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
(algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
¬ ↑round ≤ ↑bidder → self.allocation algorithm permutation round bidder = 0

field round_allocation_sum_le_one:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
(self : AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
(algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round : Fin N),
∑ bidder ∈ AppliedModelingLib.Online.theorem9EligibleBidders N round,
self.allocation algorithm permutation round bidder ≤
1

field expectedAllocation_eq_of_eligible:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
(self : AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
(algorithm : Algorithm) (round bidder : Fin N),
↑round ≤ ↑bidder →
↑round ≤ ↑bidder' →
(AppliedModelingLib.pmfExp (AppliedModelingLib.Online.uniformPermutationDistribution N)
fun (permutation : Equiv.Perm (Fin N)) => self.allocation algorithm permutation round bidder) =
AppliedModelingLib.pmfExp (AppliedModelingLib.Online.uniformPermutationDistribution N)
fun (permutation : Equiv.Perm (Fin N)) => self.allocation algorithm permutation round bidder'

field revenueBound_le_ratio:
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
(self : AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio),
AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9EligibleBidders](#)
- [AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound](#)
- [AppliedModelingLib.Online.uniformPermutationDistribution](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.toSymmetricPointwiseAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
type:
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio →
AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio

value:
fun {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
(C : AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio) =>
{ normalizedRevenue := C.normalizedRevenue, allocation := C.allocation,
normalizedRevenue_le_cappedAllocationSpend := C.normalizedRevenue_le_cappedAllocationSpend,
allocation_zero_of_ineligible := C.allocation_zero_of_ineligible,
round_allocation_sum_le_one := C.round_allocation_sum_le_one,
expectedAllocation_eq_of_eligible :=
AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.toSymmetricPointwiseAllocationRevenueCertificate._pro_1
↪ of_1
↪ C,
revenueBound_le_ratio := C.revenueBound_le_ratio }
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.uniformPermutationDistribution](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.toPointwiseAllocationRevenueCertificate

Declaration location: paper-local

Lean declaration body

```
type:
  {N : ℕ} →
  {Algorithm : Type u_1} →
  {ratio : ℝ} →
  AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio →
  AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio

value:
  fun {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
    (C : AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio) =>
    { normalizedRevenue := C.normalizedRevenue, allocation := C.allocation,
      normalizedRevenue_le_cappedAllocationSpend := C.normalizedRevenue_le_cappedAllocationSpend,
      expectedAllocation_le :=
        AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.toPointwiseAllocationRevenueCertificate._proof_1
        C,
      revenueBound_le_ratio := C.revenueBound_le_ratio }
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.uniformPermutationDistribution](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9HardDistribution

Declaration location: paper-local

Lean declaration body

```
type:
  (N : ℕ) → PMF (Equiv.Perm (Fin N))

value:
  fun (N : ℕ) => AppliedModelingLib.Online.uniformPermutationDistribution N
```

Direct retained dependencies

- [AppliedModelingLib.Online.uniformPermutationDistribution](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation

Declaration location: paper-local

Lean declaration body

```
type:
  (N : ℕ) → AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N → Fin N → Fin N → ℝ

value:
  fun (N : ℕ) (algorithm : AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N) (round bidder : Fin N) =>
  AppliedModelingLib.pmfExp (AppliedModelingLib.Online.uniformPermutationDistribution N)
    fun (permutation : Equiv.Perm (Fin N)) =>
    algorithm.prefixAllocation (AppliedModelingLib.Online.theorem9ObservedPrefix N permutation round) round
    (permutation bidder)
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm](#)
- [AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation](#)
- [AppliedModelingLib.Online.theorem9ObservedPrefix](#)
- [AppliedModelingLib.Online.uniformPermutationDistribution](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralRoundCertificate

Declaration location: paper-local

Lean declaration body

```
type:
(N : ℕ) →
  AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate N
  (AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N) 1

value:
fun (N : ℕ) =>
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralFiniteCertificate
    N).toObservedPrefixAllocationRevenueCertificate.toRelabelSymmetricPointwiseAllocationRevenueCertificate.toSymmetricPointwiseAllocationRevenueCertificate.toPointwiseAllocationRevenueCertificate.toRoundAllocationRevenueCertificate
```

Direct retained dependencies

- [AppliedModelingLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.toObservedPrefixAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm](#)
- [AppliedModelingLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.toRelabelSymmetricPointwiseAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingPointwiseAllocationRevenueCertificate.toRoundAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.toSymmetricPointwiseAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.toPointwiseAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralFiniteCertificate](#)

instDecidable_mSVV07AdWords

Declaration location: paper-local

Lean declaration body

```
type:
(p : Prop) → Decidable p

value:
fun (p : Prop) => Classical.propDecidable p
```

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution

Declaration location: paper-local

Lean declaration body

```

field payment:
{m : ℕ} →
  {Advertiser : Type u_3} →
  {Event : Type u_4} →
  [inst : Fintype Advertiser] →
  [inst_1 : Fintype Event] →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser
  Event →
  Event → ℝ

field optOwner:
{m : ℕ} →
  {Advertiser : Type u_3} →
  {Event : Type u_4} →
  [inst : Fintype Advertiser] →
  [inst_1 : Fintype Event] →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser
  Event →
  Event → Advertiser

field finalType:
{m : ℕ} →
  {Advertiser : Type u_3} →
  {Event : Type u_4} →
  [inst : Fintype Advertiser] →
  [inst_1 : Fintype Event] →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser
  Event →
  Advertiser → Fin (m + 1)

field optCurrentSlab:
{m : ℕ} →
  {Advertiser : Type u_3} →
  {Event : Type u_4} →
  [inst : Fintype Advertiser] →
  [inst_1 : Fintype Event] →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser
  Event →
  Event → Fin (m + 1)

field balanceSlab:
{m : ℕ} →
  {Advertiser : Type u_3} →
  {Event : Type u_4} →
  [inst : Fintype Advertiser] →
  [inst_1 : Fintype Event] →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser
  Event →
  Event → Fin (m + 1)

field psi:
{m : ℕ} →
  {Advertiser : Type u_3} →
  {Event : Type u_4} →
  [inst : Fintype Advertiser] →
  [inst_1 : Fintype Event] →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser
  Event →
  Fin (m + 1) → ℝ

field optCurrent_le_final:
∀ {m : ℕ} {Advertiser : Type u_3} {Event : Type u_4} [inst : Fintype Advertiser] [inst_1 : Fintype Event]
  (self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event)
  (event : Event), self.optCurrentSlab event ≤ self.finalType (self.optOwner event)

field balance_choice:
∀ {m : ℕ} {Advertiser : Type u_3} {Event : Type u_4} [inst : Fintype Advertiser] [inst_1 : Fintype Event]
  (self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event)
  (event : Event),
  self.payment event * self.psi (self.optCurrentSlab event) ≤ self.payment event * self.psi (self.balanceSlab event)

field payment_nonneg:
∀ {m : ℕ} {Advertiser : Type u_3} {Event : Type u_4} [inst : Fintype Advertiser] [inst_1 : Fintype Event]
  (self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event)
  (event : Event), 0 ≤ self.payment event

field psi_strictAnti:
∀ {m : ℕ} {Advertiser : Type u_3} {Event : Type u_4} [inst : Fintype Advertiser] [inst_1 : Fintype Event]
  (self :
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event),
  StrictAnti self.psi

field opt_exhausts_unit_budget:
∀ {m : ℕ} {Advertiser : Type u_3} {Event : Type u_4} [inst : Fintype Advertiser] [inst_1 : Fintype Event]
  (self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event)
  (advertiser : Advertiser), (∑ event : Event, if self.optOwner event = advertiser then self.payment event else 0) = 1

field slab_payment_eq_width_of_final_type:
∀ {m : ℕ} {Advertiser : Type u_3} {Event : Type u_4} [inst : Fintype Advertiser] [inst_1 : Fintype Event]
  (self : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event)
  (slab : Fin (m + 1)),
  (∑ event : Event, if self.balanceSlab event = slab then self.payment event else 0) =
  ∑ advertiser : Advertiser, if slab ≤ self.finalType advertiser then 1 / ↑(m + 1) else 0

```

Direct retained dependencies

- [instDecidable_mSVV07AdWords](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution.type Count

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} →
{Advertiser : Type u_3} →
{Event : Type u_4} →
[inst : Fintype Advertiser] →
[inst_1 : Fintype Event] →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser
Event →
Fin m → ℝ

value:
fun {m : ℕ} {Advertiser : Type u_3} {Event : Type u_4} [Fintype Advertiser] [Fintype Event]
(E : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event)
(a : Fin m) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8PreterminalAlphaFromFinalTypes m E.finalType a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8PreterminalAlphaFromFinalTypes](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
(Advertiser → Prop) →
(Advertiser → ℝ) →
List Advertiser → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser

value:
fun {Advertiser : Type u_3} (feasible : Advertiser → Prop) (score : Advertiser → ℝ) (a : List Advertiser) =>
List.brecOn a (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan._f feasible score)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [instDecidable_mSVV07AdWords](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.balanceScan

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser]
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q)
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceBalanceScore I S q) Finset.univ.toList
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceBalanceScore](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceStep

Declaration location: paper-local

Lean declaration body

```

type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision I S q
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan k hk I S q)

```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance

Declaration location: paper-local

Lean declaration body

```

type:
(m : ℕ) →
AppliedModelingLib.Online.AdWordsInstance
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)

value:
fun (m : ℕ) =>
{ budget := fun (x : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m) => 1,
  bid :=
    fun (a : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
      (q : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m) =>
        if AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEligible m a q then 1 / ↑(m + 1) else 0 }

```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightEligible](#)
- [instDecidable_mSVV07AdWords](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightBalanceRule

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) →
  AppliedModelingLib.Online.AdWordsInstance.Assignment
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m) →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m →
  Option (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)

value:
fun (m : ℕ)
  (a :
    AppliedModelingLib.Online.AdWordsInstance.Assignment
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m))
  (a_1 : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m) =>
  if h :
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).IsBalanceChoice a a_1
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightQueryTarget m a_1) then
    some (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightQueryTarget m a_1)
  else (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).balanceChoiceRule a a_1
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.IsBalanceChoice](#)
- [AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightQueryTarget](#)
- [instDecidable_mSVV07AdWords](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScan

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(m : ℕ) →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ) →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query →
  Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m)

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (m : ℕ)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (rankWeight : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ)
  (S :
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query)
  (q : Query) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance m l) S q)
  (fun (r : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) =>
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScore l rankWeight r q)
  Finset.univ.toList
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan](#)

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingStep

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(m : ℕ) →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ) →
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query →
    Query →
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query)

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (m : ℕ)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (rankWeight : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ)
  (S :
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query)
  (q : Query) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance m I) S q
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScan m I rankWeight S q)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScan](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  List Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceStep k hk I)
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceStep](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlphaTypeCount

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} → [Fintype Advertiser] → (k : ℕ) → (Advertiser → ℝ) → Fin k → ℝ

value:
fun {Advertiser : Type u_3} [Fintype Advertiser] (k : ℕ) (finalSpentFraction : Advertiser → ℝ) (a : Fin k) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4TypeCount k finalSpentFraction a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4TypeCount](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCount

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} → [Fintype Advertiser] → (m : ℕ) → (Advertiser → ℝ) → Fin m → ℝ

value:
fun {Advertiser : Type u_3} [Fintype Advertiser] (m : ℕ) (finalSpentFraction : Advertiser → ℝ) (a : Fin m) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlphaTypeCount (m + 1) finalSpentFraction a.castSucc
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlphaTypeCount](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteScan

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
  Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceTheorem8DiscreteScore k hk I S q)
  Finset.univ.toList
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceTheorem8DiscreteScore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteStep

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
  Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision I S q
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteScan k hk I S q)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteScan](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  List Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteStep k hk I)
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteStep](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8StateBefore

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (history : List Query) →
  Fin history.length →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
```

```

(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(t : Fin history.length) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteStep k hk I)
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState (List.take (↑t) history)

```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteStep](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8CurrentSlabAt

Declaration location: paper-local

Lean declaration body

```

type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) → Fin history.length → Advertiser → Fin k

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(t : Fin history.length) (a : Advertiser) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk
((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8StateBefore k hk I history t).spent a)
(I.budget a)

```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8StateBefore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8WinnerAt

Declaration location: paper-local

Lean declaration body

```

type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) → Fin history.length → Option Advertiser

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(t : Fin history.length) =>
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteScan k hk I
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8StateBefore k hk I history t)
(history.get t)).winner

```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteScan](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8StateBefore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8AlgWeightedRevenueAt

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) → Fin history.length → ℝ

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
  (t : Fin history.length) =>
match AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8WinnerAt k hk I history t with
| none => 0
| some a =>
  I.bid a (history.get t) *
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteTradeoff k
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8CurrentSlabAt k hk I history t a)
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8CurrentSlabAt](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8DiscreteTradeoff](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8WinnerAt](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.tradeoffScan

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(ℝ → ℝ) →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (psi : ℝ → ℝ)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceTradeoffScore psi I S q) Finset.univ.toList
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceTradeoffScore](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.tradeoffStep

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
( $\mathbb{R} \rightarrow \mathbb{R}$ ) →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (psi :  $\mathbb{R} \rightarrow \mathbb{R}$ )
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision I S q
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.tradeoffScan psi I S q)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.tradeoffScan](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTradeoffOccurrences

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
( $\mathbb{R} \rightarrow \mathbb{R}$ ) →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
List Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (psi :  $\mathbb{R} \rightarrow \mathbb{R}$ )
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.tradeoffStep psi I)
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.tradeoffStep](#)

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5TradeoffRunSpentFraction

Declaration location: paper-local

Lean declaration body

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
( $\mathbb{R} \rightarrow \mathbb{R}$ ) →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query → List Query → Advertiser →  $\mathbb{R}$ 

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (psi :  $\mathbb{R} \rightarrow \mathbb{R}$ )
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(a : Advertiser) =>
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTradeoffOccurrences psi I history).spent a / I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTradeoffOccurrences](#)

Paper-specific semantic prerequisites

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section2_equal_unit_budgets_simplifying_assumption

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section2_equal_unit_budgets_simplifying_assumption

Source locator: cited publication, lines 205-212

Verbatim paper-source connection

this problem mentioned in the Introduction, this is a reasonable assumption.
An online algorithm is said to be α -competitive if for every instance, the ratio of the revenue of the online algorithm to the revenue of the best off-line algorithm is at least α .
While presenting the algorithm and the proofs, we will make the simplifying assumptions that the budgets of all bidders are equal (assumed unit) and that the best offline algorithm exhausts the budget of each bidder. These assumptions will be relaxed in Section 6.

3

Lean-elaborated paper signature

$\forall \{ \text{Advertiser} : \text{Type } u_5 \} \{ \text{Query} : \text{Type } u_6 \}$
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query),
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets I $\leftrightarrow \forall (a : \text{Advertiser}), I.\text{budget } a = 1$

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Section 2 expressly says that all bidder budgets are equal and assumed unit while presenting the algorithm and proofs. The target is exactly forall advertisers a, I.budget a = 1, so it records the stated unit normalization without changing its scope.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section2_exhaustive_optimum_simplifying_assumption

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section2_exhaustive_optimum_simplifying_assumption

Source locator: cited publication, lines 205-212

Verbatim paper-source connection

this problem mentioned in the Introduction, this is a reasonable assumption.
An online algorithm is said to be α -competitive if for every instance, the ratio of the revenue of the online algorithm to the revenue of the best off-line algorithm is at least α .
While presenting the algorithm and the proofs, we will make the simplifying assumptions that the budgets of all bidders are equal (assumed unit) and that the best offline algorithm exhausts the budget of each bidder. These assumptions will be relaxed in Section 6.

3

Lean-elaborated paper signature

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(opt : AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query),
AppliedModelingLib.Online.AdWordsInstance.IsOptimalAssignment I opt ∧
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.ExhaustsEveryBudget I opt ↔
AppliedModelingLib.Online.AdWordsInstance.IsOptimalAssignment I opt ∧
∀ (a : Advertiser), AppliedModelingLib.Online.MSVV07PaperFacing.paperSpend I opt a = I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.IsOptimalAssignment](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperAssignment](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.ExhaustsEveryBudget](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperSpend](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source assumption has both components: opt is the best offline allocation and it exhausts each bidder budget. The target retains IsOptimalAssignment I opt on both sides; its supporting declaration expands that predicate to feasibility and revenue dominance over every feasible assignment. It expands ExhaustsEveryBudget exactly to forall advertisers a, paperSpend I opt a = I.budget a. Thus neither the optimality qualifier nor the exact exhaustion condition is lost.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_exact_type_spend_simplifying_assumption

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_exact_type_spend_simplifying_assumption

Source locator: cited publication, lines 556-564

Verbatim paper-source connection

before, we make the simplifying assumption (at the cost of a negligible error term) that bidders of type j spend exactly j/k fraction of their budget. Let α_j denote the number of bidders of type j . Let β_i denote the total money spent by the bidders from slab i in the run of the algorithm. It is easy to see that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$. Let $\Delta(\pi, \psi)$ be a $k - 1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$. The following lemma relates the right hand side of the LPs L and $L(\pi, \psi)$.
Lemma 5
Proof :

Lean-elaborated paper signature

$\forall \{ \text{Advertiser} : \text{Type } u_1 \} (k : \mathbb{N}) (\text{spentFraction} : \text{Advertiser} \rightarrow \mathbb{R}) (\text{finalType} : \text{Advertiser} \rightarrow \text{Fin } k),$
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5ExactTypeSpend k spentFraction finalType $\leftrightarrow$
 $\forall (a : \text{Advertiser}), \text{spentFraction } a = \uparrow (\uparrow (\text{finalType } a) + 1) / \uparrow k$

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5ExactTypeSpend](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Section 5 temporarily assumes that every bidder of source type j spends exactly j/k of her budget. Lean indexes source types $1, \dots, k$ by $\text{Fin } k$ values $0, \dots, k-1$, so $((\text{finalType } a).val + 1)/k$ is exactly the source fraction j/k . The target requires that equality for every bidder.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan

Source locator: cited publication, lines 237-241

Verbatim paper-source connection

Discrete Version of the Algorithm

When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

Note that in the special case when all the bids are equal, our algorithm works in the same way as the BALANCE algorithm of [KP00], for any monotonically decreasing tradeoff function.

[Next verbatim source excerpt]

Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}^+$ be the following (monotonically decreasing) function:

$$\psi_k(i) = 1 - e^{-(1-i/k)}$$

Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q)
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore k hk I S q) Finset.univ.toList
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source chooses a bidder maximizing $c(i)$ times $\psi_k(\text{slab}(i))$. The expanded scan filters by remaining-budget feasibility, evaluates the displayed discrete score for every finite advertiser, and returns a maximum; its tie rule is permitted by the source's unspecified tie breaking.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteBalanceChoice

Source locator: cited publication, lines 237-239

Verbatim paper-source connection

Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

[Next verbatim source excerpt]

Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}^+$ be the following (monotonically decreasing) function:
 $\psi_k(i) = 1 - e^{-(1-i/k)}$
Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
Query → Advertiser → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ) (hk : 0 < k)
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
(a : Advertiser) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign l S q a ∧
∀ (b : Advertiser),
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign l S q b →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore k hk l S q b ≤
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore k hk l S q a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The target requires the chosen advertiser to be budget-feasible and to have discrete bid-times-psi_k score at least that of every feasible advertiser. Its expanded active-slab and tradeoff dependencies reproduce the source's discrete Balance allocation rule.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_generalized_current_type_balance_algorithmSpec

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_generalized_current_type_balance_algorithmSpec

Source locator: cited publication, lines 833-842

Verbatim paper-source connection

6. A winning bidder pays only an amount equal to the next highest bid.
1, 2, 3: We say that the current type of a bidder at some time during the run of the algorithm is j if he has spent between $(j-1)/k$ and j/k fraction of his budget at that time. The algorithm allocates the next query to the bidder who maximizes the product of his bid and $\psi(\text{current type})$.
13

[form-feed in source extraction]

The proof of the competitive ratio changes minimally: Let the budget of bidder j be B_j . For $i = 1, \dots, k$, define β_{ij} to be the amount of money spent by the bidder j from the interval P_j

[Next verbatim source excerpt]

Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}^+$ be the following (monotonically decreasing) function:

$\psi_k(i) = 1 - e^{-(1-i/k)}$

Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_5} →
{Query : Type u_6} →
[inst : Fintype Advertiser] →
[inst_1 : DecidableEq Advertiser] →
(k : ℕ) →
(hk : 0 < k) →
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
(q : Query) →
(a : Advertiser) →
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan k hk l S q).winner =
  some a →
  Prop

value:
fun {Advertiser : Type u_5} {Query : Type u_6} [Fintype Advertiser] [DecidableEq Advertiser] (k : ℕ) (hk : 0 < k)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
  (a : Advertiser)
  (hchoice :
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan k hk l S q).winner = some a) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteBalanceChoice k hk l S q a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteBalanceChoice](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source allocates a new query to a bidder maximizing bid times ψ of current type. The target conditions on the finite scan winner and requires the expanded `occurrenceDiscreteBalanceChoice`, whose dependencies compute current type from spent divided by budget, use the displayed ψ_k formula, require budget feasibility, and compare the score against every feasible bidder.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsAlphaCompetitive

Source locator: cited publication, lines 198-208

Verbatim paper-source connection

The adwords problem is the following: There are N bidders, each with a specified daily budget b_i . Q is a set of query words. Each bidder i specifies a bid c_{iq} for query word $q \in Q$. A sequence $q_1 q_2 \dots q_M$ of query words $q_j \in Q$ arrive online during the day, and each query q_j must be assigned to some bidder i (for a revenue of c_{ij}). The objective is to maximize the total revenue at the end of the day while respecting the daily budgets of the bidders. Throughout this paper we will make the assumption that each bid is small compared to the corresponding budget, i.e., $\max_j c_{ij}$ is small compared to b_i , for all i . For the applications of this problem mentioned in the Introduction, this is a reasonable assumption. An online algorithm is said to be α -competitive if for every instance, the ratio of the revenue of the online algorithm to the revenue of the best off-line algorithm is at least α . While presenting the algorithm and the proofs, we will make the simplifying assumptions that

Lean-expanded paper semantic target

```
type:
{Instance : Type u_3} → (Instance → ℝ) → (Instance → ℝ) → ℝ → Prop

value:
fun {Instance : Type u_3} (onlineRevenue offlineRevenue : Instance → ℝ) (α : ℝ) =>
  ∀ (I : Instance), α ≤ onlineRevenue I / offlineRevenue I
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source defines alpha-competitiveness by requiring the online-to-best-offline revenue ratio to be at least alpha for every instance. The expanded target is exactly the universal inequality $\alpha \leq \text{onlineRevenue } I / \text{offlineRevenue } I$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType

Source locator: cited publication, lines 247-264

Verbatim paper-source connection

In this section we analyze the performance of our algorithm in the special case when all bids are equal. This is exactly the algorithm BALANCE of [KP00]. We give a simpler analysis of this algorithm using the notion of a factor-revealing LP. This technique was implicit in [MRJW77, GK98, MMSV01] and was formalized and made explicit in [JMS02, JMM+ 03]. We will see how to extend the analysis to the general case in Section 5. For another simple proof for BALANCE see [AL06].

We will assume for simplicity that in the optimum solution, each of the N players spends his entire budget, and thus the total revenue is N (the proof is similar even without this assumption, and we provide it in Section 6). Recall that BALANCE awards each query to the interested bidder who has the maximum unspent budget. We wish to lower bound the total revenue achieved by BALANCE. Let us define the type of a bidder according to the fraction of budget spent by that bidder at the end of the algorithm BALANCE: say that the bidder is of type j if the fraction of his budget spent at the end of the algorithm lies in the range $((j-1)/k, j/k]$. By convention a bidder who spends none of his budget is assigned type 1. Clearly bidders of type j for small values of j contribute little to the total revenue. The factor revealing LP for the performance of the algorithm BALANCE will proceed by bounding the number of such bidders of type j .

Lemma 1 If OPT assigns query q to a bidder B of type $j \leq k-1$, then BALANCE pays for

[Next verbatim source excerpt]

The structure of the algorithm does constrain how the LP L differs from $L(\pi, \psi)$. This is what we will explore now.

As in the analysis of BALANCE, we divide the budget of each bidder into k equal slabs, numbered 1 to k . Money in slab i is spent before moving to slab $i+1$. We say that a bidder is of type j if the fraction of his budget spent at the end of the algorithm lies in the range $((j-1)/k, j/k]$. By convention a bidder who spends none of his budget is assigned type 1. As before, we make the simplifying assumption (at the cost of a negligible error term) that bidders of type j spend exactly j/k fraction of their budget. Let α_j denote the number of bidders of type j . Let β_i denote the total money spent by the bidders from slab i in the run of the algorithm. It is easy to see that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$. Let $\Delta(\pi, \psi)$ be a $k-1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$. The following lemma relates the right hand side of the LPs L and $L(\pi, \psi)$.

Lemma 5

Proof :

Lean-expanded paper semantic target

type:

$(k : \mathbb{N}) \rightarrow \mathbb{R} \rightarrow \text{Fin } k \rightarrow \text{Prop}$

value:

```
fun (k : ℕ) (spentFraction : ℝ) (i : Fin k) =>
  spentFraction = 0 ∧ ↑i = 0 ∨ ↑i / ↑k < spentFraction ∧ spentFraction ≤ ↑(↑i + 1) / ↑k
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source assigns type j to final spent fraction in $((j-1)/k, j/k]$ and assigns zero spend to type 1. With Lean index i representing source type $i+1$, the expanded disjunction gives precisely the zero convention and interval $(i/k, (i+1)/k]$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsWeightAdjustment

Source locator: cited publication, lines 965-971

Verbatim paper-source connection

How do we actually find a good set of weights for the bidders? Here is an online heuristic might provide a quick way of computing such weights: consider the allocation of queries for some window of time under the current weights. Adjust the weight of a bidder upwards if that bidder spends less than his fair share of his budget during this time window, and downwards if he spends more than his fair share of the budget. Repeat this process after each such window of time.

In an earlier version of this paper [MSVV05], we had presented a second algorithm based on

Lean-expanded paper semantic target

type:

$\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \text{Prop}$

value:

fun (weight spent fairShare nextWeight : $\mathbb{R}$) =>

spent < fairShare $\wedge$ weight < nextWeight $\vee$

fairShare < spent $\wedge$ nextWeight < weight $\vee$ spent = fairShare $\wedge$ nextWeight = weight

Recorded source-to-declaration screening Verdict: matches

Reason: The source raises a bidder's weight after underspending and lowers it after overspending, without specifying a step size. The expanded relation records exactly those strict directions and the neutral equality case without adding a magnitude rule.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment

Source locator: cited publication, lines 965-971

Verbatim paper-source connection

How do we actually find a good set of weights for the bidders? Here is an online heuristic might provide a quick way of computing such weights: consider the allocation of queries for some window of time under the current weights. Adjust the weight of a bidder upwards if that bidder spends less than his fair share of his budget during this time window, and downwards if he spends more than his fair share of the budget. Repeat this process after each such window of time.

In an earlier version of this paper [MSVV05], we had presented a second algorithm based on

Lean-expanded paper semantic target

type:

$\mathbb{R} \rightarrow \text{List} (\mathbb{R} \times \mathbb{R}) \rightarrow \mathbb{R} \rightarrow \text{Prop}$

value:

```
fun (a :  $\mathbb{R}$ ) (a_1 : List ( $\mathbb{R} \times \mathbb{R}$ )) (a_2 :  $\mathbb{R}$ ) =>
  List.brecOn (motive := fun (x : List ( $\mathbb{R} \times \mathbb{R}$ )) =>  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \text{Prop}$ ) a_1
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment._f a a_2
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsWeightAdjustment](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source repeats the direction-only adjustment after every time window. The recursive target applies IsWeightAdjustment to each spent/fair-share pair and threads the resulting weight to the next window, with the empty list preserving the current weight.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution

Source locator: cited publication, lines 949-960

Verbatim paper-source connection

offline problem. Though this is NP-complete for large bids, in the realistic case where the bids are small compared to budgets, a $1 - \epsilon$ approximation can be obtained by linear programming [BHJ+ 04]. In practice, the query distribution fluctuates over time, varying with time of day, special events, etc. Therefore it is desirable to have a very simple, time efficient online allocation scheme.

Let us start by giving a model for the query sequence that formalizes both their statistical nature as well as their unpredictable fluctuations. In this model, the queries are drawn from an arbitrary fixed distribution for a period of time. The distribution is switched by an adversary a number of times each day. Our goal is to design an online algorithm that achieves a $1 - o(1)$ performance ratio against any fixed probability distribution, while achieving a good worst-case performance ratio when the distribution is switched suddenly (or evolves rapidly). We believe a simple modification of our algorithm achieves this. Here is a concrete proposal:

Lean-elaborated paper signature

```
field regimeAt:
{Time : Type u_3} →
{Regime : Type u_4} →
{Query : Type u_5} →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query →
  Time → Regime

field queryDistribution:
{Time : Type u_3} →
{Regime : Type u_4} →
{Query : Type u_5} →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query →
  Regime → PMF Query
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source describes fixed query distributions over periods separated by adversarial switches. The target represents a period by a regime and assigns one PMF to each regime; restricting regimeAt to a finite day with finitely many regime changes recovers every source-admissible switching sequence, while the more general type does not change behavior on that domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.queryDistributionAt

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.queryDistributionAt

Source locator: cited publication, lines 949-960

Verbatim paper-source connection

offline problem. Though this is NP-complete for large bids, in the realistic case where the bids are small compared to budgets, a $1 - o(1)$ approximation can be obtained by linear programming [BHJ+ 04]. In practice, the query distribution fluctuates over time, varying with time of day, special events, etc. Therefore it is desirable to have a very simple, time efficient online allocation scheme.

Let us start by giving a model for the query sequence that formalizes both their statistical nature as well as their unpredictable fluctuations. In this model, the queries are drawn from an arbitrary fixed distribution for a period of time. The distribution is switched by an adversary a number of times each day. Our goal is to design an online algorithm that achieves a $1 - o(1)$ performance ratio against any fixed probability distribution, while achieving a good worst-case performance ratio when the distribution is switched suddenly (or evolves rapidly). We believe a simple modification of our algorithm achieves this. Here is a concrete proposal:

Lean-expanded paper semantic target

```
type:
{Time : Type u_3} →
{Regime : Type u_4} →
{Query : Type u_5} →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query →
Time → PMF Query
```

```
value:
fun {Time : Type u_3} {Regime : Type u_4} {Query : Type u_5}
(M : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query)
(time : Time) =>
M.queryDistribution (M.regimeAt time)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution](#)

Recorded source-to-declaration screening Verdict: matches

Reason: For the source's fixed distribution in force at a time, the expanded target first selects that time's regime and then returns the regime's PMF. This is exactly the query distribution associated with the current fixed-distribution period.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeSlab

Source locator: cited publication, lines 225-233

Verbatim paper-source connection

Our algorithm rectifies this situation by taking into consideration not only the bids but also the unspent budget of each bidder. For the analysis it is convenient to discretize the budgets as follows: we pick a large integer k , and discretize the budget of each bidder into k equal

5

[form-feed in source extraction]

parts (called slabs) numbered 1 through k . Each bidder spends money in slab j before moving to slab $j + 1$.

Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .

Lean-expanded paper semantic target

type:

$(k : \mathbb{N}) \rightarrow 0 < k \rightarrow \mathbb{R} \rightarrow \text{Fin } k$

value:

$\text{fun } (k : \mathbb{N}) (hk : 0 < k) (\text{spentFraction} : \mathbb{R}) =>$

$(\min [\max 0 (\uparrow k * \text{spentFraction})]_+, (k - 1),$
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeSlab._proof_2 k hk spentFraction)

Recorded source-to-declaration screening Verdict: matches

Reason: The floor of k times the spent fraction gives the zero-based currently active slab, and the minimum with $k-1$ maps a fully spent bidder to the last $\text{Fin } k$ index. On the source's nonnegative feasible spend domain, this implements sequential slabs 1 through k , including moving to the next slab after a boundary is exhausted.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab

Source locator: cited publication, lines 225-233

Verbatim paper-source connection

Our algorithm rectifies this situation by taking into consideration not only the bids but also the unspent budget of each bidder. For the analysis it is convenient to discretize the budgets as follows: we pick a large integer k , and discretize the budget of each bidder into k equal

5

[form-feed in source extraction]

parts (called slabs) numbered 1 through k . Each bidder spends money in slab j before moving to slab $j + 1$.

Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .

Lean-expanded paper semantic target

type:

$(k : \mathbb{N}) \rightarrow 0 < k \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \text{Fin } k$

value:

$\text{fun } (k : \mathbb{N}) (hk : 0 < k) (\text{spent budget} : \mathbb{R}) \Rightarrow$

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeSlab k hk ($\text{spent} / \text{budget}$)

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeSlab](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The paper divides each budget into k equal slabs and identifies the active slab from the spent fraction. The target passes $\text{spent}/\text{budget}$ to `activeSlab`, and its expansion maps source slab numbers 1 through k to zero-based `Fin k` indices.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentFeasible

Source locator: cited publication, lines 198-205

Verbatim paper-source connection

The adwords problem is the following: There are N bidders, each with a specified daily budget b_i . Q is a set of query words. Each bidder i specifies a bid c_{ij} for query word $q \in Q$. A sequence $q_1 q_2 \dots q_M$ of query words $q_j \in Q$ arrive online during the day, and each query q_j must be assigned to some bidder i (for a revenue of c_{ij}). The objective is to maximize the total revenue at the end of the day while respecting the daily budgets of the bidders. Throughout this paper we will make the assumption that each bid is small compared to the corresponding budget, i.e., $\max_j c_{ij}$ is small compared to b_i , for all i . For the applications of this problem mentioned in the Introduction, this is a reasonable assumption.

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
[DecidableEq Advertiser] →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) → (Fin history.length → Option Advertiser) → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} [DecidableEq Advertiser]
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(A : Fin history.length → Option Advertiser) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentIsTotal history A ∧
∀ (a : Advertiser),
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend I history A a ≤ I.budget a
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentIsTotal](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source requires every arriving query to be assigned and total charges to stay within each bidder's budget. The expanded predicate conjoins an owner for every occurrence with the per-advertiser sum of assigned bids being at most that advertiser's budget.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance

Source locator: cited publication, lines 984-988

Verbatim paper-source connection

However, there is yet another way to generalize RANKING for general bids and large budgets. For a suitably large m , we can represent every bidder i with budget B_i by m bidders each with budget B_i/m and the same bid values. Now we can run the previous generalization of RANKING for new bidders. As the queries arrive and get allocated by the algorithm, the representatives of each bidder will run out of budget in the order in which they appear in the

[Next verbatim source excerpt]

In an earlier version of this paper [MSVV05], we had presented a second algorithm based on the RANKING algorithm for online bipartite matching of [KVV90]. The algorithm randomly permutes the bidders, and assigns each query to the bidder who maximizes the product of his bid for the query and the value of a particular function of his rank (position) in the permutation. The function used was the same as the function used in this paper for scaling the budgets. We claimed in [MSVV05] that this algorithm also achieves a competitive ratio of

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(m : ℕ) →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (m : ℕ)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) =>
{
  budget := fun (r : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) =>
    I.budget r.1 / ↑m,
  bid := fun (r : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) (q : Query) =>
    I.bid r.1 q }
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source replaces bidder i by m representatives with budget B_i/m and identical bids. The expanded instance uses Advertiser x in m , assigns each representative its original advertiser's budget divided by m , and preserves the original bid values; the source case specializes to suitably large positive m .

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runReplicatedRankingProposal

Source locator: cited publication, lines 984-988

Verbatim paper-source connection

However, there is yet another way to generalize RANKING for general bids and large budgets. For a suitably large m , we can represent every bidder i with budget B_i by m bidders each with budget B_i/m and the same bid values. Now we can run the previous generalization of RANKING for new bidders. As the queries arrive and get allocated by the algorithm, the representatives of each bidder will run out of budget in the order in which they appear in the

[Next verbatim source excerpt]

In an earlier version of this paper [MSVV05], we had presented a second algorithm based on the RANKING algorithm for online bipartite matching of [KVV90]. The algorithm randomly permutes the bidders, and assigns each query to the bidder who maximizes the product of his bid for the query and the value of a particular function of his rank (position) in the permutation. The function used was the same as the function used in this paper for scaling the budgets. We claimed in [MSVV05] that this algorithm also achieves a competitive ratio of

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
[DecidableEq Advertiser] →
(m : ℕ) →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ) →
  List Query →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (m : ℕ)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (rankWeight : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ)
  (history : List Query) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingStep m l rankWeight)
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Representative](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingStep](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Conditional on the source's sampled rank weights, the target processes every query occurrence with the replicated-RANKING step. The expanded dependencies maximize original bid times representative rank weight subject to the representative budget B_i/m and charge the unchanged original bid, matching the operational proposal without asserting its heuristic performance discussion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4TypeCount

Source locator: cited publication, lines 273-279

Verbatim paper-source connection

For simplicity we will assume that bidders of type i spend exactly i/k fraction of their budget, and that queries do not straddle slabs. The latter is justified by the fact that bids are small compared to budgets (e.g. taking bids to be smaller than $k/2$ of the budget). The total error resulting from this simplification is at most N/k and is negligible, once we take k to be large enough. Now, for $i = 1, 2, \dots, k-1$, let x_i be the number of bidders of type (i) . Let β_i denote the total money spent by the bidders from slab i in the run of BALANCE. It is easy to see (Figure 1) that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_3} → [Fintype Advertiser] → (k : ℕ) → (Advertiser → ℝ) → Fin k → ℝ

value:
fun {Advertiser : Type u_3} [Fintype Advertiser] (k : ℕ) (finalSpentFraction : Advertiser → ℝ) (i : Fin k) =>
  ∑ a : Advertiser,
    if AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k (finalSpentFraction a) i then 1 else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType](#)
- [instDecidable_mSVV07AdWords](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source's x_i is the number of bidders whose final spent fraction has type i . The target sums one over advertisers satisfying the expanded `IsFinalType` relation and zero otherwise, with `Fin k` index i denoting source type $i+1$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue

Source locator: cited publication, lines 579-586

Verbatim paper-source connection

We are interested in comparing the performance of our algorithm (abbreviated as ALG) with the optimal algorithm OPT. The following definitions focus on some relevant parameters comparing how ALG and OPT treat a query q :

[form-feed in source extraction]

Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .

Lemma 6 For each query q such that $1 \leq \text{type}(q) \leq k - 1$,

Lean-expanded paper semantic target

type:

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →  
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query → ℝ
```

value:

```
fun {Advertiser : Type u_1} {Query : Type u_2}  
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)  
  (d : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) =>  
  match d.winner with  
  | none => 0  
  | some a => l.bid a d.query
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source defines $ALG(q)$ as revenue earned by the online algorithm on query q . The target returns the winning bidder's actual bid for a recorded occurrence; its zero value for an absent winner is a total extension and agrees on the source-admissible total allocations.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5Delta

Source locator: cited publication, lines 560-560

Verbatim paper-source connection

Consider the i th components of the three vectors. We need to prove:

$$\alpha_1 (1 + i - 1) + \alpha_2 (1 + k) + \dots + \alpha_i$$

$$= iN/k + (\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i).$$

This equation follows using the fact that $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$.

[U+0003 control character]

[Next verbatim source excerpt]

Let $\Delta(\pi, \psi)$ be a $k - 1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$.

Lean-expanded paper semantic target

type:
{k : ℕ} → (Fin k → ℝ) → (Fin k → ℝ) → Fin k → ℝ

value:
fun {k : ℕ} (alpha beta : Fin k → ℝ) (i : Fin k) => ∑ j ∈ Finset.lic i, (alpha j - beta j)

Recorded source-to-declaration screening Verdict: matches

Reason: The source defines Δ_i as the prefix sum $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$. With zero-based $\text{Fin } k$ indices, the expanded sum over $\text{lic } i$ includes exactly the corresponding first $i+1$ components.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5IdealizedBeta

Source locator: cited publication, lines 550-569

Verbatim paper-source connection

The structure of the algorithm does constrain how the LP L differs from $L(\pi, \psi)$. This is what we will explore now.

As in the analysis of BALANCE, we divide the budget of each bidder into k equal slabs, numbered 1 to k . Money in slab i is spent before moving to slab $i + 1$. We say that a bidder is of type j if the fraction of his budget spent at the end of the algorithm lies in the range $((j - 1)/k, j/k]$. By convention a bidder who spends none of his budget is assigned type 1. As before, we make the simplifying assumption (at the cost of a negligible error term) that bidders of type j spend exactly j/k fraction of their budget. Let α_j denote the number of bidders of type j . Let β_i denote the total money spent by the bidders from slab i in the run of the algorithm. It is easy to see that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$. Let $\Delta(\pi, \psi)$ be a $k - 1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$. The following lemma relates the right hand side of the LPs L and $L(\pi, \psi)$.

Lemma 5

Proof :

$$l = b + \Delta(\pi, \psi).$$

Consider the i th components of the three vectors. We need to prove:

$$\alpha_1 (1 + i - 1)$$

Lean-expanded paper semantic target

type:

$\{k : \mathbb{N}\} \rightarrow \mathbb{R} \rightarrow (\text{Fin } k \rightarrow \mathbb{R}) \rightarrow \text{Fin } k \rightarrow \mathbb{R}$

value:

fun $\{k : \mathbb{N}\} (N : \mathbb{R}) (\alpha : \text{Fin } k \rightarrow \mathbb{R}) (a : \text{Fin } k) \Rightarrow$

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend $N \alpha a$

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source's idealized beta i equals N/k minus the counts of all earlier final types divided by k . The expanded alias reduces to section4IdealizedSlabSpend, which computes $N/k - (\text{sum over } j < i \alpha_j)/k$ under the stated rounding simplification.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5OptQueryRevenue

Source locator: cited publication, lines 579-586

Verbatim paper-source connection

We are interested in comparing the performance of our algorithm (abbreviated as ALG) with the optimal algorithm OPT. The following definitions focus on some relevant parameters comparing how ALG and OPT treat a query q :

[form-feed in source extraction]

Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .

Lemma 6 For each query q such that $1 \leq \text{type}(q) \leq k - 1$,

Lean-expanded paper semantic target

type:

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →  
  (history : List Query) → (Fin history.length → Option Advertiser) → Fin history.length → ℝ
```

value:

```
fun {Advertiser : Type u_1} {Query : Type u_2}  
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)  
  (opt : Fin history.length → Option Advertiser) (t : Fin history.length) =>  
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt l history opt t
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source defines $OPT(q)$ as the revenue OPT earns from q . The target indexes occurrences so repeated query words remain distinct and returns the actual bid of OPT's assigned advertiser; none maps to zero only as a total extension outside the source's total assignment domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun

Source locator: cited publication, lines 509-520

Verbatim paper-source connection

Here is how we proceed instead: For every monotonically decreasing tradeoff function ψ and every instance π of the adwords problem and write a new LP $L(\pi, \psi)$ for our algorithm using tradeoff function ψ run on the instance π . Of course, once we specify the algorithm as well as the input instance, the actual allocations of queries to bidders is completely determined. In particular, the number α_i of bidders of type i is fixed. $L(\pi, \psi)$ is the seemingly trivial LP obtained by taking the left hand side of each inequality in the factor revealing LP L and substituting $x_i = \alpha_i$ to obtain the right hand side. Formally:
Recall the LP L from the previous section:
$$\max c \cdot x$$

s.t. $Ax \leq b \quad x \geq 0$

[Next verbatim source excerpt]

The structure of the algorithm does constrain how the LP L differs from $L(\pi, \psi)$. This is what we will explore now.

As in the analysis of BALANCE, we divide the budget of each bidder into k equal slabs, numbered 1 to k . Money in slab i is spent before moving to slab $i + 1$. We say that a bidder is of type j if the fraction of his budget spent at the end of the algorithm lies in the range $((j - 1)/k, j/k]$. By convention a bidder who spends none of his budget is assigned type 1. As before, we make the simplifying assumption (at the cost of a negligible error term) that bidders of type j spend exactly j/k fraction of their budget. Let α_j denote the number of bidders of type j . Let β_i denote the total money spent by the bidders from slab i in the run of the algorithm. It is easy to see that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$. Let $\Delta(\pi, \psi)$ be a $k - 1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$. The following lemma relates the right hand side of the LPs L and $L(\pi, \psi)$.

Lemma 5

Proof :

$l = b + \Delta(\pi, \psi)$.

Consider the i th components of the three vectors. We need to prove:

Lean-expanded paper semantic target

type:

```
{Advertiser : Type u_3} →  
{Query : Type u_4} →  
  [Fintype Advertiser] →  
  [DecidableEq Advertiser] →  
  (m : ℕ) →  
  (ℝ → ℝ) → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query → List Query → Fin m → ℝ
```

value:

```
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] [DecidableEq Advertiser] (m : ℕ) (psi : ℝ → ℝ)  
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)  
  (a : Fin m) =>  
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCount m  
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5TradeoffRunSpentFraction psi l history) a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCount](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5TradeoffRunSpentFraction](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Compared the Section 5 passage defining α_j as the number of bidders whose final spent-budget fraction is in $((j-1)/k, j/k]$, with zero-spend bidders assigned to type 1, and defining Delta as $k-1$ dimensional, with the expanded target and displayed prerequisites. The target sets $k = m + 1$, indexes exactly the first $k-1$ types by $\text{Fin } m/\text{castSucc}$, implements the open-lower/closed-upper intervals and zero convention in IsFinalType , and sums one per bidder using the final fraction from the displayed actual tradeoff run. Its displayed claim use restricts psi by Antitone , matching the passage's monotonically decreasing scope.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab

Source locator: cited publication, lines 579-586

Verbatim paper-source connection

We are interested in comparing the performance of our algorithm (abbreviated as ALG) with the optimal algorithm OPT. The following definitions focus on some relevant parameters comparing how ALG and OPT treat a query q :

[form-feed in source extraction]

Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .

Lemma 6 For each query q such that $1 \leq \text{type}(q) \leq k - 1$,

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) →
0 < k →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query → Option (Fin k)

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ) (hk : 0 < k)
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(d : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) =>
Option.map
(fun (a : Advertiser) =>
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (d.spentBefore a) (l.budget a))
d.winner
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source says a query lies in slab i when the online algorithm pays for it from that slab. The target uses the recorded winner's spent-before fraction to obtain the active slab, which is the payment slab under the paper's no-straddling simplification, and returns none only for an unassigned extension.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5QueryType

Source locator: cited publication, lines 579-586

Verbatim paper-source connection

We are interested in comparing the performance of our algorithm (abbreviated as ALG) with the optimal algorithm OPT. The following definitions focus on some relevant parameters comparing how ALG and OPT treat a query q:

[form-feed in source extraction]
Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .
Lemma 6 For each query q such that $1 \leq \text{type}(q) \leq k - 1$,

Lean-expanded paper semantic target

type:
{Advertiser : Type u_1} → {k : ℕ} → Option Advertiser → (Advertiser → Fin k) → Option (Fin k)
value:
fun {Advertiser : Type u_1} {k : ℕ} (optOwner : Option Advertiser) (finalType : Advertiser → Fin k) =>
Option.map finalType optOwner

Recorded source-to-declaration screening Verdict: matches
Reason: The source assigns query type i according to the final type of the bidder receiving it in OPT. The expanded target maps OPT’s owner through exactly that final-type classifier, with none only extending the definition beyond source-total assignments.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6BidderSlabSpend

Source locator: cited publication, lines 840-846

Verbatim paper-source connection

P_j
 i
 $[i-1]$
 $j \beta_i$. Let α_i be the
 $k B_j, k B_j$) of his budget. Let $\beta_i =$
 P amount of money that the optimal
allocation gets from the bins of final type i . Let $\alpha = i \alpha_i$, be the total amount of money
obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become
[Next verbatim source excerpt]

[form-feed in source extraction]
The proof of the competitive ratio changes minimally: Let the budget of bidder j be B_j . For
 $i = 1, \dots, k$, define β_{ij} to be the amount of money spent by the bidder j from the interval
 P_j
 i
 $[i-1]$
 $j \beta_i$. Let α_i be the
 $k B_j, k B_j$) of his budget. Let $\beta_i =$

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) →
  AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
  (Advertiser → ℝ) → Fin k → Advertiser → ℝ

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (finalSpend : Advertiser → ℝ)
  (i : Fin k) (a : Advertiser) =>
  min (finalSpend a) (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabUpper k l i a) -
  min (finalSpend a) (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabLower k l i a)
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabLower](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabUpper](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source's β_{i^j} is bidder j 's spend in budget interval $[(i-1)B_j/k, iB_j/k]$. With zero-based i , the difference $\min(\text{finalSpend}, \text{upper}) - \min(\text{finalSpend}, \text{lower})$ gives exactly the portion of nonnegative feasible final spend in that interval and correctly splits boundary-crossing payments.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6AggregateSlabSpend

Source locator: cited publication, lines 842-850

Verbatim paper-source connection

P j
i
[i-1
j β_i . Let α_i be the
k B_j , k B_j) of his budget. Let β_i =
P amount of money that the optimal
allocation gets from the bins of final type i. Let $\alpha = \sum \alpha_i$, be the total amount of money
obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_3} →
{Query : Type u_4} →
[Fintype Advertiser] →
(k : ℕ) →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query → (Advertiser → ℝ) → Fin k → ℝ

value:
fun {Advertiser : Type u_3} {Query : Type u_4} [Fintype Advertiser] (k : ℕ)
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (finalSpend : Advertiser → ℝ)
(i : Fin k) =>
Σ a : Advertiser, AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6BidderSlabSpend k l finalSpend i a
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6BidderSlabSpend](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source defines β_i as the total money spent from budget slab i across bidders. The target sums the expanded per-bidder slab-spend amount over every advertiser, which is precisely that aggregate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6CurrentType

Source locator: cited publication, lines 833-841

Verbatim paper-source connection

6. A winning bidder pays only an amount equal to the next highest bid.
1, 2, 3: We say that the current type of a bidder at some time during the run of the algorithm is j if he has spent between $(j-1)/k$ and j/k fraction of his budget at that time. The algorithm allocates the next query to the bidder who maximizes the product of his bid and $\psi(\text{current type})$.
13

[form-feed in source extraction]

The proof of the competitive ratio changes minimally: Let the budget of bidder j be B_j . For $i = 1, \dots, k$, define β_{ij} to be the amount of money spent by the bidder j from the interval

Lean-expanded paper semantic target

```
type:
(k : ℕ) → 0 < k → ℝ → ℝ → Fin k

value:
fun (k : ℕ) (hk : 0 < k) (spent budget : ℝ) =>
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk spent budget
```

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source current type is determined by the interval between successive multiples of $1/k$ of the budget already spent. The target aliases the active budget slab computed from spent/budget, with its zero-based $\text{Fin } k$ value representing the source's one-based current type.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueByFinalType

Source locator: cited publication, lines 842-852

Verbatim paper-source connection

P_j
 i
 $[i-1]$
 $j \beta_i$. Let α_i be the
 $k B_j, k B_j$) of his budget. Let $\beta_i =$
 P amount of money that the optimal
allocation gets from the bins of final type i . Let $\alpha = i \alpha_i$, be the total amount of money
obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become
 P
 $\alpha - ij = 1 \alpha_j$

Lean-expanded paper semantic target

```
type:
{k : ℕ} →
{Advertiser : Type u_3} →
{Query : Type u_4} →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query →
(history : List Query) → (Fin history.length → Option Advertiser) → (Advertiser → Fin k) → Fin k → ℝ

value:
fun {k : ℕ} {Advertiser : Type u_3} {Query : Type u_4}
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k) (i : Fin k) =>
Σ t : Fin history.length,
match opt t with
| none => 0
| some a => if finalType a = i then l.bid a (history.get t) else 0
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source defines α_i as the amount the optimal allocation earns from bidders of final type i . The expanded target sums each occurrence's actual bid exactly when OPT assigns it to an advertiser classified in that final type.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9HardInstance

Source locator: cited publication, lines 897-918**Verbatim paper-source connection**

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b.

Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/(U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i, which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i, while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i, achieving a revenue of N. One can show that BALANCE will achieve only $N(1 - 1/e)$ revenue on this input. Now consider all the inputs which can be derived from the above input by permutation of the numbers of the bidders and take the uniform distribution D over all these inputs. Formally, D can be described as follows: Pick a random permutation π of the bidders. The queries enter in rounds in the order $Q_1, Q_2, \dots, Q_N$. Bidders $\pi(i), \pi(i + 1), \dots, \pi(N)$ bid $[U+000F$ control character] for the queries Q_i and the other bidders bid 0 for these queries. The optimal allocation for any permutation π remains N, by allocating the queries Q_i to bidder $\pi(i)$. We wish to bound the expected revenue of any deterministic algorithm over inputs from the distribution D. Fix any deterministic algorithm. Let q_{ij} be the fraction of queries from Q_i that bidder j is allocated. We have:

(
1

Lean-expanded paper semantic target

type:
(N m : ℕ) → ℝ → Equiv.Perm (Fin N) → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance (Fin N) (Fin N × Fin m)

value:
fun (N m : ℕ) (epsilon : ℝ) (permutation : Equiv.Perm (Fin N)) =>
{ budget := fun (x : Fin N) => 1,
bid := fun (bidder : Fin N) (q : Fin N × Fin m) =>
if bidder ∈ AppliedModelingLib.Online.theorem9ActualEligibleBidders N permutation q.1 then epsilon else 0 }

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.theorem9ActualEligibleBidders](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source hard input has N rounds, identical occurrences within each round, unit budgets, and epsilon bids from the permuted suffix of eligible bidders. The target constructs exactly that family; specializing m to the source's $1/\epsilon$ epsilon occurrences per round recovers the stated input, while the independent m parameter is a harmless generalization of the construction.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9HardOfflineAssignment

Source locator: cited publication, lines 897-918

Verbatim paper-source connection

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b.

Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average.

Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/(U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i, which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i, while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i, achieving a revenue of N. One can show that BALANCE will achieve only $N(1 - 1/e)$ revenue on this input.

Now consider all the inputs which can be derived from the above input by permutation of the numbers of the bidders and take the uniform distribution D over all these inputs. Formally, D can be described as follows: Pick a random permutation π of the bidders. The queries enter in rounds in the order $Q_1, Q_2, \dots, Q_N$. Bidders $\pi(i), \pi(i + 1), \dots, \pi(N)$ bid $[U+000F$ control character] for the queries Q_i and the other bidders bid 0 for these queries. The optimal allocation for any permutation π remains N, by allocating the queries Q_i to bidder $\pi(i)$. We wish to bound the expected revenue of any deterministic algorithm over inputs from the distribution D.

Fix any deterministic algorithm. Let q_{ij} be the fraction of queries from Q_i that bidder j is allocated. We have:

(
1

Lean-expanded paper semantic target

type:

$(N \text{ m} : \mathbb{N}) \rightarrow \text{Equiv.Perm}(\text{Fin } N) \rightarrow \text{Fin } N \times \text{Fin } m \rightarrow \text{Option}(\text{Fin } N)$

value:

$\text{fun } (N \text{ m} : \mathbb{N}) \text{ (permutation : Equiv.Perm }(\text{Fin } N)) \text{ (a : Fin } N \times \text{Fin } m) \Rightarrow \text{some (permutation a.1)}$

Recorded source-to-declaration screening Verdict: matches

Reason: The source offline witness allocates every occurrence of round i to bidder permutation(i). The expanded assignment maps each pair of round and within-round occurrence to some permutation of its round index exactly as stated.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9RoundBid

Source locator: cited publication, lines 897-918

Verbatim paper-source connection

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b .
Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/(U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i , which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i , while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i , achieving a revenue of N . One can show that BALANCE will achieve only $N (1 - 1/e)$ revenue on this input.
Now consider all the inputs which can be derived from the above input by permutation of the numbers of the bidders and take the uniform distribution D over all these inputs. Formally, D can be described as follows: Pick a random permutation π of the bidders. The queries enter in rounds in the order $Q_1, Q_2, \dots, Q_N$. Bidders $\pi(i), \pi(i + 1), \dots, \pi(N)$ bid $[U+000F$ control character] for the queries Q_i and the other bidders bid 0 for these queries. The optimal allocation for any permutation π remains N , by allocating the queries Q_i to bidder $\pi(i)$. We wish to bound the expected revenue of any deterministic algorithm over inputs from the distribution D .
Fix any deterministic algorithm. Let q_{ij} be the fraction of queries from Q_i that bidder j is allocated. We have:
(
1

Lean-expanded paper semantic target

type:
(N : ℕ) → ℝ → Equiv.Perm (Fin N) → Fin N → Fin N → ℝ
value:
fun (N : ℕ) (epsilon : ℝ) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N) =>
if permutation bidder ∈ AppliedModelingLib.Online.theorem9ActualEligibleBidders N permutation round then epsilon
else 0

Direct retained dependencies

- [AppliedModelingLib.Online.theorem9ActualEligibleBidders](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source says actual bidders permutation(i) through permutation(N) bid epsilon in round i and all earlier positions bid zero. The target treats its bidder argument as a permutation position, maps it to the actual bidder, and tests membership in the image of the suffix positions, yielding exactly epsilon or zero.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.assumption_click_through_rates_probability_bounds

Source locator: cited publication, lines 869-871

Verbatim paper-source connection

5: In order to model this situation, we simply set the effective bid of a bidder to be the product of his actual bid and his click-through rate (CTR), which is the probability that a user will click on his ad. We assume that the click-through rate is known to the algorithm in advance indeed several search engines keep a measure of the
↪ click-through rates of the bidders.

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_1} → {Query : Type u_2} → (Advertiser → Query → ℝ) → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2} (ctr : Advertiser → Query → ℝ) =>
  (∀ (a : Advertiser) (q : Query), 0 ≤ ctr a q) ∧ ∀ (a : Advertiser) (q : Query), ctr a q ≤ 1
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source defines a click-through rate as the probability that a user clicks the advertisement. A real-valued probability has the exact pointwise bounds $0 \leq \text{ctr } a \ q$ and $\text{ctr } a \ q \leq 1$, which are precisely the two conjuncts in the target. The source's additional statement that CTRs are known to the algorithm is not being misrepresented as one of these numeric bounds.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.assumption_epsilon_range

Source locator: cited publication, lines 203-205

Verbatim paper-source connection

Throughout this paper we will make the assumption that each bid is small compared to the corresponding budget, i.e., $\max_j c_{ij}$ is small compared to b_i , for all i . For the applications of this problem mentioned in the Introduction, this is a reasonable assumption.

[Next verbatim source excerpt]

Theorem 8 For function ψ_k defined as

$\psi_k(i) :=$

$$\prod_{j=i}^{k-1} y_j^* = 1 - (1 - \frac{1}{k})^{k-i+1}$$

the competitive ratio of the algorithm is $(1 - \frac{1}{k})$, as k tends to infinity.

Proof :

By Lemma 4, the optimal solution to $L(\pi, \psi)$ and $D(\pi, \psi)$ has value $1 \cdot y^*$. By

Lemma 5 this equals $(b + \Delta) \cdot y^* \leq N/e + \Delta \cdot y^*$ (since $b \cdot y^* \leq N/e$, from Section 4).

Now,

*

$\Delta \cdot y =$

$$\prod_{i=1}^{k-1} y_i^* ((\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i))$$

$$\prod_{i=1}^{k-1} y_i^* ((\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i))$$

$$\prod_{i=1}^{k-1} y_i^* ((\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i))$$

$$(\alpha_i - \beta_i)(y_i^* + \dots + y_{k-1}^*)$$

$$\prod_{i=1}^{k-1} y_i^* ((\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i))$$

$$(\alpha_i - \beta_i)\psi(i)$$

$$i=1$$

$$=$$

$$i=1$$

$$=$$

$$i=1$$

$$\leq$$

$$N$$

$$\prod_{i=1}^k y_i^*$$

where the last equality follows from our choice of the function ψ , and the inequality follows from Lemma 7. As k tends to infinity, we get that the competitive ratio of our algorithm is $(1 - \frac{1}{k})$.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

The finite explicit-error formalization uses a small-bids parameter in $[0,1]$.

Archival source anchor: cited publication, lines 718-773

Lean-expanded paper semantic target

type:
 $\mathbb{R} \rightarrow \text{Prop}$

value:
fun (epsilon : $\mathbb{R}$) => $0 \leq \text{epsilon} \wedge \text{epsilon} \leq 1$

Recorded source-to-declaration screening Verdict: matches approved corrected target

Reason: The excerpts discuss qualitative small bids and the Theorem 8 limit but do not literally give a universal real epsilon in the closed interval $[0,1]$. The source-map preparation record identifies assumption_epsilon_range as corrected target MSVV07-FINITE-CONVENTIONS-05, and the approval record authorizes this range as a visible finite-theorem

convention rather than an archival quantified assertion. The expanded target is exactly $0 \leq \epsilon$ and $\epsilon \leq 1$, with no hidden additional premise.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.assumption_nonnegative_bids

Source locator: cited publication, lines 198-205

Verbatim paper-source connection

The adwords problem is the following: There are N bidders, each with a specified daily budget b_i . Q is a set of query words. Each bidder i specifies a bid c_{ij} for query word $q \in Q$. A sequence $q_1, q_2, \dots, q_M$ of query words $q_j \in Q$ arrive online during the day, and each query q_j must be assigned to some bidder i (for a revenue of c_{ij}). The objective is to maximize the total revenue at the end of the day while respecting the daily budgets of the bidders. Throughout this paper we will make the assumption that each bid is small compared to the corresponding budget, i.e., $\max_j c_{ij}$ is small compared to b_i , for all i . For the applications of this problem mentioned in the Introduction, this is a reasonable assumption.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

The finite AdWords formalization assumes nonnegative bids.

Archival source anchor: cited publication, lines 198-210

Lean-expanded paper semantic target

```
type:
{Advertiser : Type u_1} →
{Query : Type u_2} → AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query → Prop

value:
fun {Advertiser : Type u_1} {Query : Type u_2}
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) =>
  AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I
```

Direct retained dependencies

- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance](#)

Recorded source-to-declaration screening Verdict: matches_approved_corrected_target
Reason: The source uses the economic notions of bid and assignment revenue but does not literally state a universal real-valued nonnegativity quantifier. The source-map preparation record identifies assumption_nonnegative_bids as corrected target MSVV07-FINITE-CONVENTIONS-05, and the approval record authorizes nonnegative_bids as an explicit visible formalization premise. The expanded target is exactly the entrywise predicate forall advertiser and query, $0 \leq I.\text{bid } a \text{ } q$, with no extra content.

Reviewer check: ☐ Matches formalized target
If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

AppliedModelingLib.Online.MSVV07PaperFacing.theorem9HardDistribution

Source locator: cited publication, lines 897-914

Verbatim paper-source connection

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b .
Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with 1/[U+000F control character] number of queries in each round. We denote by Q_i the queries of round i , which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid [U+000F control character] for each of the queries of round i , while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i , achieving a revenue of N . One can show that BALANCE will achieve only $N (1 - 1/e)$ revenue on this input.
Now consider all the inputs which can be derived from the above input by permutation of the numbers of the bidders and take the uniform distribution D over all these inputs. Formally, D can be described as follows: Pick a random permutation π of the bidders. The queries enter in rounds in the order $Q_1, Q_2, \dots, Q_N$. Bidders $\pi(i), \pi(i + 1), \dots, \pi(N)$ bid [U+000F control character] for the queries Q_i and the other bidders bid 0 for these queries. The optimal allocation for any permutation π remains N , by allocating the queries Q_i to bidder $\pi(i)$. We wish to bound the expected revenue of any deterministic algorithm over inputs from the distribution D .

Lean-expanded paper semantic target

type:
(N : ℕ) → PMF (Equiv.Perm (Fin N))

value:
fun (N : ℕ) => AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9HardDistribution N

Direct retained dependencies

- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9HardDistribution](#)

Recorded source-to-declaration screening Verdict: matches
Reason: The source distribution D is uniform over all bidder permutations. The expanded alias reduces through `uniformPermutationDistribution` to the uniform PMF on `Equiv.Perm (Fin N)`, exactly matching that distribution.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 225-239

Verbatim source input

Our algorithm rectifies this situation by taking into consideration not only the bids but also the unspent budget of each bidder. For the analysis it is convenient to discretize the budgets as follows: we pick a large integer k , and discretize the budget of each bidder into k equal

5

[form-feed in source extraction]
parts (called slabs) numbered 1 through k . Each bidder spends money in slab j before moving to slab $j + 1$.
Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .
Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}^+$ be the following (monotonically decreasing) function:
 $\psi_k(i) = 1 - e^{-(1-i/k)}$
Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.
Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

Expanded PaperInterface specification

$\forall (k : \mathbb{N}) (i : \text{Fin } k),$
`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff k i =`
`1 - Real.exp (-(1 - ↑(↑i + 1) / ↑k))`

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\text{Fin } k$
3. conclusion: `AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff k i =`
`1 - Real.exp (-(1 - ↑(↑i + 1) / ↑k))`

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3_discrete_tradeoff_formula`

Recorded source-to-Spec screening Verdict: matches

Reason: With Lean's zero-based Fin index shifted by one, the target is precisely $\psi_k(i) = 1 - \exp(-(1-i/k))$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 232-236

Verbatim source input

Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .

Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}^+$ be the following (monotonically decreasing) function:

$$\psi_k(i) = 1 - e^{-(1-i/k)}$$

Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.

Expanded PaperInterface specification

$\forall (k : \mathbb{N}), \text{Antitone } (\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff } k)$

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$

2. parameter: $\text{Fin } k$

3. parameter: $\text{Fin } k$

4. assumption: $a \leq b$

5. conclusion: $\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff } k \ b \leq \text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff } k \ a$

Lean proof endpoint (built separately)

$\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3_discrete_tradeoff_monotonicity}$

Recorded source-to-Spec screening Verdict: matches

Reason: The source explicitly calls ψ_k monotonically decreasing, which is the target's Antitone property.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 232-239

Verbatim source input

Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .

Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}_+$ be the following (monotonically decreasing) function:

$$\psi_k(i) = 1 - e^{-(1-i/k)}$$

Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.

Discrete Version of the Algorithm

When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} (k : ℕ) (hk : 0 < k)
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
(a : Advertiser),
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore k hk I S q a =
  I.bid a q *
    AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff k
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent a) (I.budget a))
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore](#)

Lean-elaborated claim roles

```
1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: ℕ
4. assumption: 0 < k
5. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query
7. parameter: Query
8. parameter: Advertiser
9. conclusion: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore k hk I S q a =
  I.bid a q *
    AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff k
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent a) (I.budget a))
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3_discrete_balance_score_formula

Recorded source-to-Spec screening Verdict: matches

Reason: The target is $I.\text{bid } a \ q$ multiplied by the discrete tradeoff at the active budget slab, exactly $c(i) \times \psi_k(\text{slab}(i))$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 214-225

Verbatim source input

A Discretized Version of the Algorithm

Let us first consider a greedy algorithm that maximizes revenue accrued at each step. It is easy to see that this algorithm achieves a competitive ratio of 21 (see, e.g., [LLN01]); moreover, this is tight as shown by the following example with only two bidders and two query words: Suppose both bidders have unit budget. The two bidders bid c and $c + [U+000F \text{ control character}]$ respectively on query word q , and they bid 0 and c on query word q_0 . The query sequence consists of a number of occurrences of q followed by a number of occurrences of q_0 . The query words q are awarded to bidder 2, and are just enough in number to exhaust his budget. When query words q_0 arrive, bidder 2's budget is exhausted and bidder 1 is not interested in this query word, and they accrue no further revenue. Our algorithm rectifies this situation by taking into consideration not only the bids but also

Expanded PaperInterface specification

```
∀ {c : ℝ},
0 < c →
  Filter.Tendsto
    (fun (epsilon : ℝ) =>
      AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3GreedyTwoWordRatio c epsilon)
    (nhds 0) (nhds (1 / 2))
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3GreedyTwoWordRatio](#)

Lean-elaborated claim roles

```
1. parameter: ℝ
2. assumption: 0 < c
3. parameter: Set ℝ
4. assumption: U ∈ nhds (1 / 2)
5. conclusion: U ∈
  Filter.map
    (fun (epsilon : ℝ) =>
      AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3GreedyTwoWordRatio c epsilon)
    (nhds 0)
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3_greedy_tight_half_example

Recorded source-to-Spec screening Verdict: matches

Reason: The two-bidder example is a tight one-half limit as the bid perturbation vanishes, exactly the Tendsto conclusion in the target.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 232-239

Verbatim source input

Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .
Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}^+$ be the following (monotonically decreasing) function:
 $\psi_k(i) = 1 - e^{-(1-i/k)}$
Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.
Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

Expanded PaperInterface specification

```
∀ (slab : ℕ → ℕ) (s : ℝ),  
  Filter.Tendsto (fun (k : ℕ) => ↑(slab k + 1) / ↑(k + 1)) Filter.atTop (nhds s) →  
    AppliedModelingLib.Sequence.SeqTendsTo (fun (k : ℕ) => 1 - Real.exp (-(1 - ↑(slab k + 1) / ↑(k + 1))))  
      (1 - Real.exp (-(1 - s)))
```

Retained governing declarations

- [AppliedModelingLib.Sequence.SeqTendsTo](#)

Lean-elaborated claim roles

```
1. parameter: ℕ → ℕ  
2. parameter: ℝ  
3. assumption: Filter.Tendsto (fun (k : ℕ) => ↑(slab k + 1) / ↑(k + 1)) Filter.atTop (nhds s)  
4. parameter: ℝ  
5. assumption: 0 < δ  
6. conclusion: ∃ (N : ℕ),  
  ∀ (k : ℕ), N ≤ k → |(fun (k : ℕ) => 1 - Real.exp (-(1 - ↑(slab k + 1) / ↑(k + 1)))) k - (1 - Real.exp (-(1 - s)))| ≤ δ
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3_discrete_tradeoff_convergence

Recorded source-to-Spec screening Verdict: matches

Reason: Given convergence of the normalized slab index, the target states convergence of ψ_k to ψ , exactly the source's $\psi_k \rightarrow \psi$ assertion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 237-245

Verbatim source input

Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times w_k(\text{slab}(i))$.
Note that in the special case when all the bids are equal, our algorithm works in the same way as the BALANCE algorithm of [KP00], for any monotonically decreasing tradeoff function.

4

Analyzing BALANCE using a Factor-Revealing LP

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] (k : ℕ)
(hk : 0 < k) (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
(a : Advertiser) {commonBid : ℝ},
(∀ (b : Advertiser), I.bid b q = commonBid) →
0 < commonBid →
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan k hk I S q).winner = some a →
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q a ∧
  ∀ (b : Advertiser),
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q b →
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent a) (I.budget a) ≤
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent b) (I.budget b)
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign](#)

Lean-elaborated claim roles

```
1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: ℕ
6. assumption: 0 < k
7. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
8. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query
9. parameter: Query
10. parameter: Advertiser
11. parameter: ℝ
12. assumption: ∀ (b : Advertiser), I.bid b q = commonBid
13. assumption: 0 < commonBid
14. assumption: (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan k hk I S q).winner = some a
15. conclusion: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q a ∧
  ∀ (b : Advertiser),
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q b →
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent a) (I.budget a) ≤
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent b) (I.budget b)
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section3_equal_bid_discrete_msvv_is_balance

Recorded source-to-Spec screening Verdict: matches

Reason: For a positive common bid, maximizing the decreasing slab tradeoff selects an available bidder with minimal current slab, which is the BALANCE behavior claimed for equal bids.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 718-773

Verbatim source input

Theorem 8 For function ψ_k defined as
 $\psi_k(i) :=$

$$\sum_{j=i}^{k-1} \frac{1}{k} y_j^* = 1 - (1 - \frac{1}{k})^{k-i+1}$$

the competitive ratio of the algorithm is $(1 - \frac{1}{e})$, as k tends to infinity.

Proof :
 By Lemma 4, the optimal solution to $L(\pi, \psi)$ and $D(\pi, \psi)$ has value $\sum_{i=1}^k y_i^*$. By Lemma 5 this equals $(b + \Delta) \cdot \sum_{i=1}^k y_i^* \leq N/e + \Delta \cdot \sum_{i=1}^k y_i^*$ (since $b \cdot \sum_{i=1}^k y_i^* \leq N/e$, from Section 4).
 Now,
 $\sum_{i=1}^k$

$$\Delta \cdot \sum_{i=1}^k y_i^* = \sum_{i=1}^k y_i^* ((\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i))$$

$$\sum_{i=1}^k (\alpha_i - \beta_i) (y_i^* + \dots + y_{k-1}^*)$$

$$\sum_{i=1}^k (\alpha_i - \beta_i) \psi(i)$$

$$\sum_{i=1}^k$$

$$=$$

$$\sum_{i=1}^k$$

$$=$$

$$\sum_{i=1}^k$$

$$\leq$$

$$N$$

$$\frac{1}{k}$$

where the last equality follows from our choice of the function ψ , and the inequality follows from Lemma 7. As k tends to infinity, we get that the competitive ratio of our algorithm is $(1 - \frac{1}{e})$.

[Next verbatim source excerpt]

Lemma 7 gives us
 $\sum_{i=1}^k$

$$\psi(i) \alpha_i \leq$$

$$\sum_{i=1}^k$$

where the choice of ψ is:

$$\sum_{i=1}^k$$

$$\psi(i) \beta_i$$

$$\sum_{i=1}^k$$

$$\psi(i) = 1 - (1 - \frac{1}{k})^{k-i+1}$$

Combining these relations we get:

$$\sum_{i=1}^k$$

$$\alpha_i$$

$$\sum_{i=1}^k$$

$$\sum_{i=1}^k$$

$$\leq$$

$$N$$

$$e$$

But the left side of the inequality above is precisely the amount of money left unspent at the end of the algorithm. This establishes that the competitive ratio is $1 - 1/e$.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

Under the paper's stated temporary unit-budget and exhaustive-optimum normalization, after the documented finite suffix correction (exponent $k-i$ and $\hookrightarrow$ unspent coefficient $(k-i)/k$), Theorem 8's Balance/MSVV competitive ratio tends to $1-1/e$ as its discretization/small-bids parameter tends to infinity.

Archival source anchor: cited publication, lines 718-773

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] (n : ℕ → ℕ)
  (l : (m : ℕ) → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser (Fin (n m)))
  (opt : (m : ℕ) → Fin (List.finRange (n m)).length → Option Advertiser),
  (∀ (m : ℕ), AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids (l m)) →
  (∀ (hbudget : ∀ (m : ℕ), AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets (l m)),
    (∀ (m : ℕ), AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets (l m)) →
    (∀ (m : ℕ) (a : Advertiser) (q : Fin (n m)), (l m).bid a q ≤ (l m).budget a / ↑(m + 1)) →
    (∀ (m : ℕ),
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentFeasible (l m)
        (List.finRange (n m)) (opt m)) →
    (∀ (m : ℕ) (a : Advertiser),
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend (l m) (List.finRange (n m))
        (opt m) a =
        1) →
    ∀ (delta : ℝ),
    0 < delta →
    ∃ (M : ℕ),
    ∀ (m : ℕ),
    M ≤ m →
    AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
    AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance (l m)
      (List.finRange (n m)))
    (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_competitive_in_small_bids_limitSpec_pr_j
      ↪ oof_1
      n l hbudget m) ≤
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences
      (m + 1) (Nat.zero_lt_succ m) (l m) (List.finRange (n m))).revenue +
    delta

```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentFeasible](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)

Lean-elaborated claim roles

1. parameter: Type u_5
2. parameter: Fintype Advertiser
3. parameter: DecidableEq Advertiser
4. parameter: ℕ → ℕ
5. parameter: (m : ℕ) → AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser (Fin (n m))
6. parameter: (m : ℕ) → Fin (List.finRange (n m)).length → Option Advertiser
7. assumption: ∀ (m : ℕ), AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids (l m)
8. assumption: ∀ (m : ℕ), AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets (l m)
9. assumption: ∀ (m : ℕ), AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets (l m)
10. assumption: ∀ (m : ℕ) (a : Advertiser) (q : Fin (n m)), (l m).bid a q ≤ (l m).budget a / ↑(m + 1)
11. assumption: ∀ (m : ℕ), AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentFeasible (l m) (List.finRange (n m)) (opt m)
12. assumption: ∀ (m : ℕ) (a : Advertiser), AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend (l m) (List.finRange (n m)) (opt m) a = 1
13. parameter: ℝ
14. assumption: 0 < delta

```

15. conclusion:  $\exists (M : \mathbb{N}),$ 
 $\forall (m : \mathbb{N}),$ 
 $M \leq m \rightarrow$ 
  AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
  AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance (l m)
   (List.finRange (n m)))
  (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_competitive_in_small_bids_limitSpec._proof_1
   n l hbudget m)  $\leq$ 
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences (m + 1)
   (Nat.zero_lt_succ m) (l m) (List.finRange (n m))).revenue +
  delta

```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_competitive_in_small_bids_limit

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: The Spec makes the unit-budget, exhaustive-optimum, occurrence-indexed small-bids limit explicit and uses the approved corrected finite tradeoff, yielding the stated 1-1/e limit.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 848-853

Verbatim source input

allocation gets from the bins of final type i . Let $\alpha = \sum_i \alpha_i$, be the total amount of money obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become
$$\sum_j \alpha_{ij} = 1$$
$$\forall i : \beta_i \geq$$

Expanded PaperInterface specification

$\forall \{ \text{Advertiser} : \text{Type } u_1 \} \{ \text{Query} : \text{Type } u_2 \} \{ k : \mathbb{N} \}$
($l : \text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query}$) ($\text{history} : \text{List Query}$)
($\text{opt} : \text{Fin history.length} \rightarrow \text{Option Advertiser}$) ($\text{finalType} : \text{Advertiser} \rightarrow \text{Fin k}$),
 $\text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType } l \text{ history opt finalType} =$
 $\text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenue } l \text{ history opt}$

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType](#)

Lean-elaborated claim roles

1. parameter: $\text{Type } u_1$
2. parameter: $\text{Type } u_2$
3. parameter: $\mathbb{N}$
4. parameter: $\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query}$
5. parameter: List Query
6. parameter: $\text{Fin history.length} \rightarrow \text{Option Advertiser}$
7. parameter: $\text{Advertiser} \rightarrow \text{Fin k}$
8. conclusion: $\text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType } l \text{ history opt finalType} =$
 $\text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenue } l \text{ history opt}$

Lean proof endpoint (built separately)

$\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_alpha_total_definition}$

Recorded source-to-Spec screening Verdict: matches

Reason: The source defines alpha as total OPT revenue; the target identifies the total-by-type quantity with the full occurrence revenue.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 852-860

Verbatim source input

$\alpha - ij=1 \alpha_j$
 $\forall i : \beta_i \geq$
 k
 X
 X
 $\psi(i)\alpha_i \leq$
 $\psi(i)\beta_i$
 i

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] {k : ℕ}
(hk : 0 < k) (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k),
AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets l →
  (∀ (a : Advertiser),
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend l history opt a ≤ l.budget a) →
  (∀ (a : Advertiser),
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k
      ((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences k hk l
        history).spent
        a /
        l.budget a)
      (finalType a)) →
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6BetaLowerBounds
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType l history opt
    finalType)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueByFinalType l history opt
    finalType)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6AggregateSlabSpend k l
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences k hk l
      history).spent)

```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6BetaLowerBounds](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6AggregateSlabSpend](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueByFinalType](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType](#)

Lean-elaborated claim roles

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: ℕ
6. assumption: 0 < k
7. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
8. parameter: List Query
9. parameter: Fin history.length → Option Advertiser
10. parameter: Advertiser → Fin k
11. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets l
12. assumption: $\forall (a : \text{Advertiser}),$
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend l history opt a ≤ l.budget a
13. assumption: $\forall (a : \text{Advertiser}),$
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k
((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences k hk l history).spent a /
l.budget a)
(finalType a)
14. parameter: Fin k
15. conclusion: (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType l history opt finalType -
 $\sum j \in \text{Finset.lit } i,$
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueByFinalType l history opt finalType
j) /
 $\uparrow k \leq$
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section6AggregateSlabSpend k l
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences k hk l history).spent i

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_beta_lower_bound_formula

Recorded source-to-Spec screening Verdict: matches

Reason: After expanding section6BetaLowerBounds, the target asserts $\beta_i \geq (\alpha\text{-sum}_{\{j \leq i\}} \alpha_j)/k$, the first displayed Section 6 relation.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 852-861

Verbatim source input

$\alpha - \beta_i = 1 \alpha_j$
 $\forall i : \beta_i \geq$
 k
 X
 X
 $\psi(i) \alpha_i \leq$
 $\psi(i) \beta_i$
 i
 i

Expanded PaperInterface specification

$\forall \{m : \text{Type } u_5\} \{ \text{Query} : \text{Type } u_6\} [\text{inst} : \text{Fintype } m] [\text{inst}_1 : \text{Fintype } \text{Query}] (\psi \text{ alpha } \beta : m \rightarrow \mathbb{R})$
 $(\text{opt } \text{alg} : \text{Query} \rightarrow \mathbb{R}) (\text{queryType } \text{querySlab} : \text{Query} \rightarrow m),$
 $\sum q : \text{Query}, (\text{opt } q * \psi (\text{queryType } q) - \text{alg } q * \psi (\text{querySlab } q)) \leq 0 \rightarrow$
 $\sum q : \text{Query}, \text{opt } q * \psi (\text{queryType } q) = \sum i : m, \psi i * \text{alpha } i \rightarrow$
 $\sum q : \text{Query}, \text{alg } q * \psi (\text{querySlab } q) \leq \sum i : m, \psi i * \text{beta } i \rightarrow$
 $\sum i : m, \psi i * \text{alpha } i \leq \sum i : m, \psi i * \text{beta } i$

Lean-elaborated claim roles

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype m
4. parameter: Fintype Query
5. parameter: m → ℝ
6. parameter: m → ℝ
7. parameter: m → ℝ
8. parameter: Query → ℝ
9. parameter: Query → ℝ
10. parameter: Query → m
11. parameter: Query → m
12. assumption: $\sum q : \text{Query}, (\text{opt } q * \psi (\text{queryType } q) - \text{alg } q * \psi (\text{querySlab } q)) \leq 0$
13. assumption: $\sum q : \text{Query}, \text{opt } q * \psi (\text{queryType } q) = \sum i : m, \psi i * \text{alpha } i$
14. assumption: $\sum q : \text{Query}, \text{alg } q * \psi (\text{querySlab } q) \leq \sum i : m, \psi i * \text{beta } i$
15. conclusion: $\text{Real.le} \uparrow (\sum i : m, \psi i * \text{alpha } i) (\sum i : m, \psi i * \text{beta } i)$

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_weighted_alpha_beta_inequality

Recorded source-to-Spec screening Verdict: matches

Reason: The target derives the displayed weighted inequality $\sum \psi_i \alpha_i \leq \sum \psi_i \beta_i$ from the per-query and accounting equalities, matching the source relation.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 825-865

Verbatim source input

In this section we show how our algorithm and analysis can be generalized to the following situations:

1. Advertisers have different daily budgets.
 2. The optimal allocation does not exhaust all the money of advertisers
 3. Advertisers enter at different times.
 4. More than one ad can appear with the results of a query. The most general situation is that with each query we are provided a number specifying the maximum number of ads.
 5. A bidder pays only if the user clicks on his ad.
 6. A winning bidder pays only an amount equal to the next highest bid.
- 1, 2, 3: We say that the current type of a bidder at some time during the run of the algorithm is j if he has spent between $(j-1)/k$ and j/k fraction of his budget at that time. The algorithm allocates the next query to the bidder who maximizes the product of his bid and $\psi(\text{current type})$.
- 13

[form-feed in source extraction]

The proof of the competitive ratio changes minimally: Let the budget of bidder j be B_j . For $i = 1, \dots, k$, define β_{ij} to be the amount of money spent by the bidder j from the interval

P_j
 i
[$i-1$
 j β_i . Let α_i be the
 k B_j , k B_j) of his budget. Let $\beta_i =$
 P amount of money that the optimal
allocation gets from the bins of final type i . Let $\alpha = \sum \alpha_i$, be the total amount of money
obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become
 P
 $\alpha - \sum_{ij=1} \alpha_j$
 $\forall i : \beta_i \geq$
 k
 X
 X
 X
 $\psi(i)\alpha_i \leq$
 $\psi(i)\beta_i$
 i

 i

These two sets of equations suffice to prove that the competitive ratio is at least $1 - 1/e$. We also note that the algorithm and the proof of the competitive ratio remain unchanged even if we allow advertisers to enter the bidding process at any time during the query sequence.

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : DecidableEq Advertiser] (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query),
AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I →
  ∀ (hbudget : AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query) {epsilon : ℝ},
  0 ≤ epsilon →
  epsilon ≤ 1 →
  AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon →
  AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
  AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
  (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec.)
  → _proof_1
  I hbudget) ≤
  AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
  (AppliedModelingLib.Online.AdWordsInstance.runAssignment
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
  (AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history))
  (List.finRange history.length)) +
  epsilon * (Real.exp 1 + 1) *
  ∑ t : Fin history.length,
  AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history) t

```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery](#)
- [AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue](#)

- [AppliedModelingLib.Online.AdWordsInstance.runAssignment](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids](#)

Lean-elaborated claim roles

```

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. assumption: Nonempty Advertiser
5. parameter: DecidableEq Advertiser
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
7. assumption: AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I
8. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I
9. parameter: List Query
10. parameter: ℝ
11. assumption: 0 ≤ epsilon
12. assumption: epsilon ≤ 1
13. assumption: AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon
14. conclusion: Real.le t
  (AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
   AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue
   (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
   (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec._proof_1
    I hbudget))
  (AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue
   (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
   (AppliedModelingLib.Online.AdWordsInstance.runAssignment
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
    (AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule
     (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history))
    (List.finRange history.length)) +
   epsilon * (Real.exp 1 + 1) *
   ∑ t : Fin history.length,
     AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery
       (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history) t)

```

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_different_budgets_and_nonexhaustive_optimum_theorem8_finite_explicit_error`

Recorded source-to-Spec screening Verdict: matches

Reason: The expanded Spec covers the source's different-budget and nonexhaustive-optimum extension with the finite explicit-error conclusion it states.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 872-879

Verbatim source input

6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query. Search engine companies charge a lower amount: the next highest bid. There are different ways of defining “next highest bid”. We can extend our analysis for two of these definitions: the next highest bid is chosen from all bids received at the start of the algorithm or only among alive bidders, i.e. bidders who still have money. It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1 - 1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi$ (fraction of money spent). Next, let us consider the second possibility. In this case, the offline

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser]
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (a : Advertiser) (q : Query),
AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I a q =
  have others : Finset Advertiser := Finset.univ.erase a;
  if h : others.Nonempty then max 0 (others.sup' h fun (b : Advertiser) => I.bid b q) else 0
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all](#)

Lean-elaborated claim roles

```
1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
6. parameter: Advertiser
7. parameter: Query
8. conclusion: AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I a q =
  have others : Finset Advertiser := Finset.univ.erase a;
  if h : others.Nonempty then max 0 (others.sup' h fun (b : Advertiser) => I.bid b q) else 0
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_bid_all_formula

Recorded source-to-Spec screening **Verdict:** matches

Reason: The target defines the next highest bid among all other initial bidders, exactly the first option in the source.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 872-882

Verbatim source input

6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query. Search engine companies charge a lower amount: the next highest bid. There are different ways of defining “next highest bid”. We can extend our analysis for two of these definitions: the next highest bid is chosen from all bids received at the start of the algorithm or only among alive bidders, i.e. bidders who still have money. It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1-1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi$ (fraction of money spent). Next, let us consider the second possibility. In this case, the offline algorithm will attempt to keep alive bidders simply to charge other bidders higher amounts. If the online algorithm is also allowed this capability, it can also keep all bidders alive all the way to the end and this possibility reduces to the first one.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser]
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (alive : Advertiser → Query → Prop)
[inst_2 : (a : Advertiser) → (q : Query) → Decidable (alive a q)] (a : Advertiser) (q : Query),
AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive l alive a q =
  have others : Finset Advertiser := {b ∈ Finset.univ.erase a | alive b q};
  if h : others.Nonempty then max 0 (others.sup' h fun (b : Advertiser) => l.bid b q) else 0
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive](#)

Lean-elaborated claim roles

```
1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
6. parameter: Advertiser → Query → Prop
7. parameter: (a : Advertiser) → (q : Query) → Decidable (alive a q)
8. parameter: Advertiser
9. parameter: Query
10. conclusion: AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive l alive a q =
  have others : Finset Advertiser := {b ∈ Finset.univ.erase a | alive b q};
  if h : others.Nonempty then max 0 (others.sup' h fun (b : Advertiser) => l.bid b q) else 0
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_bid_alive_formula

Recorded source-to-Spec screening Verdict: matches

Reason: The target defines the next highest bid as the maximum among other alive bidders (or zero if none), exactly the paper’s alive-bid option.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 877-885

Verbatim source input

It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1-1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi$ (fraction of money spent). Next, let us consider the second possibility. In this case, the offline algorithm will attempt to keep alive bidders simply to charge other bidders higher amounts. If the online algorithm is also allowed this capability, it can also keep all bidders alive all the way to the end and this possibility reduces to the first one.

7

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser]
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (a : Advertiser) (q : Query),
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive I
    (fun (x : Advertiser) (x_1 : Query) => True) a q =
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all I a q
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all](#)

Lean-elaborated claim roles

```
1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
6. parameter: Advertiser
7. parameter: Query
8. conclusion: AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive I
  (fun (x : Advertiser) (x_1 : Query) => True) a q =
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all I a q
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_alive_keep_alive_reduction

Recorded source-to-Spec screening Verdict: matches

Reason: With every bidder declared alive, the alive-bid maximum equals the all-bids maximum, which is exactly the keep-alive reduction described in the source.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 872-879

Verbatim source input

6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query. Search engine companies charge a lower amount: the next highest bid. There are different ways of defining “next highest bid”. We can extend our analysis for two of these definitions: the next highest bid is chosen from all bids received at the start of the algorithm or only among alive bidders, i.e. bidders who still have money. It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1 - 1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi$ (fraction of money spent). Next, let us consider the second possibility. In this case, the offline

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For arbitrary budgets, the all-bidder next-price rule has the finite Theorem 8 bound when each actual charge is at most epsilon times the possible winner's $\hookrightarrow$ budget, evaluated over occurrence-indexed arrivals.

Archival source anchor: cited publication, lines 875-882

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : DecidableEq Advertiser] (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(hbudget : AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets l) (history : List Query) {epsilon : ℝ},
0 ≤ epsilon →
epsilon ≤ 1 →
(∀ (a : Advertiser) (q : Query),
  AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all l a q ≤ epsilon * l.budget a) →
let J : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser (Fin history.length) :=
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
  (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids l
    (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all l))
history;
AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
  AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue J
  (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec._pr J
     $\hookrightarrow$  oof_1
    l hbudget) ≤
AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue J
  (AppliedModelingLib.Online.AdWordsInstance.runAssignment J
    (AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule J) (List.finRange history.length)) +
epsilon * (Real.exp 1 + 1) *
  Σ t : Fin history.length, AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery J t
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery](#)
- [AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue](#)
- [AppliedModelingLib.Online.AdWordsInstance.runAssignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all](#)

Lean-elaborated claim roles

```
1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. assumption: Nonempty Advertiser
5. parameter: DecidableEq Advertiser
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
7. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets l
8. parameter: List Query
9. parameter: ℝ
10. assumption: 0 ≤ epsilon
11. assumption: epsilon ≤ 1
12. assumption: ∀ (a : Advertiser) (q : Query),
  AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all l a q ≤ epsilon * l.budget a
13. conclusion: Real.le t
  (AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
   AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue
   (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
    (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids l
     (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all l))
    history)
   (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec._proof_1
    l hbudget))
  (AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue
   (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
    (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids l
     (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all l))
    history)
   (AppliedModelingLib.Online.AdWordsInstance.runAssignment
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
     (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids l
      (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all l))
     history)
    (AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule
     (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
      (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids l
       (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all l))
      history))
    (List.finRange history.length)) +
   epsilon * (Real.exp 1 + 1) *
   ∑ t : Fin history.length,
   AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery
   (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
    (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids l
     (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all l))
    history)
   t)
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_bid_all_theorem8_finite_explicit_error

Recorded source-to-Spec screening Verdict: matches approved corrected target

Reason: The Spec uses occurrence-indexed arrivals, arbitrary positive budgets, and precisely assumes each all-bidder next-price charge is at most epsilon times the potential winner's budget before giving the finite explicit-error bound.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 872-882

Verbatim source input

6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query. Search engine companies charge a lower amount: the next highest bid. There are different ways of defining “next highest bid”. We can extend our analysis for two of these definitions: the next highest bid is chosen from all bids received at the start of the algorithm or only among alive bidders, i.e. bidders who still have money. It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1-1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times$ ψ (fraction of money spent). Next, let us consider the second possibility. In this case, the offline algorithm will attempt to keep alive bidders simply to charge other bidders higher amounts. If the online algorithm is also allowed this capability, it can also keep all bidders alive all the way to the end and this possibility reduces to the first one.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

When the online model keeps every bidder alive, the alive-bidder next-price charge reduces pointwise to the all-bidder charge; under its explicit $\hookrightarrow$ effective-charge small-bids premise, the all-bidder finite Theorem 8 bound applies on occurrence-indexed arrivals.

Archival source anchor: cited publication, lines 875-882

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : DecidableEq Advertiser] (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(hbudget : AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query) {epsilon : ℝ},
0 ≤ epsilon →
epsilon ≤ 1 →
(∀ (a : Advertiser) (q : Query),
  AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I a q ≤ epsilon * I.budget a) →
let J : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser (Fin history.length) :=
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
  (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids I
    (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I))
  history;
(∀ (a : Advertiser) (q : Query),
  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive I
    (fun (x : Advertiser) (x_1 : Query) => True) a q =
    AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all I a q) ∧
AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
  AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue J
  (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec._)
   $\hookrightarrow$  proof 1
  I hbudget) ≤
AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue J
  (AppliedModelingLib.Online.AdWordsInstance.runAssignment J
    (AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule J) (List.finRange history.length)) +
epsilon * (Real.exp 1 + 1) *
  Σ t : Fin history.length, AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery J t
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery](#)
- [AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue](#)
- [AppliedModelingLib.Online.AdWordsInstance.runAssignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all](#)

Lean-elaborated claim roles

```
1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. assumption: Nonempty Advertiser
5. parameter: DecidableEq Advertiser
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
7. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I
8. parameter: List Query
9. parameter: ℝ
10. assumption: 0 ≤ epsilon
11. assumption: epsilon ≤ 1
12. assumption: ∀ (a : Advertiser) (q : Query),
    AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I a q ≤ epsilon * I.budget a
13. conclusion: (∀ (a : Advertiser) (q : Query),
    AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive I
    (fun (x : Advertiser) (x_1 : Query) => True) a q =
    AppliedModelingLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all I a q) ∧
    AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
    AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
    (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids I
    (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I))
    history)
    (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec._proof_1
    I hbudget) ≤
    AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
    (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids I
    (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I))
    history)
    (AppliedModelingLib.Online.AdWordsInstance.runAssignment
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
    (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids I
    (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I))
    history)
    (AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
    (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids I
    (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I))
    history))
    (List.finRange history.length)) +
    epsilon * (Real.exp 1 + 1) *
    ∑ t : Fin history.length,
    AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
    (AppliedModelingLib.Online.AdWordsInstance.withEffectiveBids I
    (AppliedModelingLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I))
    history)
    t
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_bid_alive_theorem8_finite_explicit_error

Recorded source-to-Spec screening Verdict: matches_approved_corrected target

Reason: The Spec proves the pointwise alive-price/all-price equality with every bidder alive, then applies the approved effective-charge finite bound on occurrence-indexed arrivals.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 866-872

Verbatim source input

4: If the arriving query q requires n_q number of advertisements to be placed, then allocate it to the bidders with the top n_q values of the product of bid and ψ (current type). The proof of the competitive ratio remains unchanged.
5: In order to model this situation, we simply set the effective bid of a bidder to be the product of his actual bid and his click-through rate (CTR), which is the probability that a user will click on his ad. We assume that the click-through rate is known to the algorithm in advance indeed several search engines keep a measure of the $\hookrightarrow$ click-through rates of the bidders.
6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query.

Expanded PaperInterface specification

$\forall \{ \text{Advertiser} : \text{Type } u_5 \} \{ \text{Query} : \text{Type } u_6 \}$
($I : \text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query}$) ($\text{ctr} : \text{Advertiser} \rightarrow \text{Query} \rightarrow \mathbb{R}$)
($a : \text{Advertiser}$) ($q : \text{Query}$),
($\text{AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates } I \text{ ctr}$).bid $a \ q = \text{ctr } a \ q * I.\text{bid } a \ q$

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

Lean-elaborated claim roles

1. parameter: $\text{Type } u_5$
2. parameter: $\text{Type } u_6$
3. parameter: $\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query}$
4. parameter: $\text{Advertiser} \rightarrow \text{Query} \rightarrow \mathbb{R}$
5. parameter: Advertiser
6. parameter: Query
7. conclusion: ($\text{AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates } I \text{ ctr}$).bid $a \ q = \text{ctr } a \ q * I.\text{bid } a \ q$

Lean proof endpoint (built separately)

$\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_click_through_rates_effective_bid_formula}$

Recorded source-to-Spec screening Verdict: matches

Reason: The target exactly sets the effective bid to CTR times the actual bid, as the paper specifies.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 869-871

Verbatim source input

5: In order to model this situation, we simply set the effective bid of a bidder to be the product of his actual bid and his click-through rate (CTR), which is the probability that a user will click on his ad. We assume that the click-through rate is known to the algorithm in advance indeed several search engines keep a measure of the
↪ click-through rates of the bidders.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : DecidableEq Advertiser] (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(ctr : Advertiser → Query → ℝ),
(∀ (a : Advertiser) (q : Query), 0 ≤ ctr a q) →
(∀ (a : Advertiser) (q : Query), ctr a q ≤ 1) →
AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I →
  ∀ (hbudget : AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query) {epsilon : ℝ},
    0 ≤ epsilon →
    epsilon ≤ 1 →
    AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon →
    let J : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser (Fin history.length) :=
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
        (AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates I ctr) history;
    AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
      AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue J
      (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpe J
        ↪ c_proof_1
        I hbudget) ≤
    AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue J
      (AppliedModelingLib.Online.AdWordsInstance.runAssignment J
        (AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule J)
        (List.finRange history.length)) +
    epsilon * (Real.exp 1 + 1) *
    ∑ t : Fin history.length, AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery J t
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery](#)
- [AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue](#)
- [AppliedModelingLib.Online.AdWordsInstance.runAssignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids](#)

Lean-elaborated claim roles

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. assumption: Nonempty Advertiser
5. parameter: DecidableEq Advertiser
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
7. parameter: Advertiser → Query → ℝ
8. assumption: ∀ (a : Advertiser) (q : Query), 0 ≤ ctr a q
9. assumption: ∀ (a : Advertiser) (q : Query), ctr a q ≤ 1
10. assumption: AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I
11. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I
12. parameter: List Query
13. parameter: ℝ
14. assumption: 0 ≤ epsilon
15. assumption: epsilon ≤ 1
16. assumption: AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon
17. conclusion: Real.le+
(AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *

```

AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
(AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates l ctr) history)
(AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec._proof_1
l hbudget))
(AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
(AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates l ctr) history)
(AppliedModelingLib.Online.AdWordsInstance.runAssignment
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
(AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates l ctr) history)
(AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
(AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates l ctr) history))
(List.finRange history.length)) +
epsilon * (Real.exp 1 + 1) *
Σ t : Fin history.length,
  AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance
  (AppliedModelingLib.Online.AdWordsInstance.withClickThroughRates l ctr) history)
t)

```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_click_through_rates_theorem8_finite_explicit_error

Recorded source-to-Spec screening Verdict: matches

Reason: The expanded Spec formalizes click-through rates as effective bids and states the corresponding finite Theorem 8 guarantee on the source model.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 863-865

Verbatim source input

These two sets of equations suffice to prove that the competitive ratio is at least $1 - 1/e$. We also note that the algorithm and the proof of the competitive ratio remain unchanged even if we allow advertisers to enter the bidding process at any time during the query sequence.

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : DecidableEq Advertiser] (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query),
AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I →
  ∀ (hbudget : AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query)
    (entryTime : Advertiser → Fin (history.length + 1)) {epsilon : ℝ},
    0 ≤ epsilon →
    epsilon ≤ 1 →
    AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon →
    let J0 : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser (Fin history.length) :=
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history;
    let J : AppliedModelingLib.Online.AdWordsInstance Advertiser (Fin history.length) :=
      AppliedModelingLib.Online.AdWordsInstance.withAvailability J0
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.availableFromEntry entryTime);
    AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
    J.offlineOptimumValue
    (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec.)
    ↪ _proof 1
    I hbudget) ≤
    AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue J
    (J.runAssignment J.balanceChoiceRule (List.finRange history.length)) +
    epsilon * (Real.exp 1 + 1) * ∑ t : Fin history.length, J.maxBidForQuery t

```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.AdWordsInstance.balanceChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.maxBidForQuery](#)
- [AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue](#)
- [AppliedModelingLib.Online.AdWordsInstance.runAssignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.withAvailability](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.instDecidable_mSVV07AdWords](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.availableFromEntry](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids](#)

Lean-elaborated claim roles

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. assumption: Nonempty Advertiser
5. parameter: DecidableEq Advertiser
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
7. assumption: AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I
8. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I
9. parameter: List Query
10. parameter: Advertiser → Fin (history.length + 1)
11. parameter: ℝ
12. assumption: 0 ≤ epsilon
13. assumption: epsilon ≤ 1
14. assumption: AppliedModelingLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon
15. conclusion: Real.le†
(AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio *
(AppliedModelingLib.Online.AdWordsInstance.withAvailability
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history))

```

    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.availableFromEntry entryTime)).offlineOptimumValue
  (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_source_unit_budget_finite_competitive_with_explicit_slab_errorSpec._proof_1
   I hbudget))
(AppliedModelingLib.Online.MSVV07PaperFacing.paperRevenue
 (AppliedModelingLib.Online.AdWordsInstance.withAvailability
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.availableFromEntry entryTime))
 ((AppliedModelingLib.Online.AdWordsInstance.withAvailability
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.availableFromEntry entryTime)).runAssignment
 (AppliedModelingLib.Online.AdWordsInstance.withAvailability
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.availableFromEntry entryTime)).balanceChoiceRule
 (List.finRange history.length)) +
epsilon * (Real.exp 1 + 1) *
Σ t : Fin history.length,
  (AppliedModelingLib.Online.AdWordsInstance.withAvailability
   (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceIndexedInstance I history)
   (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.availableFromEntry entryTime)).maxBidForQuery
 t)

```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section6_availability_theorem8_finite_explicit_error

Recorded source-to-Spec screening Verdict: matches
Reason: The expanded Spec states the source availability variant on occurrence-indexed arrivals and gives its finite explicit-error guarantee without changing the stated economic object.
Reviewer check: ☐ Matches source input
 If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 20

Source locator: cited publication, lines 960-965

Verbatim source input

We believe a simple modification of our algorithm achieves this. Here is a concrete proposal: each bidder is assigned a weight, and his effective bid for a keyword is defined to be the product of the actual bid and his weight. We modify our algorithm to use effective bids in place of bids. The main open question here is whether for any fixed distribution on queries there is always a set of weights such that this algorithm achieves $1 - o(1)$ expected competitive ratio. How do we actually find a good set of weights for the bidders? Here is an online heuristic

Expanded PaperInterface specification

$\forall \{ \text{Advertiser} : \text{Type } u_5 \} \{ \text{Query} : \text{Type } u_6 \}$
 $(l : \text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query}) (weight : \text{Advertiser} \rightarrow \mathbb{R})$
 $(a : \text{Advertiser}) (q : \text{Query}),$
 $(\text{AppliedModelingLib.Online.AdWordsInstance.withAdvertiserWeights } l \text{ weight}).bid \ a \ q = weight \ a * l.bid \ a \ q$

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.withAdvertiserWeights](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)

Lean-elaborated claim roles

- 1. parameter: $\text{Type } u_5$
- 2. parameter: $\text{Type } u_6$
- 3. parameter: $\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query}$
- 4. parameter: $\text{Advertiser} \rightarrow \mathbb{R}$
- 5. parameter: Advertiser
- 6. parameter: Query
- 7. conclusion: $(\text{AppliedModelingLib.Online.AdWordsInstance.withAdvertiserWeights } l \text{ weight}).bid \ a \ q = weight \ a * l.bid \ a \ q$

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section8_weighted_effective_bid_formula`

Recorded source-to-Spec screening Verdict: matches
Reason: The target exactly defines an effective bid as bidder weight times actual bid.
Reviewer check: ☐ Matches source input
 If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 21

Source locator: cited publication, lines 1176-1213

Verbatim source input

Bin N

Bin $0.5N$

Bin $0.1N$ Bin 1

Figure 2: The bidders are ordered from right to left. The area inside the dark outline is the amount of money generated by the algorithm. The optimum allocation gets an amount equal to the whole rectangle.

We present an example to show a factor strictly less than $1 - 1/e$ for the algorithm which gives a query to a highest bidder, breaking ties by giving it to the bidder with most left-over money. This example has only three values for the bids - 0 , a or $2a$, for some small $a > 0$. Thus, in the case of arbitrary bids, the strategy of bucketing close enough bids (say within a factor of 2) together, and running such an algorithm does not work.

There are N bidders numbered $1, \dots, N$, each with budget 1 . We get the following query sequence and bidding pattern. Each bid is either 0 , a or $2a$. Let $m = 1/a$. We will take $a \rightarrow 0$. The queries arrive in N rounds. In each round m queries are made. The N rounds are divided into 3 phases.

Phase 1 ($1 \leq i \leq 0.4N$): In the first round m queries are made, for which the bidders $0.1N + 1$ to N bid with a bid of a , and bidders 1 to $0.1N$ do not bid. Similarly, for $1 \leq i \leq 0.4N$, in the i th round m queries are made, for which bidders $0.1N + i$ to N bid with a bid of a , and for which bidders 1 to $0.1N + i - 1$ do not bid.

For $1 \leq i \leq 0.4N$, the algorithm will distribute the queries of the i th round equally between bidders $0.1N + i$ to N . This will give the partial allocation as shown in Figure 2.

Phase 2 ($0.4N + 1 \leq i \leq 0.5N$): In the $(0.4N + 1)$ th round m queries are made, for which bidder 1 bids a , and bidders $0.5N$ to N bid $2a$ (the rest of the bidders bid 0). Similarly, for $0.4N + 1 \leq i \leq 0.5N$, in the i th round m queries are made, for which bidder $i - 0.4N$ bids a , and bidders $0.5N$ to N bid $2a$.

For $0.4N + 1 \leq i \leq 0.5N$, the algorithm will distribute the queries of round i equally between bidders $0.5N$ to N .

At this point during the algorithm, bidders $0.5N + 1$ to N have spent all their money.

20

[form-feed in source extraction]

Phase 3 ($0.5N + 1 \leq i \leq N$): m queries enter in round i , for which only bidder i bids at a , and the other bidders do not bid.

The algorithm has to throw away these queries, since bidders $0.5N + 1$ to N have already spent

Formalized review target The archival source above is not asserted equivalent to this formalized target.

The approved target is the normalized continuous-limit three-phase execution with a positive phase-three residual and its exact online/offline values.

Archival source anchor: cited publication, lines 1182-1220

Expanded PaperInterface specification

```
0 < AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail ^
(∀ (owner : ℝ),
  1 / 2 ≤ owner ^ owner ≤ 1 →
    AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeSpentBefore
    AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail owner =
    1) ^
    AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue = 9 / 10 - 1 / 2 * Real.log (9 / 5) ^
    AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue = 1
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeSpentBefore](#)

Lean-elaborated claim roles

```
1. conclusion: 0 < AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail ^
(∀ (owner : ℝ),
  1 / 2 ≤ owner ^ owner ≤ 1 →
    AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeSpentBefore
    AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail owner =
    1) ^
    AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue = 9 / 10 - 1 / 2 * Real.log (9 / 5) ^
    AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue = 1
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.appendix_naive_three_phase_constructionSpec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: The target is the approval-pinned normalized continuous-limit three-phase execution with positive phase-three residual and exact online/offline values, replacing rather than literally reproducing the archival finite construction.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 1182-1220

Verbatim source input

Figure 2: The bidders are ordered from right to left. The area inside the dark outline is the amount of money generated by the algorithm. The optimum allocation gets an amount equal to the whole rectangle.

We present an example to show a factor strictly less than $1 - 1/e$ for the algorithm which gives a query to a highest bidder, breaking ties by giving it to the bidder with most left-over money. This example has only three values for the bids - 0, a or $2a$, for some small $a > 0$. Thus, in the case of arbitrary bids, the strategy of bucketing close enough bids (say within a factor of 2) together, and running such an algorithm does not work. There are N bidders numbered $1, \dots, N$, each with budget 1. We get the following query sequence and bidding pattern. Each bid is either 0, a or $2a$. Let $m = 1/a$. We will take $a \rightarrow 0$. The queries arrive in N rounds. In each round m queries are made. The N rounds are divided into 3 phases.

Phase 1 ($1 \leq i \leq 0.4N$): In the first round m queries are made, for which the bidders $0.1N + 1$ to N bid with a bid of a , and bidders 1 to $0.1N$ do not bid. Similarly, for $1 \leq i \leq 0.4N$, in the i th round m queries are made, for which bidders $0.1N + i$ to N bid with a bid of a , and for which bidders 1 to $0.1N + i - 1$ do not bid.

For $1 \leq i \leq 0.4N$, the algorithm will distribute the queries of the i th round equally between bidders $0.1N + i$ to N . This will give the partial allocation as shown in Figure 2.

Phase 2 ($0.4N + 1 \leq i \leq 0.5N$): In the $(0.4N + 1)$ th round m queries are made, for which bidder 1 bids a , and bidders $0.5N$ to N bid $2a$ (the rest of the bidders bid 0). Similarly, for $0.4N + 1 \leq i \leq 0.5N$, in the i th round m queries are made, for which bidder $i - 0.4N$ bids a , and bidders $0.5N$ to N bid $2a$.

For $0.4N + 1 \leq i \leq 0.5N$, the algorithm will distribute the queries of round i equally between bidders $0.5N$ to N .

At this point during the algorithm, bidders $0.5N + 1$ to N have spent all their money.

20

[form-feed in source extraction]

Phase 3 ($0.5N + 1 \leq i \leq N$): m queries enter in round i , for which only bidder i bids at a , and the other bidders do not bid.

The algorithm has to throw away these queries, since bidders $0.5N + 1$ to N have already spent their money.

The optimum allocation, on the other hand, is to allocate the queries in round i as follows:

- For $1 \leq i \leq 0.4N$, allocate all queries in round i to bidder $0.1N + i$.
- For $0.4N + 1 \leq i \leq 0.5N$, allocate all queries in round i to bidder $i - 0.4N$.
- For $0.5N + 1 \leq i \leq N$, allocate all queries in round i to bidder i .

Clearly, OPT makes N amount of money. A calculation shows that the algorithm makes $0.62N$ amount of money. Thus the factor is strictly less than $1 - 1/e$.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

The Appendix's approved normalized limiting execution has value $9/10 - (1/2)\log(9/5)$, a positive phase-three residual, and does not equal the archival $0.62 \leftarrow$ value.

Archival source anchor: cited publication, lines 1182-1220

Expanded PaperInterface specification

```
AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue = 9 / 10 - 1 / 2 * Real.log (9 / 5) ^
AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeServedMass = 3 / 10 - 1 / 2 * Real.log (9 / 5) ^
0 < AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeServedMass ^
2 / 5 + 2 * (1 / 10) = 3 / 5 ^ 3 / 5 ^ 31 / 50
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeServedMass](#)

Lean-elaborated claim roles

```
1. conclusion: AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue = 9 / 10 - 1 / 2 * Real.log (9 / 5) ^
AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeServedMass = 3 / 10 - 1 / 2 * Real.log (9 / 5) ^
0 < AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeServedMass ^
2 / 5 + 2 * (1 / 10) = 3 / 5 ^ 3 / 5 ^ 31 / 50
```

Lean proof endpoint (built separately)

[AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.appendix_naive_algorithm_revenue_formulaSpec_proof](#)

Recorded source-to-Spec screening Verdict: matches approved corrected target

Reason: The archival $0.62N$ is replaced by the approval-pinned continuous-limit value $9/10-(1/2)\log(9/5)$, positive phase-three mass, and explicit numerical discrepancy; that approval calls it non-archival-equivalent.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 1210-1219

Verbatim source input

Phase 3 ($0.5N + 1 \leq i \leq N$): m queries enter in round i , for which only bidder i bids at a , and the other bidders do not bid.

The algorithm has to throw away these queries, since bidders $0.5N + 1$ to N have already spent their money.

The optimum allocation, on the other hand, is to allocate the queries in round i as follows:

- For $1 \leq i \leq 0.4N$, allocate all queries in round i to bidder $0.1N + i$.
- For $0.4N + 1 \leq i \leq 0.5N$, allocate all queries in round i to bidder $i - 0.4N$.
- For $0.5N + 1 \leq i \leq N$, allocate all queries in round i to bidder i .

Clearly, OPT makes N amount of money. A calculation shows that the algorithm makes $0.62N$

Expanded PaperInterface specification

```
(( $\forall$  ( $r : \mathbb{R}$ ),  $0 \leq r \rightarrow r \leq 2 / 5 \rightarrow$  AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneRawBid  $r$  ( $1 / 10 + r$ ) = 1)  $\wedge$ 
( $\forall$  (offset :  $\mathbb{R}$ ),
   $0 \leq$  offset  $\rightarrow$ 
    offset  $\leq 1 / 10 \rightarrow$  AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoRawBid offset offset = 1)  $\wedge$ 
( $\forall$  (owner :  $\mathbb{R}$ ),
   $1 / 2 \leq$  owner  $\rightarrow$ 
    owner  $\leq 1 \rightarrow$  AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeRawBid owner owner = 1)  $\wedge$ 
   $2 / 5 + 1 / 10 + 1 / 2 = 1$ )  $\wedge$ 
(AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible fun ( $\_bidder : \mathbb{R}$ ) => 1)  $\wedge$ 
AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue = 1  $\wedge$ 
 $\forall$  (spend :  $\mathbb{R} \rightarrow \mathbb{R}$ ),
  AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible spend  $\rightarrow$ 
     $\int$  (bidder :  $\mathbb{R}$ ) in  $0..1$ , spend bidder  $\leq$  AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneRawBid](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeRawBid](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoRawBid](#)

Lean-elaborated claim roles

```
1. conclusion: (( $\forall$  ( $r : \mathbb{R}$ ),  $0 \leq r \rightarrow r \leq 2 / 5 \rightarrow$  AppliedModelingLib.Online.MSVV07Appendix.printedPhaseOneRawBid  $r$  ( $1 / 10 + r$ ) = 1)  $\wedge$ 
( $\forall$  (offset :  $\mathbb{R}$ ),
   $0 \leq$  offset  $\rightarrow$ 
    offset  $\leq 1 / 10 \rightarrow$  AppliedModelingLib.Online.MSVV07Appendix.printedPhaseTwoRawBid offset offset = 1)  $\wedge$ 
( $\forall$  (owner :  $\mathbb{R}$ ),
   $1 / 2 \leq$  owner  $\rightarrow$ 
    owner  $\leq 1 \rightarrow$  AppliedModelingLib.Online.MSVV07Appendix.printedPhaseThreeRawBid owner owner = 1)  $\wedge$ 
   $2 / 5 + 1 / 10 + 1 / 2 = 1$ )  $\wedge$ 
(AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible fun ( $\_bidder : \mathbb{R}$ ) => 1)  $\wedge$ 
AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue = 1  $\wedge$ 
 $\forall$  (spend :  $\mathbb{R} \rightarrow \mathbb{R}$ ),
  AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible spend  $\rightarrow$ 
     $\int$  (bidder :  $\mathbb{R}$ ) in  $0..1$ , spend bidder  $\leq$  AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.appendix_optimum_revenue_formula

Recorded source-to-Spec screening Verdict: matches

Reason: After normalizing the paper's unit budgets and total OPT= N to a unit continuum, the three phase-wise offline assignments, feasibility, and offline value 1 express the stated optimum allocation.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 24

Source locator: cited publication, lines 1215-1221

Verbatim source input

The optimum allocation, on the other hand, is to allocate the queries in round i as follows:

- For $1 \leq i \leq 0.4N$, allocate all queries in round i to bidder $0.1N + i$.
- For $0.4N + 1 \leq i \leq 0.5N$, allocate all queries in round i to bidder $i - 0.4N$.
- For $0.5N + 1 \leq i \leq N$, allocate all queries in round i to bidder i .

Clearly, OPT makes N amount of money. A calculation shows that the algorithm makes $0.62N$ amount of money. Thus the factor is strictly less than $1 - 1/e$.
We can modify the above example to allow bids of 0 , a and ka , for any $k > 1$, such that the

Expanded PaperInterface specification

```
AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue /  
  AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue <  
AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)

Lean-elaborated claim roles

```
1. conclusion: Real.lt+  
(AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOnlineValue /  
  AppliedModelingLib.Online.MSVV07Appendix.printedContinuousOfflineValue)  
AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Lean proof endpoint (built separately)

```
AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.appendix_naive_strict_ratio_conclusionSpec_proof
```

Recorded source-to-Spec screening Verdict: matches

Reason: The source conclusion is exactly that the displayed algorithm/OPT factor is strictly below $1 - 1/e$, as is the normalized strict inequality in the target.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 25

Source locator: cited publication, lines 1221-1224

Verbatim source input

We can modify the above example to allow bids of 0, a and κa , for any $\kappa > 1$, such that the algorithm performs strictly worse than $1 - 1/e$.
As $\kappa \rightarrow \infty$, the factor tends to $1 - 1/e$, and as $\kappa \rightarrow 1$, the factor tends to $1/2$. Of course, if $\kappa = 1$, then this reduces to the original model of [KP00], and the factor is $1 - 1/e$.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For each $\kappa > 1$, the approved normalized continuous-limit witness is feasible, has unit offline value, and has factor strictly below $1 - 1/e$; its κ -to-one $\hookrightarrow$ limit remains distinct from the separate $\kappa = 1$ BALANCE instance.

Archival source anchor: cited publication, lines 1221-1224

Expanded PaperInterface specification

```

 $\forall \{ \kappa : \mathbb{R} \},$ 
 $1 < \kappa \rightarrow$ 
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue  $\kappa =$ 
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio  $\kappa \wedge$ 
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa = 1 \wedge$ 
    (AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible fun ( $\_bidder : \mathbb{R}$ ) => 1)  $\wedge$ 
    ( $\forall$  ( $spend : \mathbb{R} \rightarrow \mathbb{R}$ ),
      AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible  $spend \rightarrow$ 
         $\int$  ( $bidder : \mathbb{R}$ ) in 0..1,  $spend\ bidder \leq$ 
          AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa$ )  $\wedge$ 
        AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue  $\kappa /$ 
          AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa <$ 
          AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)

Lean-elaborated claim roles

```

1. parameter:  $\mathbb{R}$ 
2. assumption:  $1 < \kappa$ 
3. conclusion: AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue  $\kappa =$ 
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio  $\kappa \wedge$ 
  AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa = 1 \wedge$ 
  (AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible fun ( $\_bidder : \mathbb{R}$ ) => 1)  $\wedge$ 
  ( $\forall$  ( $spend : \mathbb{R} \rightarrow \mathbb{R}$ ),
    AppliedModelingLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible  $spend \rightarrow$ 
       $\int$  ( $bidder : \mathbb{R}$ ) in 0..1,  $spend\ bidder \leq$ 
        AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa$ )  $\wedge$ 
      AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue  $\kappa /$ 
        AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa <$ 
        AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.appendix_kappa_counterexample_family

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: The unspecified archival discrete three-bid family is replaced by the approval-pinned normalized continuous feasible witness with unit offline value and factor below $1 - 1/e$; the approval expressly disclaims archival equivalence.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 26

Source locator: cited publication, lines 1221-1224

Verbatim source input

We can modify the above example to allow bids of 0, a and κa , for any $\kappa > 1$, such that the algorithm performs strictly worse than $1 - 1/e$.
As $\kappa \rightarrow \infty$, the factor tends to $1 - 1/e$, and as $\kappa \rightarrow 1$, the factor tends to $1/2$. Of course, if $\kappa = 1$, then this reduces to the original model of [KP00], and the factor is $1 - 1/e$.

Expanded PaperInterface specification

```
( $\forall (\kappa : \mathbb{R}),$   
   $1 < \kappa \rightarrow$   
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue  $\kappa$  /  
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa$  =  
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio  $\kappa$ )  $\wedge$   
  Filter.Tendsto AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio Filter.atTop  
  (nhds AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio)
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)

Lean-elaborated claim roles

```
1. conclusion: ( $\forall (\kappa : \mathbb{R}),$   
   $1 < \kappa \rightarrow$   
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue  $\kappa$  /  
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa$  =  
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio  $\kappa$ )  $\wedge$   
  Filter.Tendsto AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio Filter.atTop  
  (nhds AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio)
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.appendix_kappa_ratio_limit_infinity

Recorded source-to-Spec screening Verdict: matches

Reason: Both state that the construction's online/offline factor is the kappa-ratio and tends to $1 - 1/e$ as kappa tends to infinity.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 27

Source locator: cited publication, lines 1221-1224

Verbatim source input

We can modify the above example to allow bids of 0, a and κa , for any $\kappa > 1$, such that the algorithm performs strictly worse than $1 - 1/e$.
As $\kappa \rightarrow \infty$, the factor tends to $1 - 1/e$, and as $\kappa \rightarrow 1$, the factor tends to $1/2$. Of course, if $\kappa = 1$, then this reduces to the original model of [KP00], and the factor is $1 - 1/e$.

Expanded PaperInterface specification

```
( $\forall (\kappa : \mathbb{R}),$   
   $1 < \kappa \rightarrow$   
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue  $\kappa$  /  
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa$  =  
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio  $\kappa$ )  $\wedge$   
  Filter.Tendsto AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio (nhdsWithin 1 (Set.lci 1)) (nhds (1 / 2))  $\wedge$   
   $1 / 2 \neq$  AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue](#)
- [AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)

Lean-elaborated claim roles

```
1. conclusion: ( $\forall (\kappa : \mathbb{R}),$   
   $1 < \kappa \rightarrow$   
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue  $\kappa$  /  
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue  $\kappa$  =  
    AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio  $\kappa$ )  $\wedge$   
  Filter.Tendsto AppliedModelingLib.Online.MSVV07Appendix.derivedKappaRatio (nhdsWithin 1 (Set.lci 1)) (nhds (1 / 2))  $\wedge$   
   $1 / 2 \neq$  AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.appendix_kappa_ratio_limit_one

Recorded source-to-Spec screening Verdict: matches

Reason: The $\kappa > 1$ ratio, its right-hand $\kappa \rightarrow 1$ limit $1/2$, and its difference from the separate $\kappa = 1$ factor $1 - 1/e$ exactly express the source's limiting-versus-endpoint statement.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 28

Source locator: cited publication, lines 897-908

Verbatim source input

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b .
Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/(U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i , which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i , while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i , achieving a revenue of N . One can show that BALANCE will achieve only $N(1 - 1/e)$ revenue on this input. Now consider all the inputs which can be derived from the above input by permutation of the

Expanded PaperInterface specification

$\forall (\text{delta} : \mathbb{R}),$
 $0 < \text{delta} \rightarrow$
 $\exists (N0 : \mathbb{N}),$
 $\forall (N : \mathbb{N}),$
 $N0 \leq N \rightarrow$
`AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9BaseEqualSpreadBalanceRevenue N / ↑ N ≤`
`AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta`

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9BaseEqualSpreadBalanceRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. assumption: $0 < \text{delta}$
3. conclusion: $\exists (N0 : \mathbb{N}),$
 $\forall (N : \mathbb{N}),$
 $N0 \leq N \rightarrow$
`AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9BaseEqualSpreadBalanceRevenue N / ↑ N ≤`
`AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta`

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_base_instance_balance_revenue_claim`

Recorded source-to-Spec screening Verdict: matches
Reason: The source's BALANCE hard-instance value is an asymptotic $1-1/e$ statement; the target gives its standard $\text{delta}-N0$ upper-bound formulation.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 29

Source locator: cited publication, lines 918-931

Verbatim source input

1
if $j \geq i$,
En $[q_{ij}] \leq N - i + 1$
0
if $j < i$.
To see this, note that there are $N - i + 1$ bidders who are bidding for queries Q_i . The deterministic algorithm allocates some fraction of these queries to some bidders who bid for them, and leaves the rest of the queries unallocated. If $j \geq i$ then bidder j is a random bidder among the bidders bidding for these queries and hence is allocated an average amount
1
of the queries which were allocated from Q_i (where the average is taken over random
of $N - i + 1$
permutations of the bidders). On the other hand, if $j < i$, then bidder j bids 0 for queries in Q_i and is not allocated any of these queries in any permutation.

Expanded PaperInterface specification

$\forall (N : \mathbb{N})$ (algorithm : `AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N`) (round bidder : `Fin N`),
(`AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round bidder` $\leq$
if $\uparrow \text{round} \leq \uparrow \text{bidder}$ then $1 / \uparrow (N - \uparrow \text{round})$ else 0) $\wedge$
($\neg \uparrow \text{round} \leq \uparrow \text{bidder} \rightarrow$
`AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round`
bidder =
0)

Retained governing declarations

- [AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: `AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N`
3. parameter: `Fin N`
4. parameter: `Fin N`
5. conclusion: (`AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round bidder` $\leq$
if $\uparrow \text{round} \leq \uparrow \text{bidder}$ then $1 / \uparrow (N - \uparrow \text{round})$ else 0) $\wedge$
($\neg \uparrow \text{round} \leq \uparrow \text{bidder} \rightarrow$
`AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round bidder` =
0)

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_expected_round_allocation_bound`

Recorded source-to-Spec screening Verdict: matches

Reason: The target states $E[q_{ij}] \leq 1/(N-i+1)$ for eligible positions and zero otherwise, matching the displayed source bound with zero-based indexing.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 30

Source locator: cited publication, lines 929-938

Verbatim source input

of $N - i + 1$ permutations of the bidders). On the other hand, if $j < i$, then bidder j bids 0 for queries in Q_i and is not allocated any of these queries in any permutation. Thus we get that the amount of money spent by a bidder j at the end of the algorithm P_j^{expected} is at most $\min\{1, \sum_{i=1}^N N - i + 1\}$. By summing this over $j = 1, \dots, N$, we get that the expected revenue of the deterministic algorithm over the distributional input D is at most $N(1 - 1/e)$. This finishes the proof of the theorem.

Expanded PaperInterface specification

```
∀ (N : ℕ) (algorithm : AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N) (bidder : Fin N),
  AppliedModelingLib.Online.theorem9BidderSpendUpperBound N bidder =
    min 1 (∑ round : Fin N, if ↑round ≤ ↑bidder then 1 / ↑(N - ↑round) else 0) ∧
    min 1
      (∑ round : Fin N,
        AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round
          bidder) ≤
        AppliedModelingLib.Online.theorem9BidderSpendUpperBound N bidder ∧
        AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N =
          (∑ b : Fin N, AppliedModelingLib.Online.theorem9BidderSpendUpperBound N b) / ↑N ∧
          (AppliedModelingLib.pmfExp (AppliedModelingLib.Online.uniformPermutationDistribution N)
            fun (permutation : Equiv.Perm (Fin N)) =>
              (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralRoundCertificate
                N).normalizedRevenue
                algorithm permutation) ≤
              AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N ∧
          ∀ (delta : ℝ),
            0 < delta →
              ∃ (N0 : ℕ),
                ∀ (marketSize : ℕ),
                  N0 ≤ marketSize →
                    AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound marketSize ≤
                      AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta
```

Retained governing declarations

- [AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm](#)
- [AppliedModelingLib.Online.BMatchingRoundAllocationRevenueCertificate](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralRoundCertificate](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [AppliedModelingLib.Online.theorem9BidderSpendUpperBound](#)
- [AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound](#)
- [AppliedModelingLib.Online.uniformPermutationDistribution](#)
- [AppliedModelingLib.pmfExp](#)

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: AppliedModelingLib.Online.BMatchingIntegralPrefixAlgorithm N
3. parameter: Fin N
4. conclusion: AppliedModelingLib.Online.theorem9BidderSpendUpperBound N bidder =
  min 1 (∑ round : Fin N, if ↑round ≤ ↑bidder then 1 / ↑(N - ↑round) else 0) ∧
  min 1
    (∑ round : Fin N,
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round
        bidder) ≤
      AppliedModelingLib.Online.theorem9BidderSpendUpperBound N bidder ∧
      AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N =
        (∑ b : Fin N, AppliedModelingLib.Online.theorem9BidderSpendUpperBound N b) / ↑N ∧
        (AppliedModelingLib.pmfExp (AppliedModelingLib.Online.uniformPermutationDistribution N)
          fun (permutation : Equiv.Perm (Fin N)) =>
            (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralRoundCertificate
              N).normalizedRevenue
              algorithm permutation) ≤
            AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound N ∧
        ∀ (delta : ℝ),
          0 < delta →
            ∃ (N0 : ℕ),
              ∀ (marketSize : ℕ),
                N0 ≤ marketSize →
                  AppliedModelingLib.Online.theorem9NormalizedRevenueUpperBound marketSize ≤
                    AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_harmonic_spend_and_revenue_bound

Recorded source-to-Spec screening Verdict: matches

Reason: It combines the source's per-bidder harmonic cap, summed normalized expected-revenue bound, and asymptotic $1-1/e$ limit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 31

Source locator: cited publication, lines 897-938

Verbatim source input

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b.

Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/(U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i, which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i, while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i, achieving a revenue of N. One can show that BALANCE will achieve only $N(1 - 1/e)$ revenue on this input. Now consider all the inputs which can be derived from the above input by permutation of the numbers of the bidders and take the uniform distribution D over all these inputs. Formally, D can be described as follows: Pick a random permutation π of the bidders. The queries enter in rounds in the order $Q_1, Q_2, \dots, Q_N$. Bidders $\pi(i), \pi(i + 1), \dots, \pi(N)$ bid $[U+000F$ control character] for the queries Q_i and the other bidders bid 0 for these queries. The optimal allocation for any permutation π remains N, by allocating the queries Q_i to bidder $\pi(i)$. We wish to bound the expected revenue of any deterministic algorithm over inputs from the distribution D. Fix any deterministic algorithm. Let q_{ij} be the fraction of queries from Q_i that bidder j is allocated. We have:

(
1
if $j \geq i$,
E $\pi [q_{ij}] \leq N - i + 1$
0
if $j < i$.

To see this, note that there are $N - i + 1$ bidders who are bidding for queries Q_i . The deterministic algorithm allocates some fraction of these queries to some bidders who bid for them, and leaves the rest of the queries unallocated. If $j \geq i$ then bidder j is a random bidder among the bidders bidding for these queries and hence is allocated an average amount

1
of the queries which were allocated from Q_i (where the average is taken over random
of $N - i + 1$
permutations of the bidders). On the other hand, if $j < i$, then bidder j bids 0 for queries in
 Q_i and is not allocated any of these queries in any permutation.

Thus we get that the
amount of money spent by a bidder j at the end of the algorithm
 P_j^{expected}

1
is at most $\min\{1, N - i + 1\}$. By summing this over $j = 1, \dots, N$, we get that the expected
revenue of the deterministic algorithm over the distributional input D is at most $N(1 - 1/e)$.
This finishes the proof of the theorem.

Settled source reading Theorem 9 uses the source's q_{ij} normal form: each repeated-query round may be split among feasible visible bidders, and the lower bound is for all finite nonanticipating capacity-feasible policies on that hard family.

Expanded PaperInterface specification

```

forall (b :  $\mathbb{N}$ ),
  0 < b  $\rightarrow$ 
    forall (delta :  $\mathbb{R}$ ),
      0 < delta  $\rightarrow$ 
        exists (N0 :  $\mathbb{N}$ ),
          forall (N :  $\mathbb{N}$ ),
            N0  $\leq$  N  $\rightarrow$ 
              forall
                (randomizedAlgorithm :
                  AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceRandomizedOnlineAlgorithm N b),
                 $\neg$ forall (permutation : Equiv.Perm (Fin N)),
                  AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta <
                    AppliedModelingLib.pmfExp randomizedAlgorithm
                      fun (policy : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy N b) =>
                        AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceNormalizedRevenue policy
                          permutation
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceNormalizedRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceRandomizedOnlineAlgorithm](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [AppliedModelingLib.Online.instFintypeBMatchingTheorem9SourceHardPolicy](#)
- [AppliedModelingLib.pmfExp](#)

Lean-elaborated claim roles

```
1. parameter: ℕ
2. assumption: 0 < b
3. parameter: ℝ
4. assumption: 0 < delta
5. conclusion: ∃ (N0 : ℕ),
  ∀ (N : ℕ),
    N0 ≤ N →
      ∀
        (randomizedAlgorithm :
          AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceRandomizedOnlineAlgorithm N b),
        ¬∀ (permutation : Equiv.Perm (Fin N)),
          AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta <
            AppliedModelingLib.pmfExp randomizedAlgorithm
              fun (policy : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceHardPolicy N b) =>
                AppliedModelingLib.Online.MSVV07PaperFacing.Proof.theorem9SourceNormalizedRevenue policy permutation
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_no_randomized_online_algorithm_beats_msvv_ratio

Recorded source-to-Spec screening Verdict: matches

Reason: Selected source row ‘theorem9_no_randomized_online_algorithm_beats_msvv_ratio’: the source span ‘Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b.’ is covered by specialization of the expanded target ‘ $\forall (b : \mathbb{N}), 0 < b \rightarrow \dots$ ’. Every source-admissible sufficiently-large positive ‘b’ lies in the target’s positive-natural-‘b’ domain, so restricting the target’s universal ‘b’ binder to that source domain yields the stated source regime. The target’s ‘ $\forall \text{delta} > 0, \exists N0, \forall N \geq N0$ ’ bad-permutation conclusion is the asymptotic form of no competitive ratio strictly above ‘ $1 - 1/e$ ’, consistent with the source’s uniform-permutation/Yao proof. The displayed randomized-policy and normalized-revenue prerequisites remain indexed by that same ‘N, b’ hard family. No source-admissible case is absent from the target.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 32

Source locator: cited publication, lines 273-300

Verbatim source input

For simplicity we will assume that bidders of type i spend exactly i/k fraction of their budget, and that queries do not straddle slabs. The latter is justified by the fact that bids are small compared to budgets (e.g. taking bids to be smaller than $k/2$ of the budget). The total error resulting from this simplification is at most N/k and is negligible, once we take k to be large enough. Now, for $i = 1, 2, \dots, k - 1$, let x_i be the number of bidders of type (i) . Let β_i denote the total money spent by the bidders from slab i in the run of BALANCE. It is easy to see (Figure 1) that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

3/k
SLAB 3
2/k
TYPE 2

1/k
0

x_k

x_3

x_2

x_1

Figure 1: The bidders are ordered from right to left in order of increasing type. We have labeled here the bidders of type 2 and the money in slab 3.
Lemma 2

Expanded PaperInterface specification

$\forall \{k : \mathbb{N}\} (N : \mathbb{R}) (x : \text{Fin } k \rightarrow \mathbb{R}) (i : \text{Fin } k),$
`AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend` $N \ x \ i =$
 $N / \uparrow k - (\sum j \in \text{Finset.lto } i, x \ j) / \uparrow k$

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\mathbb{R}$
3. parameter: $\text{Fin } k \rightarrow \mathbb{R}$
4. parameter: $\text{Fin } k$
5. conclusion: `AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend` $N \ x \ i =$
 $N / \uparrow k - (\sum j \in \text{Finset.lto } i, x \ j) / \uparrow k$

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_slab_spend_beta_formula`

Recorded source-to-Spec screening Verdict: matches

Reason: Using zero-based i , `Finset.lto` is the prior-type prefix, so the target is $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 33

Source locator: cited publication, lines 270-279

Verbatim source input

for allocating q , and therefore B must allocate q to some bidder who has spent at most j/k fraction of his budget.
For simplicity we will assume that bidders of type i spend exactly i/k fraction of their budget, and that queries do not straddle slabs. The latter is justified by the fact that bids are small compared to budgets (e.g. taking bids to be smaller than $k/2$ of the budget). The total error resulting from this simplification is at most N/k and is negligible, once we take k to be large enough. Now, for $i = 1, 2, \dots, k - 1$, let x_i be the number of bidders of type (i) . Let β_i denote the total money spent by the bidders from slab i in the run of BALANCE. It is easy to see (Figure 1) that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

Expanded PaperInterface specification

$\forall \{ \text{Advertiser} : \text{Type } u_5 \} [\text{inst} : \text{Fintype Advertiser}] \{k : \mathbb{N}\},$
 $0 < k \rightarrow$
 $\forall (\text{error} : \text{Advertiser} \rightarrow \mathbb{R}),$
 $(\forall (a : \text{Advertiser}), 0 \leq \text{error } a) \rightarrow$
 $(\forall (a : \text{Advertiser}), \text{error } a \leq 1 / \uparrow k) \rightarrow \sum a : \text{Advertiser}, \text{error } a \leq \uparrow (\text{Fintype.card Advertiser}) / \uparrow k$

Lean-elaborated claim roles

1. parameter: $\text{Type } u_5$
2. parameter: $\text{Fintype } \text{Advertiser}$
3. parameter: $\mathbb{N}$
4. assumption: $0 < k$
5. parameter: $\text{Advertiser} \rightarrow \mathbb{R}$
6. assumption: $\forall (a : \text{Advertiser}), 0 \leq \text{error } a$
7. assumption: $\forall (a : \text{Advertiser}), \text{error } a \leq 1 / \uparrow k$
8. conclusion: $\text{Real.le } \uparrow (\sum a : \text{Advertiser}, \text{error } a) (\uparrow (\text{Fintype.card Advertiser}) / \uparrow k)$

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_discretization_error_bound

Recorded source-to-Spec screening Verdict: matches

Reason: Summing nonnegative per-bidder errors bounded by $1/k$ gives N/k , the stated bound after identifying N with the number of bidders.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 34

Source locator: cited publication, lines 344-456

Verbatim source input

The revenue of the algorithm is
BAL $\geq$

$$\sum_{i=1}^{k-1} x_i$$

$$+ \sum_{i=1}^k x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

$$= N -$$

To find a P lower bound on the performance of BALANCE we want to find the minimum value

$$\sum_{i=1}^{k-1} x_i$$

$$+ \sum_{i=1}^k x_i$$

that $N - \sum_{i=1}^{k-1} x_i$ can take over the feasible $\{x_i\}$ s. This gives the following LP, which

[form-feed in source extraction]
we call L. In both the constraints below, i ranges from 1 to $k - 1$.

maximize

$$\Phi =$$

$$\sum_{i=1}^{k-1} x_i$$

$$+ \sum_{i=1}^k x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

subject to

$$\forall i:$$

$$\sum_{i=1}^k x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

$$\forall i: x_i \geq 0$$

Let us also write down the dual LP, D, which we will use in the case of arbitrary bids.

minimize

$$\sum_{i=1}^{k-1} x_i$$

$$+ \sum_{i=1}^k x_i$$

$$= N -$$

$$\sum_{i=1}^{k-1} x_i$$

subject to

$$\forall i:$$

$$\sum_{i=1}^{k-1} x_i$$

$$= N -$$

$$\begin{aligned} & k-i \\ & j-i \\ & |y_j| \geq \\ & (1 + \\ & k \\ & k \\ & j=i \end{aligned}$$

$$\forall i : y_i \geq 0$$

Define A, b, c so the primal LP, L, can be written as
 $\max c \cdot x$

s.t. $Ax \leq b$ $x \geq 0$.

and the dual LP, D, can be written as
 $\min b \cdot y$

s.t. $ATy \geq c$

$y \geq 0$.

Lemma 3 As $k \rightarrow \infty$, the value Φ of the linear programs L and D goes to Ne

Expanded PaperInterface specification

```

∀ {m : ℕ} (N : ℝ) (x y : Fin m → ℝ),
  (AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).primalObjective x =
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective x ∧
    ((AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).PrimalFeasible x ↔
      AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalFeasible N x) ∧
  (AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).dualObjective y =
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualObjective N y ∧
    ((AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).DualFeasible y ↔
      AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualFeasible y)

```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualFeasible](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualObjective](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalFeasible](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.DualFeasible](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.PrimalFeasible](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.dualObjective](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.primalObjective](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. parameter: ℝ
3. parameter: Fin m → ℝ
4. parameter: Fin m → ℝ
5. conclusion: (AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).primalObjective x =
  AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective x ∧
  ((AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).PrimalFeasible x ↔
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalFeasible N x) ∧
  (AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).dualObjective y =
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualObjective N y ∧
  ((AppliedModelingLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).DualFeasible y ↔
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualFeasible y)

```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_factor_revealing_primal_dual_lp_formulas

Recorded source-to-Spec screening Verdict: matches

Reason: The target identifies the factor-revealing LP's primal objective/feasibility and dual objective/feasibility with the displayed $c \cdot x$, $Ax \leq b$ and $b \cdot y, A^T y \geq c$ formulas.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 35

Source locator: cited publication, lines 264-272

Verbatim source input

Lemma 1 If OPT assigns query q to a bidder B of type $j \leq k - 1$, then BALANCE pays for q from some slab i such that $i \leq j$.
The lemma follows immediately from the criterion used by BALANCE for assigning queries to bidders: B has type $j \leq k - 1$ and therefore spends at most $j/k < 1$ fraction of his budget at the end of BALANCE. It follows that when query q arrives, B is available to BALANCE

6

[form-feed in source extraction]
for allocating q , and therefore B must allocate q to some bidder who has spent at most j/k fraction of his budget.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser]
(l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query)
(opt : Advertiser) (optType : ℝ),
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign l S q opt →
S.spent opt / l.budget opt ≤ optType →
(∀ (a : Advertiser), l.bid a q = l.bid opt q) →
0 < l.bid opt q →
∃ (chosen : Advertiser),
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.balanceScan l S q).winner = some chosen ∧
S.spent chosen / l.budget chosen ≤ optType
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.balanceScan](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign](#)

Lean-elaborated claim roles

```
1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query
7. parameter: Query
8. parameter: Advertiser
9. parameter: ℝ
10. assumption: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign l S q opt
11. assumption: S.spent opt / l.budget opt ≤ optType
12. assumption: ∀ (a : Advertiser), l.bid a q = l.bid opt q
13. assumption: 0 < l.bid opt q
14. conclusion: ∃ (chosen : Advertiser),
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.balanceScan l S q).winner = some chosen ∧
S.spent chosen / l.budget chosen ≤ optType
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_lemma1_balance_pays_no_later_slab

Recorded source-to-Spec screening Verdict: matches

Reason: The source's Lemma 1 argument is the equal-positive-bid Balance choice: a feasible OPT owner whose current spent-budget fraction is at most the type threshold is an available candidate, so the finite maximum scan chooses a bidder at no larger fraction. 'optType' abstracts the source threshold j/k , and the winner conclusion is exactly the source statement that BALANCE pays from no later slab; it does not rely on the later no-straddling accounting convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 36

Source locator: cited publication, lines 273-330

Verbatim source input

For simplicity we will assume that bidders of type i spend exactly i/k fraction of their budget, and that queries do not straddle slabs. The latter is justified by the fact that bids are small compared to budgets (e.g. taking bids to be smaller than $k/2$ of the budget). The total error resulting from this simplification is at most N/k and is negligible, once we take k to be large enough. Now, for $i = 1, 2, \dots, k - 1$, let x_i be the number of bidders of type (i) . Let β_i denote the total money spent by the bidders from slab i in the run of BALANCE. It is easy to see (Figure 1) that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

3/k
SLAB 3
2/k
TYPE 2

1/k
0

x_k

x_3

x_2

x_1

Figure 1: The bidders are ordered from right to left in order of increasing type. We have labeled here the bidders of type 2 and the money in slab 3.
Lemma 2

$\sum_{i=1}^k x_i \leq N$

$\forall i, 1 \leq i \leq k - 1 :$

$j=1$

Proof :

By Lemma 1,
 $\sum_{i=1}^k x_i$

$x_j \leq$

$\sum_{i=1}^k x_i$
 $j=1$

$j=1$

i

Settled source reading Section 4 explicitly idealizes exact type endpoints and nonstraddling payments; the checked Lemma 2 derives its beta formula from that idealized slab definition, while its prior-Lemma-1 prefix aggregation remains the source proof splice.

Expanded PaperInterface specification

```
∀ {m : ℕ} {Advertiser : Type u_5} {Event : Type u_6} [inst : Fintype Advertiser] [inst_1 : Fintype Event]
(E : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event)
(i : Fin m),
AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow E.typeCount i ≤
AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs (↑(Fintype.card Advertiser)) i
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution.typeCount](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs](#)

Lean-elaborated claim roles

- 1. parameter: ℕ
- 2. parameter: Type u_5
- 3. parameter: Type u_6
- 4. parameter: Fintype Advertiser
- 5. parameter: Fintype Event
- 6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.Section4IdealizedBalanceExecution m Advertiser Event
- 7. parameter: Fin m
- 8. conclusion: Real.le↑ (AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow E.typeCount i) (AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs (↑(Fintype.card Advertiser)) i)

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_lemma2_factor_revealing_lp_constraint

Recorded source-to-Spec screening Verdict: matches

Reason: The source Lemma 2 requires, for every paper index $I = 1, \dots, k - 1$, $\sum_{J=1}^I (1 + (I-J)/k) x_J \leq IN/k$. With $k = m + 1$, $I = i.val + 1$, $J = j.val + 1$, $N = \text{Fintype.card Advertiser}$, and x_J given by the displayed zero-indexed typeCount, paperRoutePrefix expands to $J \leq I$, paperRouteCoeff to $1 + (I-J)/k$, and paperRouteRhs to IN/k . The displayed approved Section 4 exact-type/no-straddling convention and the execution fields supply the same finite idealized ledger, including its BALANCE score comparison, per-advertiser unit OPT expenditure, and slab/type accounting. The zero-indexing is only a relabeling; no source premise, formula, quantifier, or conclusion is changed.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 37

Source locator: cited publication, lines 456-490

Verbatim source input

Lemma 3 As $k \rightarrow \infty$, the value Φ of the linear programs L and D goes to N/e
Proof : On setting all the primal constraints to equality and solving the resulting system, we get a feasible solution $x^* \geq 0$. Similarly, we can set all the dual constraints to equality and solve the resulting system to get a feasible dual solution.
These two feasible solutions are:
 $x^*_i = Nk(1 - k^{-1})^{i-1}$

for $i = 1, \dots, k - 1$

$y_i^* = k^{-1}(1 - k^{-1})^{k-i-1}$

for $i = 1, \dots, k - 1$

Clearly they satisfy all complementary slackness conditions, hence they are also optimal solutions of the primal and dual programs.

[form-feed in source extraction]
This gives an optimal objective function value of
 $\Phi = c \cdot x^* = b \cdot y^*$
=

$\sum_{i=1}^{k-1}$
 x_i

$\sum_{i=1}^N$
 $\sum_{k=i}^{k-1} k(1 - k^{-1})^{k-i}$

$\sum_{i=1}^N$

$= N(1 - k^{-1})^k$

As we make the discretization finer (i.e. as $k \rightarrow \infty$) Φ tends to N/e .
[U+0003 control character]

Recall that the size of the matching is at least $N - \Phi - Nk^{-1}$, hence it tends to $N(1 - 1/e)$. Since OPT is N , the competitive ratio is at least $1 - 1/e$.

Expanded PaperInterface specification

$\forall (N : \mathbb{R}), \text{AppliedModelingLib.Sequence.SeqTendsTo} (\text{fun } (k : \mathbb{N}) \Rightarrow N * (1 - 1 / \uparrow k) ^ k) (N / \text{Real.exp } 1)$

Retained governing declarations

- [AppliedModelingLib.Sequence.SeqTendsTo](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. parameter: $\mathbb{R}$
3. assumption: $0 < \delta$
4. conclusion: $\exists (N_1 : \mathbb{N}), \forall (k : \mathbb{N}), N_1 \leq k \rightarrow |(\text{fun } (k : \mathbb{N}) \Rightarrow N * (1 - 1 / \uparrow k) ^ k) k - N / \text{Real.exp } 1| \leq \delta$

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_lemma3_factor_revealing_lp_optimality`

Recorded source-to-Spec screening Verdict: matches

Reason: The target is the paper's limiting objective formula $N(1-1/k)^k \rightarrow N/e$; finite initial indices do not affect the sequence limit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 38

Source locator: cited publication, lines 488-495

Verbatim source input

[U+0003 control character]
Recall that the size of the matching is at least $N - \Phi - N_k$, hence it tends to $N(1 - 1e)$. Since OPT is N , the competitive ratio is at least $1 - 1e$.
On the other hand one can find an instance of the problem (e.g., the one provided in [KP00]) such that at the end of the algorithm all the inequalities of the primal are tight, hence the competitive ratio of BALANCE is exactly $1 - 1e$.

5

Expanded PaperInterface specification

```
∀ (m : ℕ),
let l :
  AppliedModelingLib.Online.AdWordsInstance
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m) :=
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m;
have H : List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m) :=
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineHistory m;
have R :
  AppliedModelingLib.Online.AdWordsInstance.ChoiceRule
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m) :=
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightBalanceRule m;
(∀
  (A :
    AppliedModelingLib.Online.AdWordsInstance.Assignment
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m))
  (q : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)
  (a : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m),
  R A q = some a → l.IsBalanceChoice A q a) ∧
  AppliedModelingLib.Online.AdWordsInstance.historyFinset H = Finset.univ ∧
  l.revenue (l.runHistoryState R H).assignment =
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue m ↑((m + 1) ^ m) ∧
  l.offlineOptimumValue
    (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_finite_balance_tightness_instanceSpec._proof_1
     m) =
    ↑((m + 1) ^ m) ∧
  l.revenue (l.runHistoryState R H).assignment /
    l.offlineOptimumValue
      (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_finite_balance_tightness_instanceSpec._proof_1
       m) =
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue m 1 ∧
  AppliedModelingLib.Sequence.SeqTendsTo
    (fun (r : ℕ) =>
      (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance r).revenue
        ((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance r).runHistoryState
         (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightBalanceRule r)
         (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineHistory
          r)).assignment /
        (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance r).offlineOptimumValue
          (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_finite_balance_tightness_instanceSpec._proof_1
           r))
    AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.Assignment](#)
- [AppliedModelingLib.Online.AdWordsInstance.ChoiceRule](#)
- [AppliedModelingLib.Online.AdWordsInstance.HistoryState](#)
- [AppliedModelingLib.Online.AdWordsInstance.IsBalanceChoice](#)
- [AppliedModelingLib.Online.AdWordsInstance.historyFinset](#)
- [AppliedModelingLib.Online.AdWordsInstance.offlineOptimumValue](#)
- [AppliedModelingLib.Online.AdWordsInstance.revenue](#)
- [AppliedModelingLib.Online.AdWordsInstance.runHistoryState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightBalanceRule](#)

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineHistory](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [AppliedModelingLib.Sequence.SeqTendsTo](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. conclusion: (∀
  (A :
    AppliedModelingLib.Online.AdWordsInstance.Assignment
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m)
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m))
  (q : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightQuery m)
  (a : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.FiniteTightAdvertiser m),
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightBalanceRule m A q = some a →
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).IsBalanceChoice A q a) ∧
  AppliedModelingLib.Online.AdWordsInstance.historyFinset
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineHistory m) =
  Finset.univ ∧
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).revenue
  ((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).runHistoryState
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightBalanceRule m)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineHistory m)).assignment =
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue m ↑((m + 1) ^ m) ∧
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).offlineOptimumValue
  (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_finite_balance_tightness_instanceSpec._proof_1
  m) =
  ↑((m + 1) ^ m) ∧
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).revenue
  ((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).runHistoryState
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightBalanceRule m)
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineHistory m)).assignment /
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance m).offlineOptimumValue
  (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_finite_balance_tightness_instanceSpec._proof_1
  m) =
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue m 1 ∧
  AppliedModelingLib.Sequence.SeqTendsTo
  (fun (r : ℕ) =>
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance r).revenue
    ((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance r).runHistoryState
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightBalanceRule r)
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightOnlineHistory
    r)).assignment /
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.finiteTightInstance r).offlineOptimumValue
    (AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_finite_balance_tightness_instanceSpec._proof_1
    r))
  AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio

```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section4_finite_balance_tightness_instance

Recorded source-to-Spec screening Verdict: matches

Reason: The cited Section 4 sentence is an existential tightness claim: it says an instance can be found, giving KP00 only as an example, and concludes that BALANCE attains the factor-LP bound. The finite Spec constructs such a family for every m, fixes an actual deterministic Balance tie rule, covers every query, proves the offline benchmark and exact factor-LP ratio, and proves its 1-1/e limit. It does not identify the construction with KP00 or attribute its incidence data to MSVV, which is a provenance clarification rather than a mismatch of the existential conclusion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 39

Source locator: cited publication, lines 509-549

Verbatim source input

Here is how we proceed instead: For every monotonically decreasing tradeoff function ψ and every instance π of the adwords problem and write a new LP $L(\pi, \psi)$ for our algorithm using tradeoff function ψ run on the instance π . Of course, once we specify the algorithm as well as the input instance, the actual allocations of queries to bidders is completely determined. In particular, the number α_i of bidders of type i is fixed. $L(\pi, \psi)$ is the seemingly trivial LP obtained by taking the left hand side of each inequality in the factor revealing LP L and substituting $x_i = \alpha_i$ to obtain the right hand side. Formally:

Recall the LP L from the previous section:

$\max c \cdot x$

s.t. $Ax \leq b$ $x \geq 0$

Let a be a $k - 1$ dimensional vector whose i th component is α_i . Let $Aa = l$. We denote the following LP by $L(\pi, \psi)$:

$\max c \cdot x$

s.t. $Ax \leq l$

$x \geq 0$

9

[form-feed in source extraction]

The dual LP is denoted by $D(\pi, \psi)$ and is:

$\min l \cdot y$

s.t. $ATy \geq c$

$y \geq 0$

Clearly, any one LP $L(\pi, \psi)$ offers no insight into the performance of our algorithm; after all the right hand sides of the inequalities are expressed in terms of the unknown number of bidders of type i . Nevertheless, the entire family $L(\pi, \psi)$ does contain useful information which is revealed by considering the duals of these LP's.

Since $L(\pi, \psi)$ differs from L only in the right hand side, the dual $D(\pi, \psi)$ differs from D only in the dual objective function; the constraints remain unchanged. Hence solution y^* of D is feasible for $D(\pi, \psi)$ as well. Recall that this solution was obtained by setting all nontrivial inequalities of D to equality.

Now by construction, if we set all the nontrivial inequalities of LP $L(\pi, \psi)$ to equality we get a feasible solution, namely a . Clearly, a and y^* satisfy all complementary slackness conditions. Therefore they are both optimal. Hence we get:

Lemma 4 For any instance π and monotonically decreasing tradeoff function ψ , y^* is an optimal solution to $D(\pi, \psi)$.

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\} (c : \text{Fin } m \rightarrow \mathbb{R}),$
 $(\text{AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP } c \text{ l}).\text{primalObjective } x = \sum i : \text{Fin } m, c \ i * x \ i \wedge$
 $((\text{AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP } c \text{ l}).\text{PrimalFeasible } x \leftrightarrow$
 $(\forall (i : \text{Fin } m), 0 \leq x \ i) \wedge$
 $\forall (r : \text{Fin } m),$
 $\sum i : \text{Fin } m, \text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff } r \ i * x \ i \leq l \ r) \wedge$
 $(\text{AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP } c \text{ l}).\text{dualObjective } y = \sum r : \text{Fin } m, y \ r * l \ r \wedge$
 $((\text{AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP } c \text{ l}).\text{DualFeasible } y \leftrightarrow$
 $(\forall (r : \text{Fin } m), 0 \leq y \ r) \wedge$
 $\forall (i : \text{Fin } m),$
 $c \ i \leq \sum r : \text{Fin } m, \text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff } r \ i * y \ r)$

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.DualFeasible](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.PrimalFeasible](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.dualObjective](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.primalObjective](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\text{Fin } m \rightarrow \mathbb{R}$
3. parameter: $\text{Fin } m \rightarrow \mathbb{R}$
4. parameter: $\text{Fin } m \rightarrow \mathbb{R}$
5. parameter: $\text{Fin } m \rightarrow \mathbb{R}$
6. conclusion: $(\text{AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP } c \text{ l}).\text{primalObjective } x = \sum i : \text{Fin } m, c \ i * x \ i \wedge$
 $((\text{AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP } c \text{ l}).\text{PrimalFeasible } x \leftrightarrow$
 $(\forall (i : \text{Fin } m), 0 \leq x \ i) \wedge$
 $\forall (r : \text{Fin } m),$

$$\begin{aligned} & \sum_i : \text{Fin } m, \text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff } r_i * x_i \leq l_r) \wedge \\ & (\text{AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP } c \ l), \text{dualObjective } y = \sum_r : \text{Fin } m, y_r * l_r \wedge \\ & ((\text{AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP } c \ l).\text{DualFeasible } y \leftrightarrow \\ & (\forall (r : \text{Fin } m), 0 \leq y_r) \wedge \\ & \forall (i : \text{Fin } m), \\ & c_i \leq \sum_r : \text{Fin } m, \text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff } r_i * y_r) \end{aligned}$$

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_tradeoff_revealing_primal_dual_lp_formulas

Recorded source-to-Spec screening Verdict: matches

Reason: The target gives max $c \cdot x$ subject to $Ax \leq l, x \geq 0$ and its min l^*y dual with $A^T y \geq c$, the displayed $L(\pi, \psi)$ and $D(\pi, \psi)$ formulas.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 40

Source locator: cited publication, lines 597-617

Verbatim source input

Lemma 7

$$\psi(i)(\alpha_i - \beta_i) \leq$$

$$i=1$$

Proof :

N

i

We start by observing that for $1 \leq i \leq k - 1$:

X

$$\text{OPT}(q) = \alpha_i$$

X

$$\text{ALG}(q) = \beta_i$$

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] {k : ℕ}
(I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k),
(∀ (a : Advertiser), AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend I history opt a = 1) →
  ∀ (i : Fin k),
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5OptTypeFiberRevenue I history opt finalType i =
      ∑ a : Advertiser, if finalType a = i then 1 else 0
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5OptTypeFiberRevenue](#)

Lean-elaborated claim roles

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: ℕ
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
7. parameter: List Query
8. parameter: Fin history.length → Option Advertiser
9. parameter: Advertiser → Fin k
10. assumption: ∀ (a : Advertiser), AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend I history opt a = 1
11. parameter: Fin k
12. conclusion: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5OptTypeFiberRevenue I history opt finalType i = ∑ a : Advertiser, if finalType a = i then 1 else 0

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_type_fiber_alpha_accounting_formula

Recorded source-to-Spec screening Verdict: matches

Reason: Under the target's explicit full-unit-spend premise, the OPT revenue on final-type i bidders is their count, the source's alpha_i identity in the unit-budget reduction.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 41

Source locator: cited publication, lines 597-617

Verbatim source input

Lemma 7

$$\psi(i)(\alpha_i - \beta_i) \leq$$

$$i=1$$

Proof :

N

k

We start by observing that for $1 \leq i \leq k - 1$:

$$\text{OPT}(q) = \alpha_i$$

X

$$\text{ALG}(q) = \beta_i$$

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ) (hk : 0 < k)
  (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (decisions : List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query))
  (i : Fin k),
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgSlabFiberRevenue k hk I decisions i =
    (List.map
      (fun (d : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) =>
        if AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab k hk I d = some i then
          AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue I d
        else 0)
      decisions).sum
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgSlabFiberRevenue](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: ℕ
4. assumption: 0 < k
5. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
6. parameter: List (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query)
7. parameter: Fin k
8. conclusion: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgSlabFiberRevenue k hk I decisions i =
  (List.map
    (fun (d : AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) =>
      if AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab k hk I d = some i then
        AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue I d
      else 0)
    decisions).sum
```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_slab_fiber_beta_accounting_formula

Recorded source-to-Spec screening Verdict: matches

Reason: The target sums $\text{ALG}(q)$ over precisely the decisions whose query slab is i , which is the β_i accounting used in the paper.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 42

Source locator: cited publication, lines 713-731

Verbatim source input

The lemma follows from these three inequalities.
[U+0003 control character]
The final step consists of choosing the correct tradeoff function ψ as a function of the dual optimal solution $y \star$ itself, so that for every instance π , the value of the optimal solution to $L(\pi, \psi)$ is at most that of L .
Theorem 8 For function ψ_k defined as
 $\psi_k(i) :=$
$$\begin{matrix} k-1 \\ X \\ j=i \\ 1 \\ y_j \star = 1 - (1 -)^{k-i+1} \\ k \end{matrix}$$

the competitive ratio of the algorithm is $(1 - 1e)$, as k tends to infinity.
Proof :
By Lemma 4, the optimal solution to $L(\pi, \psi)$ and $D(\pi, \psi)$ has value $l \cdot y \star$. By

Formalized review target The archival source above is not asserted equivalent to this formalized target.

Theorem 8's finite tradeoff weight is the suffix sum of the displayed dual witness and equals the geometric closed form with exponent $k-i$, not the archival $\hookrightarrow$ off-by-one exponent $k-i+1$.

Archival source anchor: cited publication, lines 718-729

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\} (i : \text{Fin } m),$
`AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate i = 1 - (1 - 1 / ↑(m + 1)) ^ (m - ↑i)`

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate](#)

Lean-elaborated claim roles

- 1. parameter: $\mathbb{N}$
- 2. parameter: $\text{Fin } m$
- 3. conclusion: `AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate i = 1 - (1 - 1 / ↑(m + 1)) ^ (m - ↑i)`

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_dual_induced_tradeoff_formula`

Recorded source-to-Spec screening Verdict: matches approved corrected target
Reason: The archival display has exponent $k-i+1$; the target's $k-i$ suffix-sum closed form is exactly the approval-pinned off-by-one correction, not a literal source match.
Reviewer check: ☐ Matches formalized target
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 43

Source locator: cited publication, lines 731-773

Verbatim source input

By Lemma 4, the optimal solution to $L(\pi, \psi)$ and $D(\pi, \psi)$ has value $l \cdot y^*$. By Lemma 5 this equals $(b + \Delta) \cdot y^* \leq N/e + \Delta \cdot y^*$ (since $b \cdot y^* \leq N/e$, from Section 4). Now,

$$\begin{aligned} \Delta \cdot y &= \\ &\sum_{k=1}^{k-1} y \cdot ((\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)) \\ &= \sum_{k=1}^{k-1} (\alpha_i - \beta_i)(y_1 + \dots + y_{k-1}) \\ &= \sum_{k=1}^{k-1} (\alpha_i - \beta_i)\psi(i) \\ &= \sum_{i=1}^i (\alpha_i - \beta_i) \end{aligned}$$

where the last equality follows from our choice of the function ψ , and the inequality follows from Lemma 7. As k tends to infinity, we get that the competitive ratio of our algorithm is $(1 - 1/e)$.

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\} (\alpha \beta \gamma : \text{Fin } m \rightarrow \mathbb{R}),$
 $\sum i : \text{Fin } m, \gamma \ i * \text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDelta } \alpha \beta \gamma \ i =$
 $\sum i : \text{Fin } m, \text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual } \gamma \ i * (\alpha \ i - \beta \ i)$

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDelta](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\text{Fin } m \rightarrow \mathbb{R}$
3. parameter: $\text{Fin } m \rightarrow \mathbb{R}$
4. parameter: $\text{Fin } m \rightarrow \mathbb{R}$
5. conclusion: $\sum i : \text{Fin } m, \gamma \ i * \text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDelta } \alpha \beta \gamma \ i =$
 $\sum i : \text{Fin } m, \text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual } \gamma \ i * (\alpha \ i - \beta \ i)$

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_delta_dual_suffix_identity

Recorded source-to-Spec screening Verdict: matches
Reason: The target is the finite sum rearrangement $\Delta \cdot y = \sum_i \psi_i(\alpha_i - \beta_i)$, where ψ_i is the dual suffix sum, exactly the displayed identity.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 44

Source locator: cited publication, lines 777-785

Verbatim source input

The above analysis helped us derive the correct tradeoff function ψ together with the competitive ratio. However, the proof of the competitive ratio of the $\hookrightarrow$ algorithm is simpler, once we are given the correct ψ . We give a quick sketch the main steps of such a proof below:
From the definitions of the α and β variables, we have the following relations:
$$P$$
$$N - i - 1$$
$$j = 1 \alpha_j$$
$$\forall i : \beta_i =$$
$$k$$

Lemma 7 gives us

Expanded PaperInterface specification

$\forall \{k : \mathbb{N}\} (N : \mathbb{R}) (\alpha : \text{Fin } k \rightarrow \mathbb{R}) (i : \text{Fin } k),$
`AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5IdealizedBeta` $N \text{ alpha } i =$
 $N / \uparrow k - (\sum j \in \text{Finset.lto } i, \text{ alpha } j) / \uparrow k$

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5IdealizedBeta](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\mathbb{R}$
3. parameter: $\text{Fin } k \rightarrow \mathbb{R}$
4. parameter: $\text{Fin } k$
5. conclusion: `AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5IdealizedBeta` $N \text{ alpha } i =$
 $N / \uparrow k - (\sum j \in \text{Finset.lto } i, \text{ alpha } j) / \uparrow k$

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_simple_proof_beta_recurrence_formula`

Recorded source-to-Spec screening Verdict: matches
Reason: The target is the recurrence $\beta_i = N/k - (\text{sum of earlier } \alpha_j)/k$ stated at the start of the simpler proof.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 45

Source locator: cited publication, lines 785-799

Verbatim source input

Lemma 7 gives us
 $\sum_i \psi(i) \alpha_i \leq$
 $\sum_i \psi(i) \beta_i$
where the choice of ψ is:

Expanded PaperInterface specification

$\forall \{k : \mathbb{N}\} (\psi : \text{Fin } k \rightarrow \mathbb{R}),$
 $\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8WeightedAlphaBeta } \psi \text{ alpha beta} \leftrightarrow$
 $\sum_i : \text{Fin } k, \psi \ i * \text{alpha } i \leq \sum_i : \text{Fin } k, \psi \ i * \text{beta } i$

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8WeightedAlphaBeta](#)

Lean-elaborated claim roles

- 1. parameter: $\mathbb{N}$
- 2. parameter: $\text{Fin } k \rightarrow \mathbb{R}$
- 3. parameter: $\text{Fin } k \rightarrow \mathbb{R}$
- 4. parameter: $\text{Fin } k \rightarrow \mathbb{R}$
- 5. conclusion: $\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8WeightedAlphaBeta } \psi \text{ alpha beta} \leftrightarrow \sum_i : \text{Fin } k, \psi \ i * \text{alpha } i \leq \sum_i : \text{Fin } k, \psi \ i * \text{beta } i$

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_simple_proof_weighted_alpha_beta_inequality`

Recorded source-to-Spec screening Verdict: matches
Reason: The iff expands the named predicate to exactly the weighted alpha-beta inequality displayed in the source.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 46

Source locator: cited publication, lines 777-819

Verbatim source input

The above analysis helped us derive the correct tradeoff function ψ together with the competitive ratio. However, the proof of the competitive ratio of the $\hookrightarrow$ algorithm is simpler, once we are given the correct ψ . We give a quick sketch the main steps of such a proof below:
From the definitions of the α and β variables, we have the following relations:

$$\sum_{i=1}^N \alpha_i = \sum_{i=1}^N \beta_i$$

Lemma 7 gives us

$$\sum_{i=1}^N \alpha_i \leq$$

$$\sum_{i=1}^N$$

where the choice of ψ is:

$$\sum_{i=1}^N$$

$$\sum_{i=1}^N \beta_i$$

$$\sum_{i=1}^N$$

$$\sum_{i=1}^N \psi(i) = 1 - (1 - \frac{1}{e})^{k-i+1}$$

Combining these relations we get:

$$\sum_{i=1}^N$$

$$\sum_{i=1}^N$$

$$\sum_{i=1}^N \alpha_i$$

$$\sum_{i=1}^N$$

$$\sum_{i=1}^N \alpha_i$$

$$\sum_{i=1}^N$$

$$\sum_{i=1}^N$$

$$\sum_{i=1}^N$$

But the left side of the inequality above is precisely the amount of money left unspent at the end of the algorithm. This establishes that the competitive ratio is $1 - 1/e$.

Expanded PaperInterface specification

$\forall (N \text{ revenue error} : \mathbb{R}),$
 $\text{AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio} * N \leq \text{revenue} + \text{error} \rightarrow$
 $N - \text{revenue} \leq N / \text{Real.exp } 1 + \text{error}$

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. parameter: $\mathbb{R}$
3. parameter: $\mathbb{R}$
4. assumption: $\text{AppliedModelingLib.Online.MSVV07PaperFacing.paperMsvvRatio} * N \leq \text{revenue} + \text{error}$
5. conclusion: $\text{Real.le} \uparrow (N - \text{revenue}) (N / \text{Real.exp } 1 + \text{error})$

Lean proof endpoint (built separately)

$\text{AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_unspent_budget_bound_formula}$

Recorded source-to-Spec screening Verdict: matches

Reason: Rearranging $(1-1/e)N \leq \text{revenue} + \text{error}$ gives unspent budget $N - \text{revenue} \leq N/e + \text{error}$, the quantitative conclusion used by the source proof.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 47

Source locator: cited publication, lines 509-549

Verbatim source input

Here is how we proceed instead: For every monotonically decreasing tradeoff function ψ and every instance π of the adwords problem and write a new LP $L(\pi, \psi)$ for our algorithm using tradeoff function ψ run on the instance π . Of course, once we specify the algorithm as well as the input instance, the actual allocations of queries to bidders is completely determined. In particular, the number α_i of bidders of type i is fixed. $L(\pi, \psi)$ is the seemingly trivial LP obtained by taking the left hand side of each inequality in the factor revealing LP L and substituting $x_i = \alpha_i$ to obtain the right hand side. Formally:

Recall the LP L from the previous section:

$\max c \cdot x$

s.t. $Ax \leq b$ $x \geq 0$

Let a be a $k - 1$ dimensional vector whose i th component is α_i . Let $Aa = l$. We denote the following LP by $L(\pi, \psi)$:

$\max c \cdot x$

s.t. $Ax \leq l$

$x \geq 0$

9

[form-feed in source extraction]

The dual LP is denoted by $D(\pi, \psi)$ and is:

$\min l \cdot y$

s.t. $ATy \geq c$

$y \geq 0$

Clearly, any one LP $L(\pi, \psi)$ offers no insight into the performance of our algorithm; after all the right hand sides of the inequalities are expressed in terms of the unknown number of bidders of type i . Nevertheless, the entire family $L(\pi, \psi)$ does contain useful information which is revealed by considering the duals of these LP's.

Since $L(\pi, \psi)$ differs from L only in the right hand side, the dual $D(\pi, \psi)$ differs from D only in the dual objective function; the constraints remain unchanged. Hence solution y^* of D is feasible for $D(\pi, \psi)$ as well. Recall that this solution was obtained by setting all nontrivial inequalities of D to equality.

Now by construction, if we set all the nontrivial inequalities of LP $L(\pi, \psi)$ to equality we get a feasible solution, namely a . Clearly, a and y^* satisfy all complementary slackness conditions. Therefore they are both optimal. Hence we get:

Lemma 4 For any instance π and monotonically decreasing tradeoff function ψ , y^* is an optimal solution to $D(\pi, \psi)$.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] (m : ℕ)
(l : AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) (history : List Query)
(psi : ℝ → ℝ),
Antitone psi →
  AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids l →
  AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets l →
  AppliedModelingLib.Optimization.IsMinimizerOn
    (AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP
      AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff fun (row : Fin m) =>
        ∑ i : Fin m,
          AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff row i *
            AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun
              m psi l history i).DualFeasible
    (AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP
      AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff fun (row : Fin m) =>
        ∑ i : Fin m,
          AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff row i *
            AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun
              m psi l history i).dualObjective
  AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate
```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.instDecidable_mSVV07AdWords](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff](#)

- [AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP](#)
- [AppliedModelingLib.Optimization.IsMinimizerOn](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.DualFeasible](#)
- [AppliedModelingLib.Optimization.StandardMaxLP.dualObjective](#)

Lean-elaborated claim roles

```

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: ℕ
5. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query
6. parameter: List Query
7. parameter: ℝ → ℝ
8. assumption: Antitone psi
9. assumption: AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids l
10. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets l
11. conclusion: (AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP
  AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff fun (row : Fin m) =>
    Σ i : Fin m,
      AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff row i *
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun
        psi l history i).DualFeasible
  AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate α
  ∀ (y : Fin m → ℝ),
    (AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP
      AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff fun (row : Fin m) =>
        Σ i : Fin m,
          AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff row i *
          AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun
            m psi l history i).DualFeasible
      y →
        (AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP
          AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff fun (row : Fin m) =>
            Σ i : Fin m,
              AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff row i *
              AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun
                m psi l history i).dualObjective
          AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate ≤
            (AppliedModelingLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP
              AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff fun (row : Fin m) =>
                Σ i : Fin m,
                  AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff row i *
                  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.section5PreterminalAlphaTypeCountOfTradeoffRun
                    m psi l history i).dualObjective
              y
            )
    )
  )

```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_lemma4_dual_optimal_from_primal_dual_match

Recorded source-to-Spec screening Verdict: matches

Reason: Lemma 4 quantifies over an AdWords instance and decreasing psi and concludes that the displayed y-star is optimal for D(pi, psi). The Spec uses the approved finite AdWords domain, actual-run alpha type counts, and RHS A alpha in the unchanged-constraint dual, and states that paperRouteDualCandidate is feasible and minimizes its dual objective; this is precisely that conclusion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 48

Source locator: cited publication, lines 560-577

Verbatim source input

Let $\Delta(\pi, \psi)$ be a $k - 1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$.
The following lemma relates the right hand side of the LPs L and $L(\pi, \psi)$.

Lemma 5

Proof :

$l = b + \Delta(\pi, \psi)$.

Consider the i th components of the three vectors. We need to prove:

$$\alpha_1 (1 + i - 1) + \alpha_2 (1 + k) + \dots + \alpha_i$$

$$= iN \\ k + (\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i).$$

This equation follows using the fact that $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$.

[U+0003 control character]

Settled source reading Section 5's idealized beta identity is derived from its definition; the actual dual-induced runner comparison exposes the finite endpoint term later absorbed by the Lemma 7 accounting.

Expanded PaperInterface specification

```
∀ {m : ℕ} (N : ℝ) (alpha : Fin m → ℝ) (i : Fin m),
  AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow alpha i =
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs N i +
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDelta alpha
    (AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteIdealBeta N alpha) i
```

Retained governing declarations

- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDelta](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteIdealBeta](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\mathbb{R}$
3. parameter: $\text{Fin } m \rightarrow \mathbb{R}$
4. parameter: $\text{Fin } m$
5. conclusion: $\text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteLPRow } \alpha \ i =$
 $\text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteRhs } N \ i +$
 $\text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteDelta } \alpha$
 $(\text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteIdealBeta } N \ \alpha) \ i$

Lean proof endpoint (built separately)

`AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_lemma5_tradeoff_rhs_eq_base_add_delta`

Recorded source-to-Spec screening Verdict: matches

Reason: This is the componentwise identity in source Lemma 5. 'paperRouteLPRow' is the displayed weighted alpha left side, 'paperRouteRhs' is iN/k after the zero-based shift, and 'paperRouteDelta alpha (paperRouteIdealBeta N alpha)' is the source prefix sum of $\alpha_j - \beta_j$. Thus the target is exactly $l = b + \Delta$ under the source's stated idealization.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 49

Source locator: cited publication, lines 583-593

Verbatim source input

Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .

Lemma 6 For each query q such that $1 \leq \text{type}(q) \leq k - 1$,
 $OPT(q)\psi(\text{type}(q)) \leq ALG(q)\psi(\text{slab}(q))$.

Proof : Consider the arrival of q during the run of the algorithm. Since $\text{type}(q) \leq k - 1$, the bidder b to whom OPT assigned this query is still actively bidding from some slab $j \leq \text{type}(q)$ at this time. The inequality in the lemma follows from the criterion used by the algorithm to assign queries, together with the monotonicity of ψ .
[U+0003 control character]

Settled source reading Section 5's idealized beta identity is derived from its definition; the actual dual-induced runner comparison exposes the finite endpoint term later absorbed by the Lemma 7 accounting.

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] (k : ℕ)
(hk : 0 < k) (I : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(t : Fin history.length) (opt : Fin history.length → Option Advertiser),
AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I →
AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I →
(∀ (a : Advertiser) (q : Query), I.bid a q ≤ I.budget a / ↑k) →
  ∀ (finalType : Advertiser → Fin k),
    (∀ (a : Advertiser),
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k
        ((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences k hk I
          history).spent
            a /
            I.budget a)
        (finalType a)) →
      AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8OptWeightedRevenueAt k I history opt
        finalType t ≤
        AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8AlgWeightedRevenueAt k hk I history t +
        AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.endpointRoundingErrorAt k I history opt
          finalType
        (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8CurrentSlabAt k hk I history) t

```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.endpointRoundingErrorAt](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8AlgWeightedRevenueAt](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8CurrentSlabAt](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8OptWeightedRevenueAt](#)

Lean-elaborated claim roles

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: ℕ
6. assumption: 0 < k
7. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
8. parameter: List Query
9. parameter: Fin history.length
10. parameter: Fin history.length → Option Advertiser
11. assumption: AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids I
12. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets I
13. assumption: ∀ (a : Advertiser) (q : Query), I.bid a q ≤ I.budget a / ↑k
14. parameter: Advertiser → Fin k
15. assumption: ∀ (a : Advertiser),
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k

```

((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences k hk l history).spent a /
l.budget a)
(finalType a)
16. conclusion: Real.le+
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8OptWeightedRevenueAt k l history opt finalType t)
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8AlgWeightedRevenueAt k hk l history t +
AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.endpointRoundingErrorAt k l history opt finalType
(AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8CurrentSlabAt k hk l history) t)

```

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_lemma6_per_query_tradeoff

Recorded source-to-Spec screening Verdict: matches

Reason: The target compares the same per-occurrence OPT and BALANCE/MSVV weighted revenues at Theorem 8's dual-induced geometric tradeoff. Its nonnegative-bid, positive-budget, small-bid, and final-type inputs make explicit the source domain and type convention. Consistently with the pinned Section 5 context, it replaces the idealized pointwise equality at a finite right endpoint by the displayed one-slab endpoint error; a terminal type has zero tradeoff weight, so retaining all Fin k types does not add a source claim.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 50

Source locator: cited publication, lines 597-699

Verbatim source input

Lemma 7

$$\psi(i)(\alpha_i - \beta_i) \leq$$

$$\sum_{i=1}^k$$

Proof :

N

k

We start by observing that for $1 \leq i \leq k - 1$:

X

$$\text{OPT}(q) = \alpha_i$$

X

$$\text{ALG}(q) = \beta_i$$

$$q:\text{type}(q)=i$$

$$q:\text{slab}(q)=i$$

By Lemma 6

X

$$[\text{OPT}(q)\psi(\text{type}(q)) - \text{ALG}(q)\psi(\text{slab}(q))]$$

$$q:\text{type}(q) \leq k-1$$

$$\leq 0.$$

Next observe that

X

$$\text{OPT}(q)\psi(\text{type}(q))$$

$$q:\text{type}(q) \leq k-1$$

$$=$$

$$\sum_{i=1}^{k-1}$$

X

X

$$\text{OPT}(q)\psi(i)$$

$$\sum_{i=1}^{k-1} q:\text{type}(q)=i$$

$$=$$

$$\sum_{i=1}^{k-1}$$

X

$$\psi(i)\alpha_i.$$

$$\sum_{i=1}^k$$

And

X

$$\text{ALG}(q)\psi(\text{slab}(q))$$

$$q:\text{type}(q) \leq k-1$$

11

[form-feed in source extraction]

X

$$\leq$$

$$\text{ALG}(q)\psi(\text{slab}(q))$$

$$q:\text{slab}(q) \leq k$$

X

$$\leq$$

$$\text{ALG}(q)\psi(\text{slab}(q)) +$$

$$q:\text{slab}(q) \leq k-1$$

$$=$$

$$\sum_{i=1}^{k-1}$$

X

$\sum_{i=1}^k \text{ALG}(q) \psi(i) +$
 $\sum_{i=1}^k \text{slab}(q) = i$
 $\psi(i) \beta_i +$

Settled source reading For the Theorem 8 dual-induced tradeoff, actual-run slab accounting gives finite weighted perturbation at most $2N/k$; the $1 - 1/e$ limit is unchanged.

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] (m : ℕ)
  (l : AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
  (opt : Fin history.length → Option Advertiser),
  AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids l →
  AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets l →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets l →
  (∀ (a : Advertiser) (q : Query), l.bid a q ≤ l.budget a / ↑(m + 1)) →
  AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentFeasible l history opt →
  (∀ (a : Advertiser),
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend l history opt a = 1) →
  ∃ (finalType : Advertiser → Fin (m + 1)),
  (∀ (a : Advertiser),
    AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType (m + 1)
    ((AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences (m + 1)
      (Nat.zero_lt_succ m) l history).spent
      a /
      l.budget a)
    (finalType a)) ∧
  ∑ i : Fin m,
  AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate i *
  (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8PreterminalAlphaFromFinalTypes
    m finalType i -
    AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteIdealBeta (↑(Fintype.card Advertiser))
    (AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8PreterminalAlphaFromFinalTypes
      m finalType)
    i) ≤
  2 * (↑(Fintype.card Advertiser) / ↑(m + 1))

```

Retained governing declarations

- [AppliedModelingLib.Online.AdWordsInstance](#)
- [AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids](#)
- [AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentFeasible](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences](#)
- [AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8PreterminalAlphaFromFinalTypes](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteIdealBeta](#)
- [AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate](#)

Lean-elaborated claim roles

1. parameter: Type u_5
2. parameter: Type u_6
3. parameter: Fintype Advertiser
4. parameter: DecidableEq Advertiser
5. parameter: ℕ
6. parameter: AppliedModelingLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query
7. parameter: List Query
8. parameter: Fin history.length → Option Advertiser
9. assumption: AppliedModelingLib.Online.AdWordsInstance.NonnegativeBids l
10. assumption: AppliedModelingLib.Online.AdWordsInstance.PositiveBudgets l
11. assumption: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets l
12. assumption: ∀ (a : Advertiser) (q : Query), l.bid a q ≤ l.budget a / ↑(m + 1)

13. assumption: AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceAssignmentFeasible I history opt
 14. assumption: $\forall (a : \text{Advertiser}), \text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend I history opt } a = 1$
 15. conclusion: $\exists (\text{finalType} : \text{Advertiser} \rightarrow \text{Fin } (m + 1)),$
 $(\forall (a : \text{Advertiser}),$
 $\text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType } (m + 1)$
 $((\text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.runTheorem8DiscreteOccurrences } (m + 1)$
 $(\text{Nat.zero_lt_succ } m) \text{ I history}).\text{spent}$
 $a /$
 $\text{I.budget } a)$
 $(\text{finalType } a)) \wedge$
 $\sum i : \text{Fin } m,$
 $\text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate } i *$
 $(\text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8PreterminalAlphaFromFinalTypes } m \text{ finalType}$
 $i -$
 $\text{AppliedModelingLib.Online.MSVV07SourceLemmas.paperRouteIdealBeta } (\uparrow (\text{Fintype.card Advertiser}))$
 $(\text{AppliedModelingLib.Online.MSVV07PaperFacing.SourceRunner.theorem8PreterminalAlphaFromFinalTypes } m$
 $\text{finalType})$
 $i) \leq$
 $2 * (\uparrow (\text{Fintype.card Advertiser}) / \uparrow (m + 1))$

Lean proof endpoint (built separately)

AppliedModelingLib.Online.MSVV07PaperFacing.PaperInterface.section5_lemma7_weighted_perturbation_bound

Recorded source-to-Spec screening Verdict: matches

Reason: The source Lemma 7 is the dual-weighted sum of $\alpha_i - \beta_i$ under unit budgets, small bids, and an exhausting feasible OPT allocation. The target represents N by the advertiser cardinality, α by the actual runner's preterminal final-type counts, β by the source ideal slab formula, and ψ by the displayed dual candidate. The pinned context explicitly accounts for finite endpoint and boundary crossings by the source-supported $2N/k$ bound, so this is the selected finite-run reading rather than an arbitrary- ψ strengthening.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation